

Phil 250 Philosophy of Education

Numbered Transcript

PHIL 250 • Dr. Stephen R. C. Hicks

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Lecture 1: Ancient Greek Education

At a glance — Hicks begins with ancient Greek education and the educator's ideal.

- 1.1 Let's begin with every educator's dream. Suppose you had unlimited budget, unlimited power to make the decisions. What would your perfect school look like? You're going to be the creator, the entrepreneur, a brand new school, and it can be anything you decide. What would it be?
- 1.2 Chances are good that you'll immediately start thinking, "Whoa, there's a huge amount of stuff that I want kids to learn." The world is really complicated. The world is really interesting. There's very important things that they need to know. And so you start generating a long list. Well, we live in a very scientific and technological world, so we've got to have the sciences, physics, chemistry, biology, and so on, and some understanding of technical systems, maybe computers.
- 1.3 But there's a whole history of things that human beings have done, and all of those interesting stories. So they really need to know something about history, partly to understand where they came from. And all of the literature that is such a source of delight and instruction, I want them to be exposed to all of the great literature. They're going to have their whole lives exploring the world. Well, the world—they need to have some geography to understand just how big the world is and what is where and so on.
- 1.4 Apparently, they're going to be citizens of this country, and that means they're going to be making some political decisions. So, we need them to understand the different political systems and religious systems and cultural systems of the world, as well as their own, and art. Maybe some of them will be artists, just enjoying playing some instruments and singing, or at least being able to appreciate all of the great, and so on. And immediately, we start having this great, huge, long list of things.
- 1.5 Now, we then start to think, OK, the school day, though, is going to be short. There's only so many hours in the day. So I have to start prioritizing which of these are really important and which ones are going to be a little bit less important. Do I say, for example, that knowledge of the way the world works is the most important thing? So I'm going to be a lot of science-heavy. Or do I say, no, maybe we want students to be more creative and doing their own thing and exploring freely, so maybe we're going to be emphasizing the arts more.
- 1.6 Now, that's one set of questions. How are we going to do the teaching? It's not what are we going to teach, but how are we going to teach? We want students to be able to read and talk because we learn a lot through conversation. We learn a lot through reading. We learn a lot through attentive listening to television shows, movies, music, and so forth. So we're going to develop all of our capacities for all of these things.

- 1.7 They need to be able to speak well to enter into all of these conversations and to develop their own thoughts. Now we're thinking about reading and writing and logic. Then if we want them to be able to do the science, for example, they're going to need to know the math, which is the primary tool of organizing and measuring all of the data that we use in math and so on.
- 1.8 Are we going to do it socially, where it's primarily, say, me the parent or me the teacher, with my group of students, doing instruction. Do I put them in groups because education is a social enterprise? Or do I more likely think that a lot of education is what individuals need to think for themselves? And so students need to, in many cases, have quiet time to work on their own. And maybe if I want to cultivate their own independent judgment, and not necessarily group consensus, we are going to deemphasize the social and emphasize the individual.
- 1.9 What's our school going to look like? We'll have labs, we'll have a music room, we'll have a gymnasium. Are we going to equally emphasize physical education, mental education? Do we think that one of those is more important than the other? Do we want students to be in rows? Do we want to have flexible furniture or maybe no furniture at all? An art studio? Students come in and they make it how they want?
- 1.10 Maybe we don't even want to have a school at all. We think school is artificial, it's separate from real life. We want students to be working. Work, education, go hand in hand, working with their parents, learning how to run a household, how to run a farm, how to run whatever the family business is, or as soon as possible, integrating the students into the community that they are going to be working, in which case, school as an artificial separation seems inappropriate.
- 1.11 Immediately, we are into all sorts of questions about education, and we have to answer all of those questions in order to formulate a plan, and it's going to be a strategic plan because we know kids start at age zero, and they're probably going to be under our tutelage, if we use that word, until they are adults. But for human beings, that's a long time. So we're starting at zero. We're trying to get them ready for age three, for age seven, for age 11, for age 17.
- 1.12 And at some point, we're going to say, maybe the formal education process has ended, and they're off on their own. How are we going to have a systematic plan that covers everything we want to cover in that amount of time, systematically? And how will we know that we've done a good job? How will we assess along the way that kids are learning what we want them to learn? Maybe that's not the right question, what we want them to learn.
- 1.13 How do we know the kids aren't learning what they, because they each have their own interests and their own paths, and they start making their own choices pretty soon, that they are progressing along their own life path? Do we do written tests, oral

examinations? Is it a matter of learning by doing, demonstrating actual practical skills? What's our assessment method going to be?

- 1.14** Now, point of all of this is, just in five minutes, we've got 50 or more questions on the table, and very soon we are into the terrain of philosophy, because all of these questions lead us to thinking deeply about what it is to be a human being, what kind of world we think we are preparing young people to enter into, and what we think the most important things we want young people to get out of their lives. What are their values? What is the purpose of their life?
- 1.15** We're not going to do education thoughtfully and systematically unless we attend also to those what I'm going to call philosophical questions. Now, that's all still somewhat abstract and quick, but let's plunge in a little bit. Let me give you a definition. We'll get very formal right off the bat. Now, this is a definition of education. Just consider this one as a rough draft, because it is a rough draft. Everything in philosophy, everything in education, is controversial. So consider this definition also as controversial.
- 1.16** We will put it out there. I'll give some explanation for some of the elements in it, but then expect lots of disagreement and argument along the way. But let's go through. Let's start here. "Education is the process of learning and/or teaching the young, the knowledge and skills necessary for successful adult life in the real world." OK, I said that's a rough draft, but actually some thought went into that. Each of those words is carefully chosen. Here, there's controversy in there. We'll be exploring the controversies over the course of this eight-lecture series.
- 1.17** But at this point, I really want to just focus on what's built into this definition philosophically, to introduce the concept of philosophy and why it's important for education as well. So, to do that, let's highlight a few concepts here. We say we're preparing students for the real world. Well, kids are already in the real world from day one, from the beginning. So, this is somewhat of a term of art, because we know that we insulate children from the full impact of the real world from the beginning.
- 1.18** They're in the real world, but there are insulating barriers—the parents, the room, the house, neighborhood— between them and the rest of the world. We make extra care before they learn various things and various skills, to protect them from sometimes themselves and from even things in a safe, insulated world. But the idea here is that, at some point, all of those insulating barriers will come off. There won't be their parents' home and their parents standing between them and the real world. At some point, they will be out in the world and they will go forth. That's what we mean.
- 1.19** But what is that real world? Well, we're making a distinction between the world of fantasy, the world of imagination, and reality. So there's a distinction there. You can't just live in your own dreams and your own fantasies. At some point, you have to get out and engage with the real world. And what's that? Well, it's trees and rocks and

rivers and weather systems and roads and cars and other people and animals. And you learn the Earth is a pretty big place, but it's one planet among many.

- 1.20** So we start teaching them about animals and human beings and physical objects and how to engage with all of those and their place in the larger universe. That's all real. They need to know about it and how to deal with it. But is that it? Or is there something beyond that natural, physical world? There also is another world, say, the world of the gods or the world of God. It's not really like this world. It's a different kind of reality, and there are some interactions here.
- 1.21** But if we want to teach students, young people, what the real world, it's not enough just to teach them about physical reality. We also have to teach them about something beyond the natural world, the supernatural, taking that literally, superior to; that maybe that is the world that created this world, that governs this world, and it is populated by gods, or maybe just one god, and they have certain features, and this is the real world.
- 1.22** So the point is, just with these three words, already we are into deep questions about, that every parent and educator has to think about. What's going to go into your curriculum? Is it going to include the sciences and the social sciences, and also religion or not? If we wanted to say, and we know this is one of the big philosophical controversies, that God is real, and that's the most important fact about the world, then of course that's got to be in our education.
- 1.23** On the other hand, if we think God and the gods are just made up, but we don't want just to teach made-up fantasy stories because that's a waste of time and sends people off down the wrong track. So this is a metaphysical issue. What is the nature of reality as such? Before we make any big decisions about what we're going to include in our curriculum, that is one philosophical issue we have to take up.
- 1.24** What if we emphasized this one? We're preparing them for adult life. A human being in the fully developed sense. We know children start young and they develop. Much of the history of education assumed that human beings from infancy were just tiny adults, and you just were basically scaling up to adult life. Is that true? Or is it the case that we go through different physical and developmental stages? It's not just scaling up a small thing to a big thing.
- 1.25** There are different stages, and we have to pay attention to those developmental stages. Sometimes it's more physical stages that we are going through, like puberty. Sometimes it's more psychological stages, say, when kids go into a school and they are intensely social for maybe the first time in their lives, instead of just smaller circle of family unit. There's all kinds of psychological stuff that's going on.
- 1.26** They're starting to think about their careers when they're a little bit older and what's the difference between love and infatuation when they're a little bit older—all of these adult issues that, in many cases, are just invisible to kids when they are 3, 4, 5, 6 years

old. But at the same time, we, as people who are parenting or educating young people, we have to have a pretty clear idea what this adult is that we want to create. What is it to be an adult in the full sense?

- 1.27 Physical development. There are health issues, physiological issues, learning about diet, nutrition, hygiene, exercise. All of the psychological issues, how to use your mind, how to use all five of your senses very well, how to speak well, to know grammar, how to think well, to do logic and math. How to be able to do these things in a social context in many cases, where you can be debating and discussing very difficult, emotionally engaged issues, and keep your cool and have a productive discussion. Emotional self-management.
- 1.28 Then get a career, start your own family, see the world. What do we want to think of this adult life as? Suppose we start emphasizing this one. It's not just adult life; we want successful adult life. We want our kids to live a good life, the best possible life. But what is that? We say, well, there's lots of things that we want to accomplish in life. We'd all like to find true love. We'd like our lives to be an adventure. We like security. We want aesthetic experiences from music and books.
- 1.29 We want to get a career that's financially rewarding. Of course, there's lots of jobs that are financially rewarding, but they're not very interesting. So we want people to find not only something that's going to pay them well, but that they're going to be into it. So they get all those psychological rewards from doing the kind of work that they are doing. Again, we're thinking about what are all of the things that go into making a meaningful, happy, flourishing life, and then crafting our education system that's going to prepare the student to be able to take on that big project.
- 1.30 Knowledge and skills. I don't know, take Wikipedia, take Grok, take whatever. There's a huge amount of knowledge that's out there. You can't learn it all. What's the essential stuff there? Maybe part of the essential knowledge is a pretty short list, but it's more a matter of skills, being able to find the knowledge, having research skills, judgment skills, sorting out the important from the unimportant mental skills, being able to have an idea and develop your physical skills, to be able to bring that idea into reality.
- 1.31 Another huge project, learning and teaching. We hinted at this one before. Well, there's a spectrum here. We'll be covering this in some of our further debates. Some approaches to education say, really, learning is primarily about the teacher. The most important thing is the parents. The most important thing is the teachers. The teachers have the knowledge, they have the expertise, they have the developed character. The child is an empty vessel to be shaped and molded by the teacher.
- 1.32 And so children should listen, they should obey, they should be empty vessels into which the parents and the teachers pour. So education, at its best, is primarily about effective teaching. Other end of the spectrum wants to say, no, education is primarily

about, do I decide I'm interested in something, I'm going I'm going to pay attention to it, that I'm going to stick with it and master it?

- 1.33** Maybe people will help me along the way, but really, that's just a sideline. That's just assistance. Maybe that's some guidance. Maybe that's some resource providance. The heart and soul of education is what we each as individuals take on and do with our lives. That's another big philosophical distinction there.
- 1.34** Now, to take all of that on, now we need to motivate going back over 2,700 years. Now, all of the questions that I put in these first 10 or 15 minutes or so out there, initially it's an overwhelming list. But if we are serious about being good parents, being good educators, particularly if we are going to be professional educators, this is our life mission. And part of that has to involve systematic and philosophical thinking about what it is that we're trying to do, taking a newborn child and educate it to whatever we take the best possible for human beings is going to be.
- 1.35** I'm reminded of a personal anecdote. I'm a little bit older now, but when I was younger and my friends all started to have children, I remember the young women, especially for the first time, realizing that they were pregnant and all of them, almost without exception, this is dating myself, would make a trip to the bookstore, and they would come back from the bookstore with a huge stack of books, everything to do with pregnancy, infancy, child-rearing, and they were going to embark on an intensive self-education process, because they were going to be the best mother they could possibly be.
- 1.36** So a realization is not just going to happen by accident. It's going to take thought, foresight, education, application. There was a huge amount of stuff that needed to be thought through and processed in order for that woman to become the best mother. The guys were typically along for the project, but they were secondary at that point. But there's a role, obviously, for fathers and men in this process as well.
- 1.37** Now, the reason I want to go back to the 700s is partly because I'm a philosopher, and this is Philosophy of Education, and the Greeks were the first to start doing philosophy systematically. It's just amazing to think that they did what they did so long ago. They built up a huge culture. We talk about the golden age of Greece, the era of Socrates and Pericles and Aristophanes, and the Spartans and the Athenians, and the battles against the Persians, and the great playwrights, Aeschylus and so forth, and the Parthenon being built, and the temple to Hephaestus.
- 1.38** It really is a magnificent culture in so many different areas. One of the big questions is, why the Greeks? Because human beings have been around for hundreds of thousands of years with a pretty big brain. But this small collection of islands in this one part of the world is the first to start doing not only philosophy, but mathematics in a systematic way, physics in a systematic way, putting medicine on a scientific footing. Some of the

great plays and playwrights of all time, many of the current scripts are just recycling Greek myths and Greek stories that were told 2,700 years ago.

- 1.39 So the Greeks were magnificent. But why the Greeks? And this, I think, wants to take us to the 700s. I want to say it was in the 700s that the Greeks started telling their stories in a different way, and the content of their stories was different. Now, human beings have been telling stories to their children, telling stories to each other at festivals, over campfires, and so on. We are storytelling animals in many respects.
- 1.40 But what started happening in the 700s is that the Greeks started to write their stories down. They still became not only an oral tradition, where the orators or the rhapsodes just memorized vast tracks, and they would travel around, sometimes making a pretty good living telling stories. And those stories would then be repeated by parents to their children and passed on as part of the culture. They started to be written down.
- 1.41 And that changes things. Other cultures had some writing, few and far between. Often the writing, though, was for bureaucratic and political purposes, to keep records of taxes and who's appointed to what positions, and so forth. Small number of very elite functions. But writing down our most important stories, that is revolutionary for human beings. It started to happen around the 700s.
- 1.42 The one set of stories that were extraordinarily well-known, of course, down through the time, were the stories of Homer. All of us, even in the '20s, they are familiar with Odysseus. There are Hollywood movies every couple of years retelling the various stories of Odysseus. These are great adventure stories. So Odysseus has been off to war. Now he wants to get home, but he has to go through all of these trials and tribulations and adventures before he can get home. And they're great adventure stories.
- 1.43 So we can imagine the little kids listening with attention about what these stories are. So he's captured by this monster Cyclops, he and his men, Polyphemus, and he's trapped in a cave, and Polyphemus wants to eat them all. And so Odysseus has to be very clever and even a little bit manipulative to figure out a tricky way to get out of the cave, and he lives to fight another day.
- 1.44 There's also the story of the lotus-eaters that come to this island, where there are these people who are very relaxed and groovy, and they take in fruit of the lotus flower and various kinds of drugs, and they're very unambitious, and it's very tempting for Odysseus and his men to partake of this. But Odysseus realizes that this is a temptation here that will sap his resolve to accomplish what he wants out of life.
- 1.45 And so he needs to develop not just trickery and cleverness in dealing with Polyphemus, but resistance to certain kinds of bodily appetites. If you just let yourself become a slave to them, you won't be able to achieve your full human potential. They're sailing past a place where the Sirens are known to hang out these lovely feminine voices that call to the pirates. And many men, they know, have not been able

to resist the lure of these seductive female voices and then have crashed and died as a result of this.

- 1.46** There's another lesson here, of course, for men, about certain kinds of passions and seductions and remembering what your true ambitions are and not letting those get the better of you. And so on and so on, these stories go and they are great and wonderful. But I want to talk about how the stories of the Greeks, in some ways, were unique and set them off on a different path, which is why philosophy and philosophy of education and various other foundational disciplines really do start with the Greeks.
- 1.47** So, the Greeks are not alone, of course, in telling stories and telling very interesting types of stories. But I want to contrast the Greek stories, or some of the Greek stories, with some of the other great epic stories that have come down to us through history. Here's one from some thousand miles further to the east, from India, an ancient Hindu epic that is still with us. Again, in Bollywood, there are regular remakes of the Ramayana story and so on.
- 1.48** But as I read through this highly condensed summary of what's going on, I'm going to emphasize a few words that are philosophically charged. So let's see where we're at. So, Rama, Ramayana is a story of Rama. He's a prince, but he is, first and foremost, a god. He's an incarnation of the god Vishnu. He's assumed human form and is now living as a prince in human. He came into existence as a result of a miracle, his birth. He was forced into exile from his princely territories with his wife and his brother. And so they are living in the forests.
- 1.49** There, he encounters a demoness, a female. She finds him very attractive and she wants to seduce him. But he loves his wife. He wants to be faithful to his wife, so he resists her advances. She is the female scorned. And so bad things are going to start happening to Rama as a result of this. The demoness' brother, another demon who's a king and is offended also on her behalf, he's dishonored her, so he's going to make life difficult for Rama.
- 1.50** They kidnap Rama's wife. Rama and his brother go off to go look for her. Along the way, they meet a demigod. They meet another demigod. They join forces. These guys are on their side. Then they go into a great battle against the bad deities, the bad demons, and their allies, and defeat them, rescue the wife, happy ever after.
- 1.51** Now, what is interesting philosophically about the story? All of the beings, the central beings, are not humans, primarily. They are incarnated gods. They are demons, demonesses, deities, and so on. What this story is telling you is, we might be interested in human things. Of course, we're humans, we're interested in human things. But our most important stories are about the gods and demons and demonesses, and what matters to them and their alliances.
- 1.52** And what happens to us is primarily as a result of the decisions that they make, and in some cases, it's just an overflow of conflicts going on in the world of the gods, the

demons, and so on. All right, now remember that. And then it's to say, metaphysically, the important thing are the gods and the gods and their struggles and their values.

- 1.53** Different story. This one from Mesopotamia, a little bit older, story of Gilgamesh, say, halfway between India and the Greeks. Now, Gilgamesh, in this case, is a king, and he's a human, fully human. Then the story of Gilgamesh, though, is, because initially, he's not a very good guy. He's actually a bad guy. He's not a good king at all. But the gods have created this wild man, Enkidu, to be his adversary. They want to try to get Gilgamesh to stop doing all the bad stuff that Gilgamesh is doing.
- 1.54** Now, Enkidu starts off as a wild man, but then he has sex, and the sex here is with a civilizing being. The idea is somehow men are a little bit more wild, and it's women who civilize them in some way. There's a subtext that's done there, but it's by a prostitute, but she's a sacred prostitute, that is to say, sanctified by the gods. Enkidu and Gilgamesh have a fight. Gilgamesh wins, but even after that, they become friends, and they start off on some further adventures.
- 1.55** They go out to the cedar forest, where they kill a demon. The demon is guarding the sacred tree, and then they cut the tree down, the sacred tree in this sacred garden. As a result of that, a goddess, Ishtar, is offended, and she sends a semi-supernatural being, this bull of heaven, to be their next great challenge. They have to overcome the bull of heaven. They also sentence Enkidu to death, so he's going to die, and he does die, and this is terrible for Gilgamesh to process. So he wants to find, how can I escape death, which seems to be this terrible thing that has happened now to my friend Enkidu?
- 1.56** And then just toward the end, he meets this husband and wife, and apparently there had been a great flood that had been sent by the gods to wipe out all human beings. This husband and wife were the only survivors of this great flood, and Gilgamesh meets them, but he learns a lesson from them that, unfortunately, his quest for eternal life is not possible as a human being, and that's the moral of the story.
- 1.57** OK, so that's a great huge epic in three minutes. But philosophically, what's going on here? Unlike the Ramayana, where it's all incarnations of gods and demons and demonesses and deities and so forth, interacting with each other, in this case, it's a human who is the central figure. But the stories are this human in his ongoing battles to achieve his human values. He loves life and he wants as much of it as possible and all the good things out of life. But the gods, in many cases, are obstacles and send obstacles to him.
- 1.58** So whereas Ramayana is the gods fighting amongst themselves and humans are downstream from that, in this case, the theme metaphysically seems to be, it's all about man versus the gods. That's different. That's a different world, metaphysically. Now, let's come back to the Greeks. We've gone from India to Mesopotamia, and now we are to the Greeks, and we are in the 700s. We know the stories of Odysseus, and those are great for children and teens, and of course, adults still continue to love them.

- 1.59 But what if we get to really serious adult themes like war? And war is, I don't want to say a constant, but a regular feature of human existence everywhere around the world. Why are we fighting so much? What causes war? So what we have, in the case of the Iliad, the other great work ascribed to Homer, is an epic focusing on what was then the greatest war that any of them knew up to that point, the so-called Trojan War.
- 1.60 And what caused the Trojan war, where the Trojans and all of their allies are up against the western and southern Greeks and all of their allies, and it's this great long, sometimes brutal battle. Why did this set of conflicts happen? We start to read Homer and we learn, it says, well, of course, it all starts with Helen. We've all heard of Helen of Troy. But she wasn't initially Helen of Troy. Instead, she was just Helen, a Greek princess who had been married off to Menelaus, who was a king of one of the Spartan, one of the kingdoms on the Peloponnese.
- 1.61 She was beautiful, younger. Menelaus, though, was older, but he was a king. So we have this age-old story of the young, attractive woman being forced to marry a man she does not necessarily love, but she's married off to him for reasons of wealth, reasons of political connection, and so forth. There's a diplomatic mission from Troy that comes along to visit Menelaus' kingdom. And part of this diplomatic mission is Paris, who's young and dashing and handsome, and Helen is stuck in this loveless marriage.
- 1.62 What's going to happen? Well, what happens is what often happens. She falls in love with Paris and runs off with him back to Troy. That's the Trojan story. Or Helen is so beautiful that Paris just can't help himself. And so he kidnaps her and takes her back with him to Troy. That's the Greek version of the story. Nonetheless, we now have a conflict. Diplomatic relations have broken down. Menelaus is a cuckold. He wants revenge. He's humiliated.
- 1.63 So he appeals to his big brother, Agamemnon, who's probably the most powerful king of all of the Greeks, first among equals. "Let's go teach those Trojans a lesson, get my wife back." Menelaus, that's his argument. Agamemnon doesn't particularly care about the cockolding issue, but he's ambitious. He wants power. He wants more wealth. He wants more control. He really likes the idea of taking over Troy.
- 1.64 So he says, "OK, that's fine. But we really are also going to need Achilles if we're going to make this work, and Ajax, and Odysseus, and so on. So we have to— Trojans are pretty formidable, so we need to appeal to all of these guys." So, Agamemnon wants power, and Menelaus wants revenge, and behind all of that is love and sex and so forth, causing the war as well. But they have to get Achilles on board, and Achilles doesn't really want to go to war. He's tempted by the idea of just settling down, finding a nice woman, getting married, living out his days, raising a family.
- 1.65 But Odysseus is sent to recruit him, and Odysseus tells him, "Look, you can do that, but maybe a nice life, no one's going to remember you. You want to live the best life, and you're a man and you're a great warrior. It's the warrior life that is for you. And this is

going to be the greatest battle ever in human history. And you will be able to achieve glory such that, for centuries, people will be talking about you and singing your praises." And that argument works for Achilles, so he signs up for the war as well.

- 1.66** Now, Hector is the hero of the Trojans, family man. He's next in line to become the king, his father Priam, very upstanding, righteous individual, and also a formidable fighter. So the stage is set for this great war. But now, again, a three-minute summary of this great epic. We're trying to figure out why war happens. And what's our explanation for the war? It's all human explanation. It's because humans want love. They want to have a great sex life. Sometimes they want revenge. Sometimes they want more power. They want glory.
- 1.67** The explanation then of what's going on in this great conflict and what's driving the plot is human beings and their goals and their values. Where are the gods? So far, no gods. It's a humanistic story that is being told, and that's different. Now, of course, the gods are present in Homer, and the gods are important in Homer. I don't want to overstate this.
- 1.68** So what we then have, and I want to now do what I will frequently do during the course of this lecture series, lay out some material, but then go to the text, the important texts here, so we can read the important people, philosophers, and other educators in their own words. I want to have a few words here from Homer on the role of the gods in this whole story. Let's do that. This is toward the end of the Iliad. It's 24 books altogether. This is book 22, so we're approaching the climax.
- 1.69** Then finally, Achilles is on the battlefield, and it's agreed that it's going to be a confrontation between him, as the champion for the Greeks, and Hector, as the champion for the Trojans. And they came out, and then immediately Achilles attacked, but Hector ran. He's not sure he wants to engage with Achilles because the sense that everybody has is, Achilles cannot be defeated in battle. So he's looking for an opportunity in biding time.
- 1.70** "So they whirled three times around the city of Priam," that's Troy, Priam is the king, father of Hector, "sprinting at top speed while all the gods gazed down." I'll just pause there for a moment. What's interesting about this? The gods are watching the humans. The gods are the spectators. The humans are enacting the drama. It's not human beings wondering what the gods are up to. It's a reversal. The gods are very interested in these human creatures and what they are doing.
- 1.71** Now, some of them are more sympathetic to the Athenians, some more to the Trojans; some love Hector, some love Achilles, so the gods are divided among themselves. That also is interesting. We don't have a supernatural dimension that has one agenda, one plan, one set of values. All of the human conflicts and differences are just recapitulated at a higher level among the gods.

- 1.72** "The father of men and gods," that's Zeus, "broke forth among them now," because he's so tense about what's going to happen here. This is so exciting. We've been waiting for such a long time for this great confrontation. He says, "Unbearable. A man I love," that's Hector, "hunted round his own city walls right before my eyes. My heart grieves for Hector. Hector, who burned so many oxen in my honor." So many sacrifices Hector has done as a dutiful religious man. "Rich cuts, and now on the rugged crests of Ida, now on Ilium's heights. But now, look, brilliant Achilles courses him round the city of Priam in all his savage, lethal speed."
- 1.73** So what's going to happen? And Zeus doesn't know what's going to happen. He's conflicted in his own mind. So what he does is, he turns to the other gods who are also watching. He says, "Come, you immortals, think this through. Decide. Either we pluck the man from death and save his life, or strike him down at last, here at Achilles' hands, for all his fighting heart." So what should we decide in this case?
- 1.74** "But immortal Athena, her gray eyes wide, protested strongly. 'Father, lord of the lightning, king of the black cloud, what are you saying? A man, a mere mortal, his doom sealed long ago—you'd set him free from all the pains of death? Well, do as you please, but none of the deathless gods will ever praise you.'" So Athena is arguing with Zeus. Now, Zeus might be first among equals here, but she makes an argument against him, and she's not afraid to challenge him.
- 1.75** She points out that the other gods also have opinions on this matter, and he should take those into account. But ultimately, your decision, Zeus. "And Zeus, who marshals the thunderheads, replied, 'Courage, Athena, third-born of the gods, dear child. Nothing I said was meant in earnest. Trust me, I mean you all the goodwill in the world. Go, do as your own impulse bids you. Hold back no more.'" Athena wants Hector dead. And Zeus says, OK.
- 1.76** So he launched Athena, poised, already poised for action. Down the goddess swept from Olympus' craggy peaks, and she then engages in some trickery, and eventually then Achilles is able to kill Hector, and we're into the denouement of the story. But what is also significant about this is that the gods are involved, but what kind of gods do we have here? Gods who are indecisive in some cases, but also gods who listen to each other and engage in discussion and debate.
- 1.77** And even though Zeus is the chief god, it's not the case that he is all-knowing and all-powerful, and he knows exactly what the right thing to do is. He's kind of like parents. He's kind of like political leaders. He's kind of like human religious leaders. And in this respect, we might then think that part of the moral of the story is, if you want to be like a god, even if you are in a position of authority, like Zeus is in a position of authority, part of that is listening to your peers, taking their counsel, and having a space within which they can sometimes raise criticisms and counterarguments to what you are thinking about doing.

- 1.78** That's a different kind of religious message to be extracted from this story, but it's also a story that seems primarily to be focused on human beings and their stories as well. That's a big shift. That's why I want to suggest, out of this comes a different way of thinking about human beings and what's possible for them, how they should relate to each other, understanding why things happen the way they do in the world, and of course, how we should relate to the gods. But it is interesting that the gods in this story are very interested in the humans, and they admire the humans. In many cases, they want the best for the humans. That's a different kind of world.
- 1.79** Now, one more story. This is Arachne and Athena, a little piece of artwork I pulled off the internet that I thought captured it very nicely. But coming back to this theme of parents as educators, as the first educators, and in many cases, it is stories that we are telling children to bed or over the dinner table or more formally at bigger parties and festivals and so on.
- 1.80** But this one speaks to me as a bedtime story, the story of Arachne, and we know arachnid is the generic word for spiders. But we can imagine a little kid going off to bed, seeing a spider somewhere in the room, being a little bit scared of the spider. And the mother or the father, probably the mother, is going to tell the kid a story, to soothe the child, and the kids go, "Where do spiders come from, anyway?" And this is where spiders come from.
- 1.81** The story of Arachne and Athena, and it's an ancient Greek story. The one I'm going to read you is actually a later Roman retelling of the story, but it's nicely compressed. So here's the story. Let's go. Arachne was a lovely young woman and a wonderful weaver. People traveled great distances to see her works at the loom. Her skilled fingers wove detailed multi-colored rugs and tapestries. Her skill was a work of art, and people paid much money for her creations.
- 1.82** Eventually, all of the attention went to her head. Arachne started to become boastful of her talents. "Athena, the great goddess, has given you an amazing gift, Arachne," the villagers would often say to her. This made Arachne angry. "Athena did no such thing. I taught myself to weave. No one can weave as well as I, not even Athena, who invented weaving."
- 1.83** Athena, the goddess of wisdom, watched boastful Arachne from high on Mount Olympus. One day she'd had enough of Arachne. Athena disguised herself as an old woman and went to visit Arachne. "I hear that Athena has given you a great gift, the skill of weaving," said the old woman. "I am the best weaver, but Athena has nothing to do with how good I am. Her skill is no match for mine," stated Arachne. "You are a talented weaver, Arachne, but a foolish girl. You should ask Athena for forgiveness," the old lady said, becoming angry.
- 1.84** "What? Ask forgiveness? You are the foolish one. I am telling the truth. And if Athena is offended by my claim, she is more than willing to pay me a visit. I would show her what

real weaving looks like," Arachne said confidently. With that, the old woman filled with rage and transformed back into the magnificent goddess, Athena. All of the village people gathered around the powerful goddess and fell to their knees to honor her. All of the people except Arachne, who seemed unimpressed by Athena's presence.

- 1.85** "You think you are better than I, Arachne? Well, let the competition begin," Athena proclaimed. Everyone gathered to watch the weaving contest. Arachne and Athena both began to weave. Their fingers moved fluidly across the colorful threads. Athena wove glorious pictures of the gods and goddesses performing kind and heroic deeds. They were the most beautiful images the mortals had ever seen.
- 1.86** Arachne's weavings were also gorgeous and perfectly constructed. They were also images of the gods, but portrayed them as angry and foolish. Athena was enraged when she saw how Arachne had depicted the gods. She was even more infuriated—that's interesting—more infuriated when she realized that her own skill was only slightly better than Arachne's. "You are too boastful and rude, Arachne. How dare you make fun of the gods?" Athena, beside herself, ripped Arachne's weavings to shreds. Then she grabbed a stick and hit the girl repeatedly with it.
- 1.87** At that moment, Arachne ran from Athena. "Oh, no, you won't run from me," Athena shouted. "I will make sure that you, your children, and your children's children suffer." She magically altered Arachne. Arachne began to shrink until her body was a small black bee. She sprouted eight legs and grew black hair. Arachne became the world's first spider. She scurried to the highest place she could find and began weaving a web. "Now you will be able to weave all day long," Athena said haughtily. "But from now on, no one will care about your talents. In fact, your delicate woven webs will be destroyed when people see them."
- 1.88** All right. How's that for revenge? Multigenerational revenge and so on. OK, so let's go back to the little bedroom. Kids are fascinated by this story. It's a great story. What would the conversation be like, as a result of this story? You're the little kid and you hear this story for the first time. Do you feel sorry for Arachne? Do you feel that she got what she deserved? Do you admire the two for their great weaving skills? Or is that somewhat undercut by the fact that Athena seems to lose her temper a lot?
- 1.89** It's maybe a little bit over the top when it comes to punishing Arachne for whatever sins, she was boastful, that Arachne has. But was it really over the top? After all, Athena is a goddess. Who are human beings to question the goddesses and their standards of judgment and fairness and so on? Are we allowed to do that?
- 1.90** Or we say, no, we are allowed to do that, and we should do that because, after all, if the gods are supposed to be better than us, the goddesses too, we're supposed to look up to them, and they're supposed to set standards for us, for what we're supposed to do. But it does seem to be the case that, even though we would like to think about the

gods that way, in lots of cases, the gods don't seem to be particularly upright and admirable.

- 1.91** In fact, they seem to be prey to the same kinds of not only strengths and magnificence and gloriousness that human beings can do, but also to the same weaknesses and foibles that human beings are poised to, just on a much larger scale. And now, notice what we are doing. We're asking very philosophical questions about the gods and humans, standards of justice and fairness and morality, and where we get them from.
- 1.92** And just as a result of this kind of conversation, also, we are encouraging the child to enter into this conversation and to start thinking about those kinds of questions. That's a very different kind of storytelling exercise, too. "Here are our stories, memorize these stories, and don't question them," which is characteristic of many other traditions around the world.
- 1.93** I want to suggest that something special started to happen in the Greeks. They became more philosophical; obviously, they became more literate. In a certain way, their politics also changed. We know Greeks were the first to start demonstrating significant democratic political institutions, all of which require lots of discussion and argumentation back and forth, and the idea that we should be able to challenge our leaders, and they should be on our side, and we're going to hold them up to certain standards and so on; such that, as their culture transformed, by the time we get to the 400s, something pretty golden and amazing has occurred.
- 1.94** Now, that said, I wanted to introduce a concept, now that we've come back to the 400s, the concept of Socratic. Because if the story of Arachne and Athena, or the story of Odysseus and his battles, or the conversations about why the Trojan war happened, and comparing all of the heroes and the villains in that story, we're all participating in these stories and thinking about them. We're doing this partly as individuals, but also partly socially.
- 1.95** This is what we now call Socratic method. Of course, the first great genius of Socratic-style education was Socrates, who lived in the 400s, and we'll be getting more into him very soon. But here the suggestion is that education does require this social interaction between older people and younger people. The older people have wisdom and they fought through various things, and they have know-how, but nonetheless, they are not perfect.
- 1.96** They might be looked up to, and we are striving to be better and become more godlike if we possibly can. But the way we are going to do that with our young people is by getting them into the process very early, by telling them the stories that are important to us that contain the important messages, but also teaching them how to think, how to discuss, how to challenge, how to raise a question that might seem a little bit dangerous, but raise it even so.

- 1.97** And how maybe even to disagree with their parents, but in a constructive way, so that, of course, parents, like the gods, are also not perfect. Kids can realize that and strive to learn from the mistakes of their parents. It's not simply, what are we going to teach, but how are we going to teach it? And this becomes part of the culture. And the what is lots and lots of things. There's history, there's religion, but there's also philosophy that is built into all of these stories. The crucial part of deep education is going to be philosophical.
- 1.98** What really is the nature of the world? How do we know that the gods are on our side or not on our side? What really counts as knowledge? As we get further along, are these stories real? Is the Arachne story real? Or is that just a made-up story to teach certain lessons? If we try to extend that to Zeus and the other gods, does that mean we don't really believe in the gods anymore? And that starts to sound a little dangerous.
- 1.99** But there are questions here that are philosophical questions, about the cosmos that we are living in, what it is to be a human being. What's possible to us? Is it the case that, yes, there are gods, and they're just going to beat us down no matter what, so why bother even trying? Or is it the case that the gods really are special and they're more like role models for us? Yes, we cannot become gods ourselves, but we can come pretty close. We can be like Arachne and develop ourselves.
- 1.100** It's not just a matter of gods give us gifts or they don't give us gifts; we can decide as human beings to take our capacities and become excellent in athletics, excellent in weaving, excellent in music, excellent in whatever we put our minds to, to the point that even the gods will admire us. That's a philosophical framework that is unique in human history.
- 1.101** I want to end this opening lecture by saying a few words about the philosopher who I think represents the culmination of this tradition within the Greeks. I don't want to overstate the line that I've been pushing about the Greeks so far. There are lots of different Greek stories about the gods, about human beings, emphasizing the Greeks disagreed among themselves, very many different schools of thought about all of the important issues. But this certainly was one of the main ones.
- 1.102** It says, as Aristotle does in the 300s, the world is the natural world. It's a magnificent place. We can learn it, we can learn it very well. We can understand it scientifically. We can understand it rationally. We can use that knowledge to study ourselves as part of the natural world, to learn what our capacities are, and they are considerable to develop those capacities to live a very full, flourishing, "eudaimonistic", to use the Greek word, life.
- 1.103** And that education properly is exactly that philosophical set of principles, operationalized in young people in a system of pedagogy, whatever that is. But just a few words from Aristotle on this tradition. Some of the most famous lines from him start from the beginning of the Metaphysics, his great work on the nature of reality,

one of the most influential books written. But notice how he starts, the point that's directly speaking to education, because if education is about anything, it's about knowledge.

- 1.104** Right from the beginning, Aristotle says, "All men, by nature, desire to know." It's part of human nature to want to know. Now, just by contrast, we know this is a little bit of foreshadowing, we know that some schools of philosophy and education say people don't want to know. They like being ignorant. They don't really have strong desires to develop in the cognitive direction. And Aristotle is saying, no, there are those people, but they are damaged. Something's happened to them.
- 1.105** To be a human being is to have certain capacities for learning, for self-development. That's your nature. And what we want to do is develop those, so that they are as your second nature; your developed full capacity is even more awesome and glorious. It's a natural desire to know. An indication of this is the delight we take in our senses, all five of them. Little kids, they love using their senses and seeing what they can do and exploring the world sensorially with all five of them.
- 1.106** Parents are horrified, but with little kids, everything goes in the mouth. We're using our advanced knowledge of nutrition and hygiene to say, "No, lots of things should not go in the mouth," but kids, they don't do that. They will smell anything, they will touch anything, they will look at anything. They are desirous of doing all of this. "For even apart from their usefulness, they are loved for themselves and above all others, the sense of sight."
- 1.107** Now we're talking about knowledge, natural. We have a strong desire for it, but also the sense of sight, because that's the one that puts us in contact with physical world, the natural world. This is the one that Aristotle mentions first. Knowledge, first and foremost, starts with the senses, and it starts with observing the world around us, exploring the world around us. That's natural. That's where we should begin. And that will become an even more philosophically pregnant remark when we start to contrast some of the later schools coming along a little later.
- 1.108** Well, the nature of knowledge. This is a book, *De Anima*, or on the soul, or on the psychological, because we are physical, material beings, but we also have a psychology, or a psyche, or sometimes, for more religious, we'll think of it as a soul or a spirit. And the relationship between our minds and bodies is, of course, an important issue. Aristotle has the first great treatise on psychology here.
- 1.109** But notice what, again, he says here about knowledge, because knowledge of ourselves, because we're pretty special, and knowledge of our minds, the thing that elevates us beyond the animals and possibly our souls, is extra special. "Knowledge we regard as a fine and worthwhile thing." None of this ignorance is bliss. Knowledge. "And one kind as more so than another, either in virtue of its accuracy," more and more

accurate knowledge, "or in virtue of its being concerned with superior and more remarkable things."

- 1.110** So we figure out what the really important things are and acquire knowledge about them. That's going to be very important. And some things are more important than others. Now, again, a remark that's going to be philosophically pregnant when we start contrasting a little bit later, so just remember this. In many cases, it works on psychology in the soul. There will be a development that says, well, the thing that rises us above the animals is our mind or our soul or our spirit. And we should, of course, be interested in that. But that means we really shouldn't be interested in the physical, lower side of things. It's only the higher things.
- 1.111** And one remark about Aristotle, as a Greek in this tradition that we are developing here, is to say no, just the fact that we are interested in higher things doesn't mean that we're not also really interested in the lower things as well. It's not so much of a value judgment as well. It's important here that Aristotle was a great biologist, loved dissection, collecting all kinds of plants and animals and figuring out how they worked, and he wrote—most of his works actually are in the biological sciences, and of course, huge amounts in philosophy and politics as well.
- 1.112** But notice what he says about this, because a lot of times we'll develop sensory aversions to various physical sorts of things, and we say, "Oh, no, I don't want to be engaged in it." Aristotle says, don't be childish. "We should not be childishly disgusted at the examination of the less valuable animals, for in all natural things there is something marvelous." That says something about the spirit of Aristotle. The entire natural world, from the greatest to the smallest, in all of its variations, is interesting and a proper subject for human investigation. And I want to know it all.
- 1.113** Now, he's also famous for then saying, we're not only interested in studying nature, how the planets move, what's the sun really, are there gods beyond or not? But also human beings. We are special in all of creation and also special to ourselves. What makes it possible for human beings to become the best that they possibly can be? And this also has to be a part of education.
- 1.114** It's not just knowing the world and being able to engage with the entire world. It's also knowing yourself and developing yourself into the best kind of human being. And so character training, skill training, is going to be a fundamental part of education. Now, this is a table form here, but when you read the list of Aristotelian character traits that are important, they also are very Greek in this sense.
- 1.115** Partly this is because a good part of education is learning the stories of Homer, thinking about Odysseus and how he was able successfully to find his way back home and to overcome all of those obstacles, to meditate on the Iliad and the causes of war, and all of the deep issues about human psychology, so that you can then enter into that sphere as well, and say, "Should I be more like Achilles, more like Odysseus or Hector?"

Who is my role model going to be, so I can grow up to be the best human being?" And female role models as well, Arachne, Athena, and so on.

- 1.116** What's interesting, at least on this table here, what Aristotle does is somewhat scientifically, or certainly systematically, say, well, look, take human life, which is very complicated, but we can organize it into different areas. We have to deal with conflict. We have to deal with friends. We have to deal with lovers. We have to deal with children. We have to deal with wealth, business, all these different kinds of people, all these different kinds of situations.
- 1.117** We figure out all of the types of situations that we as human beings have to deal with and say, what have we learned as human beings about what enables a human being best to deal with that situation and flourish in that situation? Having identified that, then we can, as parents and educators and, of course, as individuals ourselves, figure out how I can become the kind of person who can have the character to be able to handle that situation.
- 1.118** So, just to take one example here, we're emotional beings. Sometimes we're fearful, sometimes we're confident. Sometimes we're in conflict situations. Sometimes we are going to succeed, sometimes we are going to lose. That's life. That's a huge part of life in all sorts of areas. So we are going to be in conflict situations, win-lose situations, and we are going to have strong emotional reactions in those cases. What do I need to be able to do in those situations?
- 1.119** Then Aristotle says, well, what we've learned is courage. Courage is the character trait that enables human beings to master their emotions in situations of conflict, to be able to act appropriately in that situation. Now, that's a very abstract situation, but we do know that sometimes there are people who will—they are in a conflict situation. It's a dangerous situation, and they are afraid, but they let their fear dominate them, and they run away when, really, the best and proper thing to do in that situation is not to give into your fear, to master your fear, and deal with the conflict.
- 1.120** That's a real human circumstance. What's the difference between people who are giving into their fears and people who are able to master their fears? When we start dealing with little children, we know which outcome we want. How can we figure out how to raise them so that, when the circumstances arise, starting with small-scale childlike issues, where they're dealing with their fears, middle-size adolescent issues, all the way up to the big adult issues. And this has to be a big part of our education.
- 1.121** And we go through, systematically, all of those emotions, all of the policies of judgment, all of the actions. We make it habitual. So we take human nature and make it even better, a second nature that is more mature. We know sometimes people in the conflict situation, just because Aristotle is famous for his doctrine of the mean, that the best path, you identify it. There's always a deficiency and there's always an excess, and it's a matter of judgment to be able to figure out what the right one is.

- 1.122** We know sometimes people are in conflict situations and they're full of bravado. I'm indestructible. And so they just, in that conflict, they just run into battle and they just get totally destroyed because, here I was, we're in fighting against the Persians or whatever, and there's a whole Persian army over there, but there's three of us scouting party and we see them over there. I really want to kill some Persians. And so I go running down there and I get killed because I'm an idiot in that case. It's not that I'm afraid. In that case, I've got a deficiency of fear or a deficiency of judgment.
- 1.123** So, part of it is not only mastering your emotions, but it's also being able to rationally assess conflict situations and say, three against 100, uh-uh. Live to fight another day. You're not a coward if you retreat, if you gather your troops and so forth. So you have to train your emotions, train your rationality, and make that habitual, so that, of course, judgment is always going to be required, but nonetheless, you can act as you need to do. And this requires that parents become philosophers, moral philosophers in this case, and it requires that professional teachers also become philosophers in order to be able to do so.
- 1.124** One closing interesting historical anecdote about Aristotle's great student, Alexander. Aristotle did much of his career—he was from the north, Macedon originally, but the great philosophy school was in Athens, so he went there and studied with Plato. We'll get to Plato pretty soon. But then politics got ugly and there was tension between Macedon and the southern, more Greek states. Also, the king of Macedon, Philip, had a son, quite precocious, a young guy named Alexander, and he wanted, as a good father should, to have the best possible education for his son.
- 1.125** So he brought a small handful of carefully chosen tutors to teach Alexander everything he needed to know. But he brought back from Athens, Aristotle, who then had the reputation of being the greatest philosopher in the world. And I think that probably was true and probably still is true. Brought him back. And so some indication of how Aristotle would take his formal philosophy, which is somewhat abstract, and operationalize it in an education system.
- 1.126** We can see it in what we know of his method of education and content of his education for Alexander, who was a teenager at this point. The subjects that we know were on the list were rhetoric, how to speak, how to speak well, how to speak articulately, how to speak compellingly, how to use logic and emotion. Here we are grooming the young prince to become the king. The king has to be a kind of leader, and so obviously important.
- 1.127** Philosophy, big-picture, important issues. Principled, strategic. His job is going to be in politics. He's going to be the next king, leader of the military. So study of politics, all the different political systems, a lot of history has to be involved. All the examples of different kinds of political systems, what worked, what didn't work.

- 1.128** Physics. You're living in the world, particularly you want to know the way the world works. But if you're going to be in the military, you need to know rivers and mountains and weather, all the subject matter of physics. Biology. Animals, plants, humans to the extent that they are animals, their horses, and so forth. Medicine, literature, the other arts, because we want to enjoy life as well and appreciate beautiful things and things that are aesthetically sublime.
- 1.129** Human psychology. You're dealing with human beings. We're all humans, but we're also all different. Become a master of human psychology and their emotions. As far as we know, from the method of instruction, there was a lot of conversation. Aristotle was famous for being peripatetic. So history's school of philosophy was nicknamed the Peripatetics because, at his lyceum—they've recently rediscovered the site of it in eastern Athens—they had columns, shaded columns all around, and he would like to walk around in the shaded columns, talking and lecturing out loud and engaging in conversation back and forth with his students.
- 1.130** So that's a natural way of learning. Obviously, a lot of reading. We are now a highly literate culture. Enormous value for concentrated study on words as recording important thoughts. Learning how to observe. Aristotle is a scientist. How do you read a landscape? How do you read a person's face? How do you look at this plant? How do you figure out what all of the parts of this animal's body are for? And that, as far as we know, is what Aristotle did with Alexander, because he had specialized tutors for his military training, for his gymnastics training, for horse riding, for weapons and so forth.
- 1.131** An indication, rather, of a sophisticated and hugely important for later development of Western educational philosophy coming out of Aristotle. And with that, let me just say, this is one culture and one thread of educational philosophical development. One thing we do know is there's lots of them. Everything is controversial, everything that we have said so far today. It will be vigorously contested by different schools of philosophy, and we will march our way through them.
- 1.132** But I do want to signal it with this quotation from a Renaissance thinker, della Mirandola, from his Oration on the Dignity of Man. He says, "One cannot reasonably attach himself to one school or philosopher, unless he has previously become familiar with them all." So, a big project, but that's what we're here for. I'll stop there.
- 1.133** Dr. Hicks, you started us out in 700 BCE and really studying in the Greeks, and you went into the Socratic method of how we think. Thirty-five years ago, when I was a kid, it seemed like it was a drill of memorization facts. Or are you proposing here that the philosophy of education is changing, and it's leaning more towards how we think?
- 1.134** The second part of your question is about our generation, so to speak. As you also point out, we're two and a half millennia from getting there, and a lot has changed over the course of the century. So Socratic, some schools of philosophy we're going to look at

hate it, think it's completely inappropriate. And there have been long centuries when it was, in fact, out of fashion.

1.135 The thing I would say just quickly about our generation, or maybe just the last century or so, is that one of the clichés about the modern world is everything that happens faster. And we also do have a very open culture with influences from all over the world, including an immigrant-friendly culture of people bringing their cultural influences. So, American education, since that's where we are right now, in one sense, has just been a big inconsistent mess over the course of the last century.

1.136 And there have been, because of the pace of change in modern times, fashions. That's a bit flip to call them fashions. So let's say, methods of doing education that become trendy and ensconced. They can come in pretty quickly, but they can also get out pretty quickly. If you're talking about a span of 35 years, that's important for you and I in our lifetime, but in one sense, that's a very quick transition. Even in that 35 years, I want to say it hasn't been only drill or only Socratic. There have been lots of other possibilities. All I can say is, stay tuned, and we'll have seven more lectures to answer your question.

Lecture 2: From Plato to Augustine

At a glance — From Plato to Augustine across late antiquity.

- 2.1 All right, let's turn to 431 BCE now, still in Greece. But in 431, a great tragedy starts. It's the outbreak of the war between Athens and Sparta, Athens and all of its allies, Sparta and all of its allies. There had been tension and outbreaks previous decades, but this is the big one, one we now call the Peloponnesian War, most famously written about by Thucydides, who was a general on the losing side, then dismissed from his post, and then went on to write his famous history of the war.
- 2.2 But it was devastating, not only because wars generally are devastating, but the two great former allies who largely, jointly, are partly responsible for why Greece had to achieve the magnificence that it had in its golden era, the 400s, now at each other's throats and killing each other, a long, multi-decades war, partly attrition, and eventually the Spartans win, the Athenians lose. But in one sense, Greece is never the same as a result of this devastating war.
- 2.3 One of the interesting questions of history is why some cultures rise and become great, but then also why great cultures decline and fall. This is one of the big lessons. But at the same time, this is the life span of two of the great philosophers of all times. They are ones we've all heard of. Socrates on the left. Socrates was a teacher of Plato, Plato on the right. Then Plato, in turn, became the teacher of Aristotle, whom we covered some in the previous lecture.
- 2.4 But both of these men were alive during the Peloponnesian War, and both partly their thinking and their personal experience, obviously as well, is reflective of what went on in this terrible, terrible war. So this is our context. I want to turn to Plato, specifically. We have both Socrates and Plato together. They were different historical individuals.
- 2.5 Socrates never wrote anything down. This was part of his cognitive philosophy, we might call it, and he thought learning happened best in conversation, not just one person lecturing to a bunch of other people, not people on their own, not even necessarily from reading, though you can learn a lot, but he thought there was something cold and a little bit dead about the written word. Conversation is alive. Learning can happen partly because the student has to be on.
- 2.6 You can't just hide out in the back of the classroom with your ball cap down low and have a little nap while the professor drones on. The professor is having a conversation with you, knows when you're paying attention, knows when you're not. And so, the student is on, is energized, is actively involved in the process, but also from the teacher's side, they can be customized. What is the student not understanding? Do we need another example here, a different explanation? Do we want to explore this road and so forth?

- 2.7** So he thought that was the best method of learning. And of course, it could be that he just didn't like writing. But nonetheless, he did not write anything down. When Socrates eventually was put on trial by some Athenians, he had made lots of enemies, personal and philosophical. Other philosophical and religious factions got after him, successfully prosecuted him, had him sentenced to death.
- 2.8** Plato was outraged, thought Athens had essentially executed their greatest philosopher for being a great philosopher. But whether philosophy is a good thing or not is, of course, a contentious philosophical issue. And Plato then was in part motivated by the death of Socrates, and the martyrdom, as he saw it, of Socrates, for philosophy to start writing. So he starts writing accounts of conversations that he knew Socrates had had with other leading citizens in politics, in military, in religion, foreign dignitaries, foreign thinkers, and so forth.
- 2.9** So it starts to become difficult, as Plato's writings occur to determine, because Plato continues to write about Socrates and to use Socrates as his main character in these dialogues, whether these are entirely historically accurate to Socrates' views, or whether at some point Socrates starts to become more of a character, a literary device, a mouthpiece—that's a little bit harsh—but for Plato's views, rather than strictly Socrates'. So there's ongoing interpretive issues about what's Socrates and what is Plato, even though on many issues we do know they were simpatico.
- 2.10** In Plato's great work or his greatest work, *The Republic*, it's an enormous 10-volume work, it definitely is something that should be part of everyone's philosophical education, where Plato lays out his metaphysics, his theory of knowledge, his view of human nature, his understanding of morality and politics and religion and the arts, it's everything all in one. And it's brilliant philosophy and it's brilliant writing as well.
- 2.11** But along the way, he tells a story that he wants us to reflect on as an important story, that if we were going to do philosophy, particularly moral philosophy, and a part of raising children is about their moral character and training them, this is what we need to reflect on. The story is called the myth of Gyges, and it's not original to Plato, but he took it and did some philosophical upgrading, so to speak, to it.
- 2.12** So Gyges was a young shepherd boy, a youth, and his job was to, basically in a small village, way out in the hills somewhere, look after the sheep for the village, to in the summertime take them out into the hills where they can graze and do sheep things and look after them. And then in the wintertime, he would bring them back into town and basically be out of a job while, minimally working while the sheep are kept in confinement during the harshness of the winter.
- 2.13** So the story goes that one summer, Gyges is way up in the hills with his sheep somewhere, and he is poking around in some rocks and he finds this opening to a cave. And he goes and he explores inside this cave and he finds this skeleton, this guy who's been dead for a long time, such that all of the flesh has gone away and only the

skeleton remains, except that, on the finger, there is this ring, and it is a well-made gold ring with a stone on it. The guy has been there for so long that even whatever clothes he was wearing have worn away. And so, there are just bones and this metal ring.

2.14 So of course, Gyges takes the ring. Anyone surprised at that? They say, "No, no, Gyges wouldn't do that. He would just leave the ring where it was and leave the cave." No, no, he took the ring. He takes it out and he starts to play around. He puts it on, and it looks nice. Then he's playing, and it's got a collet. It's the thing that you mount the stone on. He's spinning it around and realizes that whenever he turns the stone so it's facing palm inwards, he becomes invisible. Then if he turns the stone back again, he becomes visible again. So he plays around with that and he realizes he can become invisible anytime he wants.

2.15 All right, this is the myth part. But what happens next? Now, if I said, "Well, he took the ring off, went back in the cave, put it back on the finger, and left," no, that's not going to happen. Don't be ridiculous. Of course, what happened is, Gyges basically did what anybody would do in that situation. You have a ring that would let you be invisible anytime you want. What would you do?

2.16 Well, what Gyges did was, first thing is, said goodbye to the sheep, back to town, and then, as the story goes, a crime wave ensues. Stuff goes missing, nice stuff from all kinds of people's places, and it's a big mystery. How could this stuff just be disappearing? More darkly, young women with whom Gyges had never had much luck, poor, smells like a sheep, no girlfriends, now—this is the dark part—rapes start to occur. Any woman who is left unattended.

2.17 And then perhaps even darker, people end up dead. Complete mystery how this happens. Just suddenly, out of nowhere, a rock crashes their skull open. A knife stabs them in the back. And then, as the story is developed by Plato and others, Gyges basically steals and rapes and kills his way to the top, kills the king's wife, or sorry, kills the king. He has other plans for the king's wife, marries the king's wife, and becomes the new king himself, with all of the resources at his disposal, plus, of course, his magic ring and so on.

2.18 OK, that's the myth, and what does it teach us? Now, suppose we say, well, who is Gyges? Well, he's a guy. What do we suppose— look, Gyges is just a guy, and basically, he's a stand-in for any guy. Because what would any guy do in that situation? Basically, what Gyges would do. Maybe you would steal different stuff, but you'd still steal stuff. Maybe you'd rape different women and kill different people, but you'd still do those same things.

2.19 So the point is, we as human beings, we have all of these desires for things that we want to do, and we don't really care how we're going to get those things. And the only reason we're not doing these stealings and rapings and murderings is because we're afraid we're going to get caught. But if we took that away, we had the magic ring, we

could do whatever we want, and we knew we would not be caught, that's what we would do. Every human being is immoral, antisocial, a menace to society, to the extent they have the power to get away with acting on whatever desires they wanted to do so.

2.20 Now, that is a story about human nature now, not just an interesting old Greek myth. If we upgrade it to a philosophical story, this is the human condition. Then we start to say, well, what does this imply for our politics? What does it imply for our social relations? What is it going to imply for education? Because if we start to think of our little children as little Gygeses, then the tenor of our education is going to be somewhat different. How can we get them, well, not to do those sorts of things? And that's a different kind of problem.

2.21 Now, another myth. Plato is a great storyteller. This is also from his Republic, and it works with the myth of Gyges. It's partly human nature is a matter of our emotions and our desires and what we want and what we're willing to do to get whatever it is that we want. A part of our human nature is also cognitive. Our knowledge, beliefs, opinions, truth versus falsity, and so on. Plato has another famous myth he calls the myth of the cave, that is meant to illustrate this set of issues.

2.22 We have a picture here, and we're going to go to the Republic and read a little bit, but I want you to keep this picture in mind. So it's a cave, and down at the bottom, there are some people chained, and they are in the shadows, and they are looking at shadows cast on the wall of the back of the cave. But then we realize that those are just shadows being cast by puppets or marionettes that are being carried by some strange characters behind a way. There's a fire built partway up the cave route that's casting light on these cutout, puppet-y types of things.

2.23 Then we see some people who are climbing up to get out of the cave, and then they're out into the real world, and the sun is shining, and there are trees and birds and so on. Keep that image in mind. Here's a different representation, more in Greek style of the same image—fire, cutout, person watching the shadow on the wall. Now let's go to Plato's Republic and read the account of the myth of the cave and the philosophical work that Plato wants us to have do.

2.24 So at this point, in this long, long work, Socrates, this is perhaps a character Socrates, not so much the real historical Socrates, is having a conversation with two younger students, very bright young men, Glaucon and Adeimantus, and here he's speaking with Glaucon. And the text goes like this, Socrates speaking.

2.25 "And now I said, let me show in a figure," that's a drawing, "how far our nature is enlightened or unenlightened." So we're interested in being enlightened and unenlightened. Where are we? What's our nature as human beings? "Behold, human beings living in an underground den," or cave, "which has a mouth open towards the light and reaching all along the den, here they have been from their childhood, and

have their legs and necks chained so that they cannot move, and can only see before them, being prevented by the chains from turning round their heads."

- 2.26** OK, so this is quite brutal. From childhood, they have been chained at the bottom of a cave. "Above and behind them, a fire is blazing at a distance, and between the fire and the prisoners, there is a raised way, an obstacle, and you will see, if you look, a low wall built along the way, like the screen which marionette players have in front of them," or puppets over which they show the puppets. Glaucon says, "I see." Socrates carries on. "Some of them are talking, others silent." Glaucon says, "You've shown me a strange image, and they are strange prisoners." Socrates says, "Like ourselves." I, I being Socrates, reply.
- 2.27** And that's an invitation. So in which way are these strange prisoners like us? This is a representation of the human condition. In some sense, we are prisoners from childhood at the bottom of a cave. So how do we unpack this metaphor? And now, Socrates carries on the story after we've absorbed it so far.
- 2.28** "Look again and see what will naturally follow if the prisoners are released and disabused of their error. At first, when any of them is liberated and compelled suddenly to stand up and turn his neck round and walk and look towards the light, he will suffer sharp pains." Let's pause there. So, first step of enlightenment. So, notice the passive voice here. Sometimes casual readers of Plato will say, "Oh, the myth of the cave, it's about self-liberation and a quest for enlightenment in the higher world."
- 2.29** You notice that Plato quite clearly in the original language says it's not a self-liberation. There's someone else who liberates them. "If the prisoners are released, if any of them is liberated and compelled suddenly to stand up." So it's not saying that we want naturally to free ourselves and stand up and rise up and become enlightened. The first step involves someone else who breaks our chains, makes us stand up, compels us to turn around and look towards the light. And that's important. So we'll keep that in mind as well.
- 2.30** "He will suffer sharp pains." Well, we all know this from first experience. When we're in the dark, we're looking in shadows, if suddenly we have to look at a light source, even if it's just a fire, it hurts. But metaphorically, enlightenment and the path of education is painful. It's not something that we want to do. It's not something that we start and initiate by ourselves. It's not something that we enjoy. It's something that is forced upon us, that we are made to do, and it's painful. And that's built into this metaphor.
- 2.31** But progress has been made. "Then he will be able to conceive someone is saying to him, what he saw before was an illusion, but that now when he is approaching nearer to being and his eyes turn toward more real existence, he has a clearer vision. And you may imagine further that his instructor is pointing to the objects as they pass and requiring him to name them. Will he not be perplexed?"

- 2.32** So your eyes adjust to the flame, to the light; the pain starts to subside, and then you have a realization. All of these things that you saw on the wall of the cave, you said those are real things. Cutouts of dogs and horses and cats and human beings, and some of them are talking and some of them are barking, you realize those are just shadows on the wall. Those aren't the real thing at all. But now what you can see is these cut-out things, the puppets, the marionettes, and that they're being pulled. There are actors back there who are making all of these sounds and voices and so on.
- 2.33** Your knowledge has improved. That is just shadows. Then these are the things that are—are they real, or are they only more real? And further education, of course, is going to be necessary. "But even so, if he is compelled to look straight at the light, will he not have a pain in his eyes which will make him turn away to take refuge in the objects of vision?"
- 2.34** What we will do is, the teacher says, "No, I want you to become more enlightened. You have to get used to this hard fire light. Look into it," but the student doesn't want to. "It hurts. I want to go back. I want to go back to the comfort where it's safe, where it doesn't hurt." And so on. That has to be overcome.
- 2.35** But next step, because the teacher at this point knows that these are just puppets behind the wall. This is just a fire, and fire is, of course, real light, but it's not as real as higher light. There's something more to be ascended to. And so the teacher has to drag the person as he puts it upwards and out of the cave.
- 2.36** "And suppose once more that he is reluctantly dragged up a steep and rugged ascent, and held fast until he's forced into the presence of the sun himself. Is he not likely to be pained and irritated?" Let's pause again. We drag the person out of the cave into the higher world, and then we look up into the sky, and what do we see? We see the sun, the sun itself. The sun is the source of all light, the source of all life perhaps, the source of all enlightenment.
- 2.37** And now you are in a position to see you have become enlightened. This is the real world, the higher world, and this lower world is just a cave full of shadows and illusions and fake things. But this entire process that I've been made to go through to become enlightened, it's not something I wanted to do. It was forced upon me by someone who knew better what I needed to do, and made me do it. It was painful and it was hard and it was difficult, but I was made to go through the process and now I have achieved enlightenment.
- 2.38** Now, finally, he sees the sun. Initially, of course, you can't look at the sun itself directly, it's too much to bear. But then he will, as part of his enlightenment, realize that the sun is the source of all reality, source of all light, source of all enlightenment. "He will proceed to argue that this is he," the sun, "who gives the season and the years and is the guardian of all that is in the visible world." OK, so good-news story. Enlightenment has been achieved.

- 2.39 But then, Plato says, plot twist. What if you remember all of your friends and family back down in the cave, still chained, still probably unwashed, still believing all sorts of false and illusory things, and they have no idea of the wonderfulness of this higher, enlightened world that you have now experienced for yourself firsthand?
- 2.40 Now, you could say, "Well, too bad for them," not in a callous way, but I have achieved enlightenment, and this is great. I don't want to go back in the stinky old cave. But no, there's a moral point here about obligation and duty and the cycle of life. "And when he remembered his old habitation," that is to say, down in the cave, "and the wisdom of the den," that's sarcastic, the wisdom, right, down in the den, "and his fellow prisoners, do you not suppose that he would felicitate himself on the change, and pity them?"
- 2.41 So you have achieved enlightenment. You didn't want to achieve enlightenment, but someone better than you, who knew better, made you achieve enlightenment. And now you realize that you have a responsibility based on pity for all of the unenlightened people down below, to rescue them as well.
- 2.42 Now, this is also an interesting part of the metaphor, because when we think about parenting and educating and becoming teachers, what is your motivation for becoming a teacher? If you start to say things like, well, it's because I love to see little children happy and develop and flourish right and so forth, that's not what Plato is talking about. The word he uses is pity. You feel pity for those who are still in an inferior state of enlightenment.
- 2.43 All right, it's best for me if I were just to stay up in this beautiful higher world, contemplate the sun and all of the beautiful real things that are here. But I feel I have an obligation to give all of that up, to go back into the dark cave. And what's it like? When you reverse the process and now you're not going from darkness into the light, you're now going from bright light into the darkness, again, it's hard on your eyes. You're banging into things. It takes a while for you to get used to.
- 2.44 So you go stumbling down into the cave. You say, "Hey, everybody, I have achieved enlightenment." And then, meanwhile, you're tripping all over yourself. They're going to think, "This person is confused, weird, wishful thinking." They're not going to pay attention to you. You start telling them about these wonderful, glorious things in a higher world that they have never seen or imagined. And you tell them that everything that they think is true and real and important, that that is all just illusion, not important at all.
- 2.45 "Come on, follow me. Let's go up and escape from this smelly old cave." And what will they say? "No." They will resist you. They will make fun of you. They will mock you, and worse. Plato's words, "And is there anything surprising in one who passes from divine contemplation to the evil state of man," back down in the cave, "misbehaving himself

in a ridiculous manner if, while his eyes are blinking and before he has become accustomed to the surrounding darkness, he is compelled to fight in courts of law."

- 2.46** They're going to argue with him over all sorts of things, "or in other places, about the images or the shadows of images of justice, and is endeavoring to meet the conceptions of those who have never seen absolute justice." So it's not just on day-to-day things as well, but really important things. You get involved in legal disputes now that you are enlightened, but you're down in the cave. You're the one who really knows what justice is, what the truth is. And you're going to be forced to go to courts of law down in, but cave courts of law, based on ignorance and unenlightenment. And so, injustice will continue.
- 2.47** What's the only thing that you can do in that circumstance? And what are you justified in doing in that circumstance? You are justified in doing to them what was done to you. Compulsion. Someone forced you to stand up. They took their hands and laid them on you and raised you up, grabbed you and turned you around and said, "Look, there's the truth." And they dragged you against your will toward the higher reality, the higher truth, the higher being. And that's how it has to happen.
- 2.48** So if we put the myth of the cave together with the myth of Gyges, we have a different understanding philosophically about the world, about human nature. It's not a good-news, at the outset, understanding of human nature. We're all like Gyges, with these acquisitive and antisocial desires, and we'd all do all kinds of nasty things to each other if we thought we could get away with it. We're all ignorant of true reality. We all think that the things down here are important and real.
- 2.49** What we really need are enlightened individuals who have the right character and the right knowledge to make us do what we don't necessarily want to do. That is the true path. This is the direction that Plato, in his philosophy, wants us to go. And already we have our first great contrast between the Aristotelian and the Platonic, the Aristotelian from the last lecture and the Platonic now being introduced in this lecture.
- 2.50** I want to carry on with a few other quotations extracted from Plato's writings, different works. So far we've been emphasizing the Republic. Here from a work on the theory of knowledge. We're really interested in knowledge, and education has to be about knowledge and truth. And so we have a long investigation in the nature of truth and knowledge. We have this statement.
- 2.51** "We ought to try to escape from Earth to the dwelling of the gods as quickly as we can. And to escape is to become like god, so far as this is possible. And to become god is to become righteous and holy and wise." So that is a suggestion of the true path of knowledge. And again, the contrast is to say, I'm interested in the natural world. We like to look at it. We like to investigate it, have adventures, and live our lives in this world, and of course, the higher world. Here, the emphasis is on getting away from the natural world, from the Earth, to the higher world as fast as we possibly can.

- 2.52** That's the object of knowledge. We are, as human beings, on this knowledge quest, where we have a body as part of knowledge, sensory, physical knowledge, or is it something more pure, intellectual, mental, spiritual? We have an indication of a separation of mind and body, not mind working with body, but an antagonism and a different destiny for mind and body.
- 2.53** From a work called the *Phaedo*, this is actually the death scene from Socrates, Socrates speaking. "If we are ever to have pure knowledge of anything, we must get rid of the body and contemplate things by themselves with the soul by itself." The body is a distraction, the senses are a distraction, because they tie us to the physical world. True knowledge is of the higher world, and this requires that only the part of us that can connect with the higher world, our soul, makes that connection, not our bodies.
- 2.54** "No pure knowledge is possible in the company of the body. So long as we are alive, we shall continue closest to knowledge if we avoid as much as we can all contact and association with the body, and instead of allowing ourselves to become infected with its nature." Notice that. The things physical and things bodily are an infection. That metaphor, real knowledge. "To purify ourselves from it until God himself gives us deliverance."
- 2.55** When we have gone through a proper course of education, we've purified ourselves and we've reached a higher state of knowledge, what will we realize about what's important and what's not important, what the real, say, meaning of life is? Again, from the *Republic*. "When a man's desires have been taught to flow in the channel of learning and all that sort of thing, they will be concerned, I presume, with the pleasures of the soul in itself, and will be indifferent to those of which the body is the instrument, if the man is a true and not a sham philosopher."
- 2.56** Lots of philosophers will tell you, be interested in the mind, in the soul, in the body, the whole package. They are sham philosophers, not real philosophers. Real philosophers are only concerned with the soul and knowledge of the higher world. They realize that physical things and things of the natural world really are of no value, of little value, if that.
- 2.57** Another, driving in this point. "Do you think that a mind habituated to thoughts of grandeur and the contemplation of all time and existence can deem this life of man a thing of great concern?" What about human life itself? Is that important? Is it great? Is it significant? If you're a real philosopher, you're contemplating all things that are grand and wonderful, but no, it's impossible. This life is not important to a philosopher. It's the higher, the beyond.
- 2.58** Now, we have a big problem, of course, like the enlightened ones going back into the cave, because the cave dwellers, that's the rest of human beings, 99.9% of them, no doubt, have no interest in true philosophy. One way that comes out is not only in family life and friendships and relationships, but also in politics. How are we going to not only

get our own lives, but our family lives, our political lives? It has to have the right kind of person in charge.

- 2.59** Here's Plato's famous line about political power, that only true philosophers should be allowed to exercise political power. We're not going to be able to solve our political problems until we have, the famous phrase is "philosopher kings." Kings being the politically power one, philosophers being the ones who have the knowledge and the character to exercise that power responsibly and in the right direction.
- 2.60** "Unless either philosophers rule in our cities or those whom we now call rulers and potentates engage genuinely and adequately in philosophy, and political power and philosophy coincide, there is no end, my dear Glaucon, to troubles for our cities, nor, I think, for the human race." It's going to be a hierarchical political structure with the philosophers who have the knowledge and the character in power. Then and only then will politics and social structures go the right way.
- 2.61** Now, of course, our question is, we start to scale that down, how are we going to raise children? What's the organization of the household going to be? What's the organization of the school going to be, such that we prepare students for their proper function in society, and perhaps one in a thousand will have the potential to become philosopher-kings themselves? And so the educational project is also a part of Plato's Republic and some of his other writings as well.
- 2.62** But one interesting angle on this is the place of Homer and the other poets in Plato's official political regime and in the place of education. And toward the end of the Republic, he remarks with this, another famous line, that there is a quarrel, an ancient quarrel, between poetry and philosophy. And poetry is used a little broadly here. It means not only rhythmical, metrical poetry, but all literary works. And he also applies this argument to music and the other art forms as well.
- 2.63** There's a differentiation then between the philosophical approach to understanding the world, and providing guidance to human beings as they go about their lives, and what the artists, what the poets do. He says there's a quarrel between them. It's an either-or. Plato comes down definitely on the side of the philosophers. What do the poets do? Are they concerned with truth? No, they are not. They just make stuff up. Philosophers are the ones who are considered with truth.
- 2.64** A big part of this is all of the stories that the poets make up about the gods, having the gods do all kinds of ridiculous and crazy things. So Zeus is going around raping all of these women, cheating on his wife, and so on. We see how Athena behaves with Arachne, losing her cool, losing her temper, and flying off the handle and all these over-the-top punishments and so on and so forth. And all of these are poems.
- 2.65** They're great stories that make the gods look foolish or ridiculous. And maybe they give some comfort to human beings, saying, "See, we don't have to try so hard because the gods aren't really that great after all." And that's an immoral lesson. Not only is it false

about the way the gods really are, but it's not teaching children and other people proper moral lessons.

- 2.66** Same thing can be said about various kinds of music. What happens when people start music? Well, they typically like to move their bodies around. If you have men and women dancing together, they're showing their bodies to each other, and that leads to all sorts of things. Then drinking is typically involved in these sorts of festivals. It just becomes a celebration of immorality. That's what the poets are all about, partying and making stuff up and giving false depictions of everything and encouraging people to go down the wrong path.
- 2.67** Homer. Bad guy, very bad guy. I'll save that one for a moment. We talked about the Iliad, and after Hector is killed by Achilles, there's this very tender scene where Hector's father comes and begs Achilles to give him the body of his son back, so he can take it and do proper funeral rites. And Priam is, when he finally gets his body, just broken down and crying and inconsolable at the death of his son.
- 2.68** And Plato's reaction is this, look, is Priam a manly man? When you're a man, you have to be tough, you have to be strong. Bad stuff happens in this world. And what you do not do, and what we should not be teaching young people through these stories of Homer and the others, is to have supposedly upright men in position, and this is Plato's language, "wailing like women," crying. No, man, suck it up. Homer and the other poets, bad. Wrong for the city, advocate of censorship of all of the poetic and musical works on political grounds, but certainly in the case of education.
- 2.69** What will we let children be exposed to? It's not going to be Homer and it's not going to be Hesiod and many of the other poets as well. Strictly philosophically vetted works for truth and for good character. Children, we know, might grow up to be like Gyges if their desires are left untamed. So there's a certain amount of strictness and obedience that is necessary there.
- 2.70** Children also don't really want to learn. They don't want to go to school. They don't want to study their books. They just want to play. And what do they do when they play? Well, they start acting like poets. They start making up their own games, making up their own rules, and so forth. And there's a strong suggestion in Plato that that also is something that needs to be controlled and regulated.
- 2.71** One of his very later works called The Laws, he has a character called the Athenian Stranger, and they talk about lots of things, but they do get to education of children, and there's this about games. Yes, we want games. Kids can learn through games, but they have to be the right kind of games.
- 2.72** "I assert that there exists in every state a complete ignorance about children's games and how they are of decisive importance for legislation." I'll skip down a little bit. "For when the program of games is prescribed and secures that the same children always

play the same games and delight in the same toys in the same ways and under the same conditions, it allows the real and serious laws to remain undisturbed."

- 2.73** What children need to learn is, games are not just about fun and doing whatever you like and being experimental and entrepreneurial. We're training them to be citizens, and the citizens are going to be governed by laws, and the laws say there's one right thing to do and one right way to do it, and that has to be instilled in education from the beginning in games. No variation, so very different tenor to the philosophy and cast to the education.
- 2.74** I want to fast forward a few centuries. Greece went into decline, eventually conquered by the upstart Romans, and the Romans became the great empire of the ancient world. Lots of interesting stories there. We're going to skip over most of them. But there is an interesting phrase about the relationship between Roman military and political might, and Greek philosophy, arts, and culture. And the phrase goes like this. It says that while the Romans conquered the Greeks, the Greeks captured the Romans.
- 2.75** So the idea here is, the Romans physically controlled the Greeks and lots of other territories as well, through their military might. But they were so impressed by the power and versatility and flexibility of all of the Greek accomplishments in culture, philosophy, arts, science, that they adopted that wholesale. It really was significantly Greek culture that became Roman culture.
- 2.76** Many of the Roman gods were just Greek gods with new names, and all of the Greek schools of philosophy and ways of doing theater and sciences became part and parcel of mature, cultured, educated Roman society. Roman experiment lasts for many centuries. Rome also goes into a decline phase. We're starting to have a hard time holding its empire together. It breaks into an East and West. Eventually, it's invaded successively by various barbarian tribes, as they considered them, from the north.
- 2.77** Then, in the 300s, a hugely significant event occurs as one of the Roman emperors converts to Christianity. The next significant phase in development, where we're getting away from traditional Greek schools of philosophy and education into something new, starts to happen in the 300s. Now, there are lots of different religions kicking around all over the Roman Empire, dozens and dozens of them. By and large, the Romans, they did persecute here and there, but for the most part, they were happy with saying, "We've conquered you. You can keep your gods. That's fine. Just after, on certain rituals, recognize the Roman gods and so forth."
- 2.78** The Jews and the Christians had always has been troublesome in that regard. And so certain amounts of persecution were visited upon them by the Romans, and they were seen as uppity and problematic on that particular score. So the conversion of Constantine to Christianity in the early 300s was huge because then, suddenly, Christianity is not just one of many religions kicking around. It becomes the official state

religion of the most powerful political unit in the ancient world. And that is a game changer, both politically, culturally, and of course, for the religion.

- 2.79** Things start to occur quickly then in the 300s for the institutionalization of Christian thinking. Fairly quickly, the Christians realized they needed to figure out what their holy scriptures were, what books were going to be in the Bible, and what books were not going to be in the Bible. This is now almost three centuries after the death of Jesus. There are lots of gospels, lots of accounts of the life of Jesus, lots of letters written by Paul and other followers and so on. What are we going to do with all of the Hebrew scriptures and different versions of those kept kicking around?
- 2.80** So, there's a council forming what is then the official Christian set of scriptures and what are apocrypha, not official parts of the doctrine. There's also the issue of what we are going to do with people who are dissenters to what is now Roman religion, which is Christianity. What do we do with all of these different religions that are part of the Roman Empire? That's partly a religious decision. Are we going to tolerate them as, by and large, the Romans did or not? And if we are not, then what is religiously and politically the proper response to the various other religions going to be?
- 2.81** And another huge thing, and this is going to take us to our next great philosopher, Augustine, is more intellectual and theological because there are all of these schools of philosophy kicking around, the Greeks and the Romans, very sophisticated, developing all of these schools of Stoicism and Aristotelianism and Platonism and Neoplatonism and Cynicism and so forth. They're very sophisticated, and most of them are pretty critical of Christianity.
- 2.82** From their perspective, you have to believe all these really weird things, miraculous, spooky, virgin births and gods that are three in one, et cetera, et cetera. There was a lot of fun being poked at Christian doctrine in the 300s by the still-extant, various other schools. The problem that the Christians had at this point is, they did not yet have an intellectual capable of addressing those philosophical arguments, very sophisticated, at a proper level. That is what St. Augustine is going to contribute, his great signal contribution.
- 2.83** Christianity, for the first few centuries, was largely an oral tradition, and oral traditions tend to become pretty scattered. It also was not necessarily very systematic. It breaks off here and picks up there. It's this letter, it's this set of poems and so on. Some of them are written down clearly, but they're in parables. What exactly was Jesus saying, literally? What was he saying metaphorically? How are we supposed to interpret all of this stuff?
- 2.84** And not very much work had been done on figuring out what Christian doctrine is, specifically, literally, systematically, at a high level, so that we can argue against the Aristotelians and the Stoics and all of the other very sophisticated Greek and Roman schools of thought. Augustine was a genius, brilliant philosophically, brilliant

theologically, and his contribution, such as why he was sainted and the great degree of his influence, was for that intellectual contribution.

- 2.85** One of my professors in graduate school, a medieval philosophy specialist, made the strong claim that the most important philosopher in human history was this guy, St. Augustine—not necessarily that he agreed with him, but just in terms of his fundamental importance and influence on subsequent Western intellectual tradition. And that's another interesting controversial point worth arguing about. But let's get into some of Augustine.
- 2.86** He's most known for these two works, a work called *The City of God*, where he systematically goes through all of the major philosophical issues that are also theological issues, so develops Christian theology, its positions, and its answers to all of the criticisms from the major competing schools of thought.
- 2.87** Then the *Confessions*, which is really an autobiography written later in his life, where he talks through his life and how originally he was not a Christian. He was rather a rotten little kid, by his own description, but eventually became a Christian, and his personal journey and an invitation to let that personal journey also become your journey. It's partly also an evangelical conversion-type of doctrine as well.
- 2.88** Let's plunge in a little bit. The important thing here also is going to be that this is written in Latin. It's not the important thing, but an important thing. It's not written in Hebrew. It's not written in New Testament Greek, as we now call it, or in Aramaic. It is now written in Latin, which is the language of the Romans. It's now the official language. And so this is meant for the new political and social reality of the new world.
- 2.89** All right, so let's go and have some time where we go through Augustine's *Confessions*. So, preface, where he dedicates the book almost as a letter that he's writing to his son, so his son remembers him and knows the family history, but then he knows his son will use it as a piece of marketing. We can use that language also for the tradition. Book 1, beginning with an invocation to God, of course.
- 2.90** "Great art thou, O God. Great is thy power, infinite thy wisdom. I seek thee, O Lord, in the faith that you gave me and through the Son who inspired me. But how shall I call upon my God?" Now, partly it's worth reflecting on, this is now an infinite God. This is not the God of the Greeks who are limited in their power. This is a unified God, rather than having all of the various divine or supernatural powers split up and possessed by different gods. It's a much more integrated, unified theology that we are working in.
- 2.91** "And how shall I call upon my God?" How can I come to know my God? How can I speak to you when I am a man and a man in a body? "Still, dust and ashes as I am, allow me to speak before thy mercy, from whence I came to this life or death, life-in-death, or is it death-in-life?" And that's a provocative phrase. I am alive, but living in death. Is that the right way to put it? Or is it that I actually am dead, but in a living state? How would we put this?

- 2.92** There's not only that issue, but the moral depravity of our condition. "When a man cries, 'Woe, the sins of men,' thou showest him mercy, for thou made the man, but not the sin. But what of the infant, mewling, jealousy, the anger?" This is a hard issue. Because according to Christian doctrine, God is an infinite perfect being who created the world and created man. And so we would expect an infinitely powerful, all-knowing, perfectly good being to create perfectly good things.
- 2.93** But the world that we actually live in is full of corruption and bad things. Where did all of the corruption and badness come from? Also, we as Christians, we say we are made by God. Sometimes we say we are made in God's image. It's one of the phrases. But we also know that human beings are sinful and nasty and do all sorts of disgusting things as well. So if we're made in the image of God, then we would be saying that God also is nasty and sinful and disgusting, but that's not possibly true. How do we interpret this "made in the image of God" phrase?
- 2.94** Even so, we still have the problem, if God made us, then why are we not perfect? Or like angels say, why are we so awful? Where does this sin come from? It can't be from God. We might say, oh, well, there's the devil. The devil is this evil being who just infects the world and human beings with sin. But then if God made everything, then did God make the devil? And so we've just relocated the problem to another issue. Where does this come from?
- 2.95** And it seems that this sin is at the origin of humans, something fundamental built into us that we also have to believe as human beings. We can't believe that we were born good and that, at various points later in our development, we became evil. And we might then say, well, then I decided to do some evil things, and that's where evil came from. But God made me good and perfect. The Christian doctrine is original sin, right from infancy.
- 2.96** "For in thy sight, not even the day-old infant is free from sin." Newborn baby, right out of the womb, sinner. Where did the sin come from? That's the problem. But it's asserted as true. And the sin carries on. "I grew out of infancy, or rather, boyhood grew into me, and I grew up, and now I was a chattering boy," so apparently he was very talkative, one of those kids when he was in school. "Oh my God, I was sent to a school to learn tricks of speech for the sake of honor and foolish wealth."
- 2.97** Classical education in the Greek model—how do I become an esteemed citizen and a prosperous citizen? That's what they were teaching me, all the wrong stuff. "When I learned not, I was flogged, as our forefathers thought proper." So I was beaten. "And I was punished for playing." Kids want to play. Proper thing to do? Not when you're in school. "And why? So that I might learn as a man to play more shameful games."
- 2.98** So there's a hypocrisy. The parents are all being disobedient and doing their own thing, but they're teaching kids, "Oh, you shouldn't be doing that," and beating them when they do it. But this is the world that we are. But of course, the kids are disobedient. "I

sinned against my parents and my teachers." So I was a sinner from infancy; I was a sinner as a boy. And it gets worse because I had heard about Christianity.

2.99 "Even as a boy, I had heard of eternal life promised to us through the Lord our God." His mother was a Christian. She taught him Christianity, but he was an evil brat, didn't listen, didn't want to listen, didn't believe. "I had no love of learning, and hated to be driven to it, for no man does well against his will." Another interesting point. Think of Gyges here. "So I, so small a boy, yet so great a sinner, was punished as thou ordained. I learned to read. I followed the wanderings of Aeneas and the sadness of Dido, who slew herself for love."

2.100 So it was a great Roman epic written by Virgil. Tragic love story, death, tugs at the emotions, but no redeeming moral values from the Christian perspective in this story. "Yet, oh my God, light of my heart, I did not love thee, and thus committed fornication against thee, while those around me cried out, 'Well done.' The friendship of this world is indeed fornication of against thee." So I engaged in improper relations with all of the pleasures of this world. To enjoy this world is to sin against God. That's not what God wants us to do.

2.101 "O woeful torrent of human custom, drowning the sons of Eve." And here again, Homer, bad. "Which of our gowned masters," the teachers, "could admit that Homer's fictions transfer human evils to the gods, and so lead men to imitate their foulness?" God and the gods are perfect. They're good. It is human beings that are sinful and evil. Yet these poets, for whatever nefarious motivations they have, they take human badness, human sins, human evils, and they make up stories about the gods, transferring them to the gods.

2.102 And then, as he suggests here, therefore, we come to be less respectful of the gods. They do it, too. And so we become less moral in our own behavior. Homer, out. As I got older, it gets worse. "I shunned the bright path of friendship and pursued the fleshly passion of fornication." Women. "If I had me a guide to turn my tide to the shore of marriage, as thy law prescribes, O Lord, or I might have heeded that voice from the clouds, 'It is good for a man not to touch a woman,' and with joy become 'a eunuch for the kingdom of heaven's sake.'"

2.103 So there I was a teenager. I had heard Christian doctrine. When it comes to sexual passions and having sex, there's two options. One is the best option. Don't touch women. Leave women alone. Dedicate yourself entirely to God and the higher world. It is good for a man not to touch a woman. That's the first option from Saint Paul.

2.104 The other option is to say, well, it's not very realistic for the vast majority of human beings who are slaves to their physical passions and so on. So what we need to do is institute marriage and say, OK, you can have sex, but only with one person, married to that person, and then other conditions will be added along. Try to contain it in a

particular institution. He knew that this is right, but he disobeyed, ignored all of that, because he wanted to be a slave to his passions.

- 2.105** Stealing. "Theft is punished by thy law and by man's." Every kid learns quickly, stealing is wrong. "Yet a group of young scoundrels, I among them, did steal away a load of pears from our neighbor's trees." There are some nice-looking pears. Let's go steal some of those pears. Yeah, let's do it. You steal the pears. He said, OK, fine. They taste good. It's fun. And Augustine points out, it's actually even worse than that. Because I was stealing them not just because I wanted to have some tasty pears, but I wanted to steal. I liked breaking the law. I liked the transgression.
- 2.106** "It was an evil act, and I loved it for being evil." That's what we're dealing with. This is how bad human beings are. "The theater captivated me." It's not just Homer, it's all of those other playwrights. "Now, what real compassion can there be in watching faked emotion?" It's fake. It's just made up. They're not really feeling that. They truck in illusion, not reality. "Yet I was moved to tears by fictions and so became infected with a foul itch, the scratching of which led to inflammation, swelling, and putrefaction."
- 2.107** I learned Cicero, the great orator, the great rhetoric, the great lawyer, and through him I discovered philosophy and the magnificence of what the Greeks and the Romans had accomplished in philosophy and in rhetoric and their legal system and so forth. At the same time he had learned Christianity, that philosophy was a temptation and not to be taken up with. It's the wrong path.
- 2.108** This is quoting now from Colossians. "But I did not know then the words of thy apostle, 'Beware lest any man spoil you through philosophy and vain deceit.'" You want to know truth. You want to acquire goodness and character. You will not get there from philosophy. You will get there from the one true faith. Do not let yourself be distracted by all of those worldly philosophies.
- 2.109** Things start to turn around a little bit through the tragic death of a friend. They're still relatively young. And so for everything, it's just childhood, play, lots of good sex and stealing and doing whatever you feel like and so forth. And then a friend dies, and that becomes an immediate step into adulthood and seriousness, death.
- 2.110** "A dear friend of childhood, who I had turned away from true faith, was taken unto thee after a fever. My heart was utterly darkened by this sorrow, and everywhere I looked, I saw death. At that time, I marveled that other mortals went on living since he whom I had loved as if he would never die was now dead." And so now I'm going to have to think about death, and I, too, am going to die. And how do I think about this issue? Is life just going to be one big party? Or am I going to finally get serious?
- 2.111** "Lovely things grow to perfection. They wax old and perish. This is the way of things." But there is something that is eternal. "'Do I ever pass away?' asks the word of God," and it does not. God is eternal. The word of God is eternal. So there is something

possible that can transcend the temporary attractions of this world, all of which are doomed to perish. And so then I realized, I don't need to worry about the philosophers.

2.112 "What did it profit me to untangle Aristotle's categories or naughty volumes of rhetoric without pious faith?" I can study the faith or I can study the philosophers like Aristotle here. Notice Aristotle is on the not-list with Homer and the others at this point. He mentions the influence of the bishop Ambrose, who's a very serious, very intelligent guy, and he increases Augustine's ability to interpret scripture. How do we deal with the naughty issues of what's allegorical and what's literal.

2.113 Notices that Ambrose is struggling with his vow of chastity, vow of celibacy. Augustine is struggling with that one as well. But through Ambrose, he comes to find the works of Plato, and he goes, aha, not all of the Greeks are bad. There's one exception, and that is the work of Plato. "Thou didst obtain from me certain books of the Platonists, and therein I found, not in the same words, but to the selfsame effect, 'In the beginning was the word, and the word was with God.' Truly, these books came from thee." Plato must have been inspired by God.

2.114 And so what we can possibly do, since we need to get Christianity theologically sophisticated and take on the Greeks, is integrate it with Plato, because Plato was a genius. And so what Augustine is eventually going to do is forge a synthesis of Christian doctrine with Platonic philosophy, and that's going to be the source of his influence.

2.115 "Having learned from the Platonists to search for the incorporeal truth," and then finally, the moral lesson and the transformation that's going to come, he starts to realize more how sinful he is, how evil he is, how little control he seems to have over his passion. He's growing more and more in his understanding of Platonic philosophy and the possibility of eternal truth that Christian religion offers to him.

2.116 But he goes through this stage where he is like a dual creature, tormented and divided within himself. Part of him wants to just plunge into an orgy of flesh, and part of him wants to have a pure spirit and reunite with God. He goes back and forth 20 times every single day, and he doesn't seem to be able to control himself. And he realizes that he, as a human being, is just not strong enough. He's going to need some sort of extra help to guide him, to make him take the final leap.

2.117 And so there's a famous scene where he is in a garden and he hears a voice of some children playing nearby and they say, "Pick up and read, pick up and read, pick up and read." And he picks up a copy of the Bible. It was close to him. It's a sign from God, opens a passage at random.

2.118 "Not in rioting," this is the passage he stumbles across, "not in rioting and drunkenness, not in chambering and wantonness, not in strife and envying, but in the Lord Jesus Christ, I had no need to read further. My heart was infused with something like the light of full certainty, and all the gloom of doubt vanished away." And he had his experience, and now he was resolved and able to carry on his path.

- 2.119** And then one more slight excerpt here, and then we'll make some closing remarks. This is toward the end of the Confessions. "I was saved," I being Augustine here. But think about what it took for me to become saved. Then consider the vast majority of other people and what they are like. Are they as intelligent? Are they as committed to goodness and character or not? And Augustine is quite pessimistic in this passage.
- 2.120** "All men would rather rejoice in truth than in falsehood." That is, if you put it abstractly to them, yeah, of course I want the truth. "Why, then, does truth generate hatred?" Try to tell them the truth about Christianity, and they hate you. You're like Plato's guy who escaped the cave, going back into the cave. They mock you. "And why does thy servant who preaches truth come to be an enemy?" I have been converted, but now my job is the rest of humankind, who seem to hate the truth and hate the person who is the messenger of the truth. What are we going to do with them?
- 2.121** What are we going to do, particularly, in the context now of Christianity being the official Roman religion? Here we come to the issue of lots of religions, lots of philosophies kicking around, but there can only be one government. The laws apply to everyone. There is only one true religion, the Christian religion. This is church-state integration. How are we going to get everybody else to convert to Christianity?
- 2.122** Now, we can try voluntary methods. We can say, with enough patience and tolerance, everybody is going to come around. Augustine says, absolutely not. No way. Not going to happen. Human beings are too depraved. Most of them are too stupid, to put it bluntly. But there is, as he puts it here, no salvation outside the church. There is one way, one truth. Just as, when one was in the cave, it required someone to force you to ascend to the light, we are going to have to use force, physical compulsion, to convert people to the true faith.
- 2.123** "Many must first be recalled to their Lord by the stripes of temporal scourging." Take a whip and you whip people, put stripes on their back, where the flesh is flayed open. It's going to be necessary. "Like evil slaves," that's what they are. They are slaves to this world, slaves to their body, slaves to their passion. "In some degree, like good-for-nothing fugitives." They are bad people running away from their master, and their true master is God.
- 2.124** We are a fugitive from true religion, and how you deal with runaway slaves, runaway bad people, people who have escaped from prison who properly belong there, those methods are going to be necessary. Now, he said he resisted this for quite a while, that he thought the beauty and truth and purity and the promise of eternal life that Christianity offers would bring people in. But he says he realized that that was just not an accurate understanding of human nature. That's not what people are like.
- 2.125** "I had not yet discovered the depths of evil to which their impunity would dare to venture." And this is even from my own personal experience. I realize what a rotten little kid I was, Augustine, growing up. There are lots of other kids like me, and even

worse, more evil, and they grow into become even more evil human beings. We have to use compulsion with respect to them.

- 2.126** This is how it works. Most people will just blow you off. They won't pay attention. You start to argue with them. You try to reason with them. You try to show them all of the positive incentives. They're not interested. They have their own passions, their own beliefs, and they're not going to take you seriously.
- 2.127** But if you can torture this person, you get his attention, you strap him down, you make his mind focus on pain and the alternative to pain, stripped down to its essentials. Then you get their attention, they will take it seriously. If a man, this is the direct quote now, "sees that it is unrighteousness for which man suffers, he may be induced, from the consideration that he is suffering and being tormented most fruitlessly, to change his purpose for the better, and may at the same time escape both the fruitless annoyance and the unrighteousness itself."
- 2.128** We can think of a thing; you want your kid to do something. The kid doesn't want to do it, isn't paying attention. So you give the kid a smack, you get his attention. And the kid knows another smack is coming if he doesn't do what you told him to do. He will give that serious consideration. Scale that up to the biggest of all big questions. Do you believe in the one true faith? Torture is the way to get it done.
- 2.129** Now, it might sound like we are being sadistic, that we are being brutal. Augustine goes at great lengths to say, no, we are torturing people because we love them. It is Christian love. Because here's the alternative. You as a Christian know the truth. If this person dies not a Christian, what is going to happen to him? He's going to burn. He's going to go to hell. Nasty things are going to happen to him forever. He has one chance to get good with God, and that chance is right now. That's the alternative, eternal suffering or salvation.
- 2.130** If you don't care about people, then you'll say, "I'll leave him to his own devices. He can burn in hell. That's fine. Forever." But that's not a loving response. If you really love someone, you don't want them to burn in hell. You want them to have the one possible chance to get right with God. And if it takes torture for a few hours, what is a few hours of torture in the light of forever? It's insignificant. And a loving Christian will not flinch from torture in that point.
- 2.131** One more. You have to do it for the proper motivation. The right people also have to do it. It doesn't mean that anybody can go out and start torturing their neighbors to get them to convert. This has to be an official. Salvation is within the church. The church is now a function of the state, so it's only authorized state officers who are going to be able to do this torturing. So it's not a free-for-all. OK. I just do, on this quotation, so it's official here.
- 2.132** "Only the state can persecute. Let the kings of the Earth serve Christ by making laws for him and for his cause," and this application of forceful methods to get people to

convert. "Kings serve God by enforcing with suitable rigor such laws as ordain what is righteous and punish what is the reverse." So there will be one law. It will be religious law. It will be based on Christian doctrine, and it will be enforced absolutely.

- 2.133** Now, our interest in this course is in education. So the immediate application then is, if we take that grand philosophical and theological scheme and we scale it down to the household, scale it down to the school room, scale it down to the one-on-one parent-child education, what is that going to look like? Here we have a model of education, extracting everything from this discussion so far.
- 2.134** The first thing is, children need to learn to follow the rules. Eve in the garden, the first sin was disobedience. She broke the rules. So the first lesson has to be, follow the rules. Kids don't typically want to follow the rules, so you have to impose that upon them. They're naturally unruly, disobedient, just like Eve, just like Augustine was as a kid. Obedience, obedience, obedience.
- 2.135** It's not for me to question the rules, to come up with alternatives to the rules, to decide when the rules apply and when they don't apply. The rules are the rules. I have to require self-discipline. We are naturally, notice the word naturally, by human nature, unruly, disobedient. We want to be free, do our own things, be anarchists, maybe even do naughty things and bad things, just because we want to do them. So, discipline has to be learned. It starts by imposed discipline, but as children learn self-discipline, following those rules is important.
- 2.136** We should expect that education is not fun, it's not a matter of play, it is painful. We're trying to get out of the cave. We have all of these bodily desires. When they don't get satisfied, we get frustrated, and we have to deal with that frustration and endure that frustration. We have to follow this hard course toward the upward truth. It's difficult, it's painful, but we have to do it. And of course, we have our teachers making us do it.
- 2.137** And none of this idea that there's lots of different poetic stories and lots of schools of philosophy, and we should study them all; rather, we know what the truth is. There is one that is the basis of our education. Everything else is either false or redundant, in which case we set it aside.
- 2.138** I'll leave you with one more quotation from St. Augustine, the Latin is, "Per molestias eruditio." That comes down to us with two translations. There's a slight possibility of interpretation. What do we mean by molesting people? One is to say, true education begins with physical abuse. We should expect a lot of spanking, a lot of using the rod, a lot of shaming, a lot of punishment of various sorts, to keep people on the right track. The same thing then is going to be scaled up politically. That's the kind of society.
- 2.139** Alternative translation is a little bit softer. True education begins with hardship. Education is not fun, it's not pleasurable, it's not anything like that. It is hard, it is difficult. We should expect, but not necessarily, physical abuse. There are different schools of philosophy, different schools of education coming out of the Augustinian

tradition, about how harshly or softly we interpret this. But nonetheless, we have staked out a very different philosophical territory in this Plato-Augustine world, compared to the Homer-Aristotle world. Let's stop there.

- 2.140** With the myth of Gyges, I had a bunch of questions because I thought, what about the opposite viewpoint with, if the Socratic method, with Plato, did they ever talk about how human nature being good?
- 2.141** You're putting your finger right on, squarely on one of the great fundamental issues. We take human beings and we make a fundamental value judgment about them. Do we think they are basically good or basically bad? Then the third option actually is that they're neutral. They're not born good or bad. They're born with some capacities that can be developed for good or developed for bad. All three of them are articulated. So far, we've seen two of them being articulated.
- 2.142** When Aristotle says, "All men by nature desire to know," he's articulating the position of we're naturally curious. We like to learn. We like to use our senses. We like to use our eyes. Kids like to develop their bodies and, "Look mommy, what I can do, or look, Daddy, what I can do." So it's a much more optimistic assessment of human nature. And then in Plato, and more extremely in Augustine, we get the opposite, that kids are nasty and disobedient and unruly and temper tantrums.
- 2.143** We can make up lots of stories. I always think of one, couple of years ago, I was at the beach and I saw a little girl, she was six years old, building a sandcastle. It's just like human nature. You take kids to the beach and they naturally start building things. Then she got the sandcastle built and she just stepped back. Then her brother was playing, turned around, and saw what she had built. You saw the change in his eye. And what happened next? Can you predict?
- 2.144** Did he come over and say, "Oh, that's a beautiful sandcastle you made. Can I help make another one with you?" No, he came over and stomped right on the sandcastle. Now, Aristotle would say, I'm putting air words in Aristotle's words, that we're all really like the little girl. We like to play and build stuff, and we expect other people to come along and help us build stuff. Other people will say, like Plato, no, we're actually like the little boy. We'd like to destroy other people's stuff. And whenever we have the power, because he's the big brother, she's the little sister, we are going to do it.
- 2.145** Then other people will want to say, it's not either-or, but we are born neutral. This is partly where the environment you are in, what parenting is like, what your schooling is like, whether your capacities for good or your capacities for bad, get encouraged, taught, reinforced, and so on. Now, you're right then to say, well, so far, all we have is stories, right? I can tell the myth of Gyges, I can talk about the kids at the beach, playing. You can tell stories about people who have power who do all sorts of wonderful things with their power.

- 2.146** So how do we figure out what human nature really is? And that's the ongoing challenge and debate. I think the best we can do in this short question period just will be to say that some people will say, "It's obvious to me that it's one or the other." Another interesting wrinkle, possibly we'll have some time to talk about this later in the semester, is to say, how much of these sorts of assessments about human nature being essentially good or trainable to the good, or human beings being naturally bad and needing the badness to be suppressed and goodness imposed upon them is a matter of philosophers or observers of human nature actually looking at people and drawing evidence from actual human behavior?
- 2.147** What people actually do with the power that they do, and how much of it is actually autobiographical. What philosophers and a lot of these commentators on human nature do is study the person they know the best, which is themselves. They know themselves from the inside. If I look inside and I realize that I'm really a pretty nasty guy, that partly I don't want to think that I'm an unusually nasty guy, it's a bit self-serving for me to say, "Well, I'm just like everybody else, and everybody else is just like me."
- 2.148** Or just to generalize from my circumstance, "Well, there's nothing unusual about me to everybody else." Then in that case, what you were calling philosophy really is more an exercise in autobiography, but you're just elevating it to philosophy. The people who are on the other side of the debate, they find it hard to believe that there really are evil people out there in the world. Sometimes they seem a little bit naive or a little bit optimistic, that what's really going on there is that they know themselves best, and they know that they are fundamentally decent human beings.
- 2.149** They want to be productive, and make the world a better place, and find true love, and they enjoy seeing other people succeeding in their lives as well. That just feels completely natural to them. So, when they become more philosophical, they are also doing philosophy as expressing autobiography. "Everybody must be like me, and people who are bad, there must be something wrong, something unusual happened in that case."
- 2.150** And we are still arguing about this, and also arguing about what would actually count as the right kind of evidence for a claim of what's actually built into human nature. So, great question.

Lecture 3: Renaissance Educational Visions

At a glance — Renaissance educational visions, starting in 1085 Spain.

- 3.1 All right, let's begin in the year 1085 in Spain. The year 1085 is significant in military, political, and religious history because Toledo, a city-state in the center of the peninsula there, reconquered from the European/Christian perspective from the Muslims and Arabs. Much of southern Spain had been controlled by the expansionist Arabs, expansionist Muslims, all across North Africa on up into, and creation of some magnificent architecture and schools of learning all over southern Spain.
- 3.2 But it was always from the European and Christian perspective, obviously a bad thing to have those guys there. Many attempts to push them further south. And in 1085, Toledo was retaken. This was important for our story because Toledo had been a major center of learning under the Arabs and Muslims who had made great efforts at preserving many of the Greek and Roman authors, translating them into Arabic, making many copies and so forth. And over this course of the centuries in Europe, most of those works had been lost.
- 3.3 Partly, it was a matter of neglect. There was a more general decline in learning, decline after the collapse eventually of the Roman Empire, particularly the Western Roman Empire. And partly, it was driven for disinterest in classical education, in classical learning. The Greeks and the Romans were largely pagans, so we're not going to read their books. We're certainly not going to make the effort of copying all of those manuscripts and so forth.
- 3.4 Perhaps many of them we will simply destroy because they should not be something that people are exposed to. It's too much of a temptation. The intellectual tenor through much of the Arab world and much of the more tolerant parts of the Muslim Arab world, extending all the way into Persia, was respectful of Greek and Roman learning, and many significant schools of Greek and Roman learning were maintained, and some first-rate commentaries and extensions of Greek philosophy, Greek medicine, Greek mathematics were made in those times.
- 3.5 So when Toledo was reconquered, some of the more open-minded Christian scholars were excited to find that the Arabs had left behind them this huge treasure trove of manuscripts. Not many of them in Greek, not many of them even in Latin, many of them had been translated into Arabic. But there then became a center of attraction, and many scholars from all over Europe, Christian Europe, came to Toledo to be exposed to these manuscripts, and there developed an informal school of translators, sometimes called the Toledo School of Translators.
- 3.6 It was never a formal institution of teaching and learning, but nonetheless, the bishop who was running the area was encouraging and funding teams of scholars to preserve as many of these manuscripts as possible, to start making copies of the important ones,

and even to translate them into a language, out of Arabic, out of Greek, and into Latin. And this was huge because the small number of Greek and Roman authors that had been preserved and were known to Christian Europe was suddenly expanded by who knows how many fold. And this is enormously exciting for anybody interested in a fuller breadth of human learning.

- 3.7 Now, one of the big questions, though, was, we are still Christian, and however open-minded we are to being interested in reading the Greeks and other pagans and so forth, what are we going to do with all of these works? Is there something blasphemous or irreligious about paying attention to these texts? At the same time, we recognize now that the Greeks had done some pretty amazing things, and the Romans had done some pretty amazing things.
- 3.8 It's a little bit shocking to us to realize that they did all of this without the benefit of Christianity. How is it possible for these magnificent civilizations? It might be blasphemous, but it might also be the case there's something that we can learn from the Greeks, learn from the Romans, and should we consider going down that road? So this question starts to be poised, opposed, rather, explicitly in the latter part of the 10-hundreds, early part, rather, of the 11-hundreds.
- 3.9 As we might expect, there's going to be strong opinions and strongly diverging opinions about what we, as Christians, should do about these Greek and Roman authors. Now, for the first kind of answer, make an immediate shift north and east a little bit to Paris, which is also coming out of what I think is fairly called the Dark Ages, that four or five centuries of European history. Controversy over that, but I think it's true.
- 3.10 Europe did start to wake up again after the year 1000, and there started to be centers of learning, centers of scholarship, centers a little bit more open to other types of philosophy, and a more serious attempt at rehabilitating and expanding not only intellectual life, but education in particular. And one of the hotspots was Paris.
- 3.11 One of the very interesting controversies about what we Christians should do with respect to those Greeks occurred just a little bit north of Paris, in Saint-Denis, where there was a great new cathedral being built under the supervision of Abbot Suger. And this is also the era in which, partly inspired by the Romans and their masterful abilities in engineering—they could build anything and it would last forever—can we start to build bigger buildings? Of course, the most important buildings for us are going to be our churches and especially our cathedrals.
- 3.12 We start to see more ambitious, more engineeringly sophisticated cathedrals being built, including the one in Saint-Denis. The origin of the Gothic style, where you have great soaring windows and then pointed arches, is at this time. Now, one of the things that's interesting here is that the Gothic engineering style enables you to have much more open spaces instead of having just stone walls and lots and lots of columns. Suddenly with these arches, you can expand the space and you will have open space.

- 3.13** The question then is, what should we do with these open spaces? Immediately there was a controversy there because Suger and some of his allies are saying, well, what we should do is do something that's beautiful and pleasing to the eyes, because people like to look at things. Maybe even when we're bringing all of these people into church for various kinds of services, they can't read. So we're not going to be reading scripture; they're mostly illiterate.
- 3.14** They're going to be listening, of course, to the priest, but the priest is most of the time going on in Latin, and they don't know Latin, so they get bored, and this is a common problem. Why don't we give them something to look at which will be pleasing to the eyes and beautiful, but something also elevating? What we will do is, we will learn and relearn and increase our ability to do stained glass to fill up all of these spaces, but put various illustrations from Bible stories in the now more open spaces. So, great cathedrals to the glory of God with these wonderful stained glass additions as well.
- 3.15** And that's immediately controversial, partly because it's an appeal to the eyes, and the eyes are a part of the body. But also, even if we say it is biblical stories or biblical parables that we are putting in there, it runs up against a very commonly interpreted piece of scripture that says we should not be doing anything in the visual arts. Painting, statuary, representative, figurative work is sacrilegious. And the quotation at stake here is from Exodus, one of the earliest books of the Bible, where the commandment is, "Thou shalt not make unto thee any graven image."
- 3.16** Now, partly it was, people would make images of animals or human beings and then start to worship them. And that was inappropriate because you were worshiping the wrong thing. God is not an animal. God is not even a person or physical body. Also, it was interpreted that it was sacrilegious because anything sacred, anything spiritual, anything godly is not physical. The attempt to take something that is purely spiritual and capture it in physical form is to debase it. It's to bring it down. And you should not be doing that.
- 3.17** We should see and experience God with the soul and only with the soul, not anything physical. So a huge debate at Saint-Denis, one of the pioneering debates about whether we are going to allow stained glass windows. And the answer was, we can do it, yes. That then is to say, we're going to reincorporate some Roman engineering. We're going to allow that pagan appeal to the eye as long as it's in the service of the true faith.
- 3.18** Now, that other side of the debate did not just go away. They continued to resist any decoration, ornamentation, as violating biblical commandment. But nonetheless, this is one answer. We're going to try to accommodate some of the things from the Greeks and integrate them with Christian teaching. That will be expressed in the architecture.
- 3.19** It will also be expressed in the music. Can we reintroduce and expand the range of music that is allowable? Or is music another appeal to the senses? It's an appeal not to the eyes, but to the ear. We also know that people like to move their bodies when they

are listening to music, and so it starts to become more bodily and physical and sensual. So maybe music is also something that we should be suspicious of.

- 3.20** Certainly dancing, we're not going to allow dancing. But what about other kinds of music? Can we compromise a little bit and expand a beautiful music and beautiful stained glass windows and maybe beautiful statuary in these magnificent new cathedrals? We'll take the best from the pagans, Greeks, and Romans, and the best from Christianity, and perform a synthesis of those two.
- 3.21** University of Paris had been founded, expanded, from originally a cathedral school to being more interested in universal education, including education in not only Christian writings, but also in pagan authors as well. And Aristotle became enormously popular among students and some of the younger professors as well. And it partly became a political battle of old guard versus new guard.
- 3.22** The most famous person associated with this is certainly Thomas Aquinas, and he is the next great philosophical genius we need to mention in this story so far. On stature level, there was Socrates, Plato, Aristotle, Augustine, and now Aquinas. The importance of Aquinas is going to be as second generation, third generation, after School of Translators in Toledo had done their work in various other centers, some works have been reimported into Europe, brought from the Far East.
- 3.23** As all of that percolated, there came to be lively interest in all of these works, but especially in Aristotle's broader philosophy. Thomas Aquinas, when he was a young scholar, a brilliant mind, became extraordinarily interested in Aristotle and conceived the ambition of integrating Christianity, which he thought was absolutely true, with Aristotelian philosophy, which he thought was pretty close also to being absolutely true. But how you're going to reconcile two things that often seem opposed to each other, that is his great project. But he's going to take it on.
- 3.24** Now, he is operative in the 1200s, and I like this image of him. He was a monk, but he also then became a professor after he became a student and was appointed professor at University of Paris. And popular there, had a gathering around him. But the image is symbolic. On the one hand, we have him holding a church, and the church is symbolic and representative of Christianity and the truths of Christianity. In his left hand, he is holding a book. The book is open, and if you look closely, it is Greek script. So it is Greek philosophy, and if you look really closely, it's a work of Aristotle. So these are the two equally important things, and I, Aquinas, stand for both of them.
- 3.25** And in his major work, his most famous work, the Summa Theologica, it's a brilliant exposition of all issues in philosophy, all issues in theology, particularly the ones where Christianity is making strongly provocative claims that philosophers need to be able to attend to, where he sympathetically says, "There is this position. Here are the arguments for it." If there's two good arguments for it, he tells us what those are, or three, or four. And here are the objections to those arguments. He presents them

clearly and fairly. "And here's how I would respond to those objections, clearly and fairly."

- 3.26** So it's thorough, it's original, it's brilliant, and it's also teaching us something about method, that if you're going to be a true thinker and you're really going to come to know God, even if you're as a Christian thinker, it's not simply going to be blind faith or seeking mystical insight in a cave or getting the body attention for the body. It's going to be through reasoning, following the model of Aristotle, where we look at the evidence, the arguments, the counterarguments, and we think that Christian doctrine is going to come out as true and validated at the highest philosophical level, if we commit to this process.
- 3.27** Now, here's a quotation from Aquinas exactly on this point, where we see Augustine arguing against the scientists, the natural philosophers, and the other pagan authors, as contrary to Christian religion, and their arguments as fallacious, and particularly their belief that it is through argumentation that we should be able to figure out what the truths are. It is a matter of faith. It has to be a faith commitment, an emotion-driven; you feel the life, you have that conversion experience.
- 3.28** Then you have certainty in your heart that infuses your whole being. It's not an intellectual matter. It's not a philosophical pagan matter at all. Augustine disagrees, and he's taking then Christian religion in a very different philosophical direction. His most famous oft-used parts of the Summa are his proofs of the existence of God.
- 3.29** He said, "Well, of course, we know that Christian religion is true, and we can believe that on faith. We hear it from our mothers and fathers and so on. But equally so, it can be proved. And there are five ways that you can prove it. And I'm going to give you five proofs for the existence of God." And I will guarantee you now, if you engage in debates over arguments for and against the existence of God, at least five out of the seven arguments that you will encounter come directly from Aquinas. They are timeless in that particular direction.
- 3.30** So the point here is, it's not faith, it's not mystical insight, it's not just trusting authorities. Every human being who has a working mind can attend to these proofs and has the rational capacity to come to see that God exists. Very first argument. Notice what he does here. "The first and most manifest way is the argument from motion." Things move. And then he goes on. That's a fact about the world. "It is certain and evident to our senses that in the world, some things are in motion."
- 3.31** So as a cognitive point, this is a philosophical point. We start with the evidence of the senses. We observe the world. We observe the natural world. On the basis of studying the natural world, we are going to reason our way to the existence of God, and everybody can do that. It's not, again, something irrational, emotionalist, tradition, faith, and so on. Very different approach to Christianity here.

- 3.32** What about this traditional insistence that we've seen in one strand of Christianity on obedience, discipline, things being imposed upon you? What we find in Aquinas is a very Aristotelian emphasis on independence and self-governance. You don't come to God or to anything important by being forced to do it, that what it is to be a human being is to be self-governing. Yes, we're all under the governance of God, but we have to find our own way there.
- 3.33** "The highest manifestation of life consists of this, that a being governs its own actions. A thing which is always subject to the direction of another thing is somewhat of a dead thing." So, this is Aquinas in the 1200s, reflecting on what is going on in pretty much every Christian school run by nuns and monks and other people everywhere you go. The kids are cowed, beaten, straight rows. They don't dare do various things. Overstating? Maybe, maybe not. But this is what he is reacting to. We're not raising dogs. We are raising human beings, and they are of a certain self-governing direction.
- 3.34** Also, this quotation indicates the tenor in which Aquinas has taken Christianity in a different direction. "The partial happiness we can hold in this life, a man can secure for himself." What is the point of life? And one of the strands of Christianity that we have seen and we're familiar with, has just said, look, life is not about happiness. Life is about duty, pain, suffering, sacrifice, possibly even becoming a martyr for the cause. This world is a veil of tears, and we should turn our back upon it. Perhaps we will get some sort of happiness in the next life, but not in this life.
- 3.35** Aquinas indicating, no, this life matters. Of course, the next life matters as well. We're going to integrate those. True eternal happiness in the next life, that's what we should be striving for, of course. But God made this world. God made us. He made us a certain way. God is a good God. He wants us to have some measure of happiness as well. He gave us bodies. The bodies then are to be used in a godly way.
- 3.36** But nonetheless, pleasure and happiness is part of the proper functioning of our bodies, just as Aristotle argued many centuries ago. And it's not simply a matter of hope that some being, God, is going to save you or some external beings are going to make you happy. Human beings have agency, self-control, and they can make for themselves their way in the world and achieve some happiness in this world. Again, a different integration of Christianity this time, not with Plato, but with Aristotle.
- 3.37** Now, that's one answer, and you start to see it developing, 10-hundreds, 11-hundreds, 12-hundreds, and so forth, as we now call the high middle age. It's the idea that we should have Christianity accommodating itself with the Greeks, pulling the best out of the pagans and integrating that with the truth that we know Christianity, and that becomes an important ongoing tradition.
- 3.38** Now, the second answer, of course, is going to be the opposite, the old-school answer now that says absolutely not. Augustine had it right. We cannot compromise with most of these pagan authors, the exception being Plato. But Aristotle certainly is wrong, and

he's the worst of them because he's the best of the pagans, aside from Plato. In him, we find, so to speak, using an old Arab proverb, "the camel's nose under the edge of the tent."

- 3.39** The problem there is, you have a camel's nose because it smells something good inside your tent. Starts with the nose, but pretty soon you have a whole camel taking up the tent, and we can't go there. So whack the camel's nose, whack the Aristotle, can't let it in, no compromise possible. At the University of Paris, as Aristotle and some of the Arab commentators on Aristotle, who were quite brilliant in their work, became popular, there was a ban on Aristotle's physical works, and various other new universities followed suit, saying, no, we need to get rid of those.
- 3.40** This tradition also carries on as the accommodation of the pagan authors, the Greeks and the Romans, more beautiful stained glass windows, more beautiful clothing, people wearing more jewelry, not particularly humble and obedient and so forth. There would be waves of reaction against all of that and crackdowns, that we are becoming worldly. We're trying to delight the senses. We're trying to delight the body. We need to get back to a pure, simpler, godly direction in life.
- 3.41** And so, aside from banning certain works in the universities, there would be certain styles of fashion, certain music, statuary, and so forth being banned. The phrase that comes down to us from this period is called the bonfire of the vanities, when religiously enthusiastic people of this more Augustinian stripe would assume a certain amount of social power and they would, as a mob, go through people's homes and anything that was pagan, gather it up. Beautiful clothes, beautiful jewelry, beautiful books, and so on. Take it to the central square and burn it in a huge bonfire.
- 3.42** And Savonarola, this is in the late 1400s, is probably the most famous representative of this tradition. So we have the integrationists with Aquinas, and the dualists and oppositionists, using Savonarola here as an example, and they are two of the streams. Now, this stream carries on. It is an important part of the story of philosophy, the story of history, and certainly the story of education. As we get into the 1500s, there's a huge split, as we know, within Christianity.
- 3.43** The Catholics, many of them becoming more worldly. Augustine was a saint, but then Aquinas, after he died, also became sainted. And the school of philosophy inspired by Thomas Aquinas, sometimes they're called the Scholastics, sometimes they're called the Thomists, integrating Aristotle and Christianity. The Protestant reformers, the major ones among them, were incredibly hostile to the followers of Aquinas.
- 3.44** There can be no compromise with Aristotle, not for specifically Catholic reasons, but also we want to reform Christianity, get back to pure, original Christianity. And that means getting rid of the pagan authors, especially, as Martin Luther puts it here, Aristotle. I just have to read a couple of passages here, give you a sense of the flavor of the rhetoric and how strong the tempers are at the time.

- 3.45 "Any potter," that is to say, anybody who makes pots, "has more knowledge of nature than these," these are the books of Aristotle, "which I can only believe the devil has introduced. His book on ethics is the worst of all books. It flatly opposes divine grace and all Christian virtues." So from the Lutheran perspective, it's either Christianity or Aristotle, and it's a complete dualistic opposition, philosophically, morally, and then certainly in terms of education.
- 3.46 And then this one, Martin Luther was one of the best insulters in all of history. "Aristotle is truly a devil, a horrid calumniator." That's a liar. But calumniator has a better ring to it, doesn't it? Yes. "A wicked sycophant." A sycophant is someone who sucks up to another person. "A prince of darkness, a beast, a most horrid imposter on mankind, one in whom there is scarcely any philosophy, a public and professed liar, a goat, and his followers are locusts, caterpillars, frogs, and lice."
- 3.47 OK. So, no accommodation, certainly no integration. Instead, we will, to the extent that we can, ban these works. They need to be stamped out of existence. That's what you do with lice, you exterminate them. Now, as the Protestants, there are many Catholics also still opposed to the reintegration of the pagan authors. One of the interesting subplots comes over a compromised position that wants to say, in part, the pagans were right, and that what we have done as Christians has too much emphasized higher authority and higher obedience.
- 3.48 We said God has all the power and everything trickles down, and we basically have no power at the bottom. That there is a truth to this self-governance point that comes out of much of pagan philosophy. The human beings do have some power. They have some power over their own soul and their actions, some responsibility for their moral character and their destiny.
- 3.49 And so, the salvation of one's soul is not simply a matter of God's going to choose to visit grace upon you and you have nothing to do about it, or you will be taught by the religious authorities who have the right truth, if you're lucky enough to be exposed to them, that really it's going to be a matter of you, by your own lights, coming to see the truth of God's word and self-educating along at that point.
- 3.50 Now, what this then means is, from this subset of Protestant reformers, is that the older Catholic way of keeping all of the books in Latin, which only the educated priesthood could read and superior educated members of the aristocracy could read, is a form of information control. If we really believe that every individual has his or her own soul, and needs to know the word of God so he or she can make his or her own soul-saving decisions, then we have to put the word of God in the hands of each individual person.
- 3.51 What that means then is, we have to take the word of God and translate it from Latin into the languages that people are more likely to have a chance of reading. Translate it into English, translate it into German, translate it into French, Italian, and so on. There

came to be a movement of translators taking scripture and translating it into all of the local languages. This, too, became enormously theologically and politically charged.

3.52 This is William Tyndale, who is one of the English martyrs for the cause of this particular self-education in religion approach to learning. He had been an Englishman enthused with the truth of Christianity. England, at this point, had recently, under King Henry VIII, broken from the established church in Rome. I don't know if you know this entire soap opera. That's a bit flip, but King Henry and his six wives, he desperately needs an heir, hopefully a male heir, and he just can't get a male heir.

3.53 And he needs a divorce from his first wife, so he can marry this other woman because first wife is not giving him a male heir. Maybe he'll have a chance with number two. But the problem is that the Pope in Rome won't give him a divorce. The Pope in Rome won't give him a divorce because the Pope is actually controlled by the King of Spain. The King of Spain is the uncle of the Queen of England. And so it's a political mess and a religious mess.

3.54 The only way we can solve this problem, King Henry says, is, I will issue my own divorce. The only way I can issue my own divorce is if I'm not only the King of England, but head of the Church. And that means we have to break from the Catholic religion and form the Church of England. And that will then give me the political and the religious authority, and so on.

3.55 But at this point, Henry is a good Catholic, along this point about, no, you don't let the English peasants, who are mostly not very smart and not very educated. They're my subjects; I love them. But theology is difficult. We can't let them take over theology. Their salvation of their soul should reside with the theologically educated, the politically educated, and so forth. So Tyndale was translating the Bible into English, having it printed on the continent, with the idea that it would be smuggled back into England in large numbers and distributed to people, so they could read their own thing.

3.56 Now, this is an important move for education, because if everybody is going to read the Bible, because we have a religious mission, but that means you're going to have to teach everybody to read. Once people start to read, they start to think, and they start to think for themselves, and they start to interpret the Bible, and they get a little bit smarter. Of course, they're going to start to have arguments about what's in the Bible, and your interpretation and my interpretation are going to be a little bit different.

3.57 But just in the act of arguing, which is going to be messy, and we're going to hurt each other's feelings, and we might not like each other anymore, we're again going to get smarter. And so, people then will also want to start reading other books, commentaries on the Bible, and then everything is going to be opened up. So there's a huge amount at stake here over who's going to control information and what form English education is going to take.

- 3.58** And King Henry VIII says, "Tyndale is a bad guy. This is the wrong road to take." And he sends all the king's men and all the king's horses—borrowing the nursery—they hunt for him for years. He's eventually actually hunted down, hiding somewhere, I think, in what is now Belgium. And the king's men kill him, and then they bring his body back to England and put it on public display as a lesson. Don't do what Tyndale does. That's not what we do here.
- 3.59** So, nasty times, and this is a non-accommodation. We're not going to go down that self-education pagan route. Now, that's King Henry VIII under the Church of England. We've got Savonarola and Catholic bonfires of the vanities. This is in Geneva, just a short story. Michael Servetus, one of the more inspired, active, younger guys, interested in just about anything, traveled all over Europe. He wanted to read every book he could get his hands on, interested in Christianity, but also interested in Eastern religions, interested in magic, interested in natural philosophy and science and mathematics and the whole shebang.
- 3.60** And he made the mistake, at a certain point, of going back to Geneva, where John Calvin was first among equals among the Protestant reformers who had taken over that particular city there, and Servetus was arrested on charges of heresy and executed. We have Catholics tormenting Protestants, Protestants tormenting free thinkers, and so on in all possible combinations over the course of this set of centuries here.
- 3.61** One of the most famous executions takes us back to Rome. This is Giordano Bruno, again, somewhat of a free thinker, a pantheist in his theology, but spent a lot of time outside of Italy because it was not particularly safe there for him to be a free thinker in the way that he wanted to be, partly because he was one of the early enthusiasts of the new Copernican model of the solar system, now to be called the solar system—the idea being that the Earth is not at the center of God's great creation and that everything rotates around the Earth and the heavens are way out there somewhere.
- 3.62** But rather, Copernicus suggested, the sun is actually at the center and all of the planets rotate around the sun. And Giordano Bruno thought this made good sense, and he became an enthusiast and an advocate for it. And of course, he was interested in lots of other things and was heretical on various other issues. But when he went back to Rome, In 1593, he was arrested, thrown in a dungeon, and then, seven years later, found guilty of heresies, including the Copernican heresy, and executed.
- 3.63** And a few years later, the Catholic Church, as more and more Copernican, sun-centered advocates came along, they issued an official codex saying, "No, this view is wrong. It doesn't matter what the scientists are saying, what the natural philosophers are saying. We know that model is wrong." So this is an official censorship.
- 3.64** "Proposition is to be forbidden that the sun is immovable at the center of the heaven and that the Earth is not at the center of the heaven and is not immoveable but moves by a double motion." That is to say, the Earth is spinning on its own axis, and that gives

us day and night. But at the same time, the other motion is to go around the sun once a year, and so on. That's false. All that whole shebang, package of beliefs is false.

3.65 Anybody who believes that, heretic, and the full force of theological and political imposition on advocates of that. This, of course, has huge implications for education. Point being, now we have two factions fighting it out over the centuries. We're going to integrate with the Greeks, or we have to totally banish the Greeks and everything that is downstream from that.

3.66 Now, there is a third alternative, and I want to introduce this one by a slightly indirect route. Here we are, back in Italy, a little bit north of Rome now. In the first three years, 1501 to 1504, when young Michelangelo, still in his 20s, imagine that, sculpts this. It's the David, one of the most famous statues in the world. I believe it is 17 feet high and it stands on a pedestal. It is magnificent.

3.67 But what is interesting about this, aside from strictly aesthetic considerations, that we might want to entertain? Well, we have a youth, a young man, and he is nude. What kind of a statement are you making, if you portray the human being not only as beautiful physically, but nude? Instead of saying the human body is disgusting, nasty, shameful, to be hidden, we should cover ourselves up, this is in a different philosophical universe.

3.68 What we have is a young artist who is now being inspired by classical Roman and classical Greek statuary, some of which, every decade at that point, was being rediscovered as Rome and various other parts of Greece and Italy were being rehabilitated, rediscovered in early archaeological enterprises. So that's already to make one statement about the human condition. Human beings are beautiful. And nudity is not only acceptable, but something to be celebrated. And there had typically been a focus on female beauty, but even male beauty is to be added to the list of potentially beautiful things.

3.69 Now, this is, of course, called the David. And when we think of the David, and we know our biblical history, we're supposed to say, oh, this is a story about David who's going into battle against Goliath, this big bad enemy guy from the other side. And David is going to use his sling to throw a rock, and he kills Goliath. All right, so it's a famous biblical story. But notice what we have here then is an integration.

3.70 We have a biblical story, out of the Judeo-Christian tradition, but it's portrayed in a Greek philosophical and aesthetic way. It's an integration. It's also interesting that, most of the time, when the David and Goliath story was told or represented in statuary, it would be after the fact. The artist would have Goliath's body or Goliath's head, and sometimes David would have his foot on it, indicating triumph and so on.

3.71 But in this case, what we have is David looking and gathering his resolve before he's going to go into battle against Goliath. Is he going to win? Is he going to lose? I don't know. But it's scary to go into battle against Goliath because he's Goliath. Who am I?

I'm the youth David, who's still a relatively young man. But I am going to do it, and he's standing, thinking, making a decision, and you can see the resolution in his eyes.

- 3.72** "I am going to, whatever the courage element is, take this step forward." Illustrating not Christian virtues of humility, piety, and obedience, but this is very pagan. Resoluteness, courage, dignity. I am going to take on this battle and seek my victory in the world. But it's a biblical story, integrated with a Greek message.
- 3.73** One other element, I don't know how far down this art historical road we want to go, but it is interesting that David is looking to the future, because if you think about this from an education perspective, much of education in the West up to this point had been to say, look, all of the truths have already been stated and discovered in the past. God wrote the Bible. If you believe that there's something to the pagan Greek authors, that they're amazing, well, Plato wrote his stuff, Aristotle wrote his stuff, Augustine wrote his stuff. It's all in the past.
- 3.74** All of the good stuff is in the past. Our job is to look to the past, learn the lessons of the past, and bring them into our future. Carry on the tradition. But this is also signaling something newer and something modern, that we are looking to the future. Now, yes, of course, we're going to draw on the past, and the past is very important here. But nonetheless, we're not looking to the past, we're looking to the future, and the future is going to be what we make, as a result of our decisions.
- 3.75** Now, we might say this is an integration, and this is a little mind-reading about Michelangelo here. We have a biblical story, and we have this Greek godlike male body. Which one matters more to Michelangelo? Is he really all about that lovely old biblical story, David versus Goliath? Or is he really about, man, human beings are beautiful? Is his focus religion or is his focus on the human being? And it's not too much of a stretch to say he's really actually interested in the human. He is a humanist, even though, of course, accommodating with Christianity.
- 3.76** Now, this then is to say, this is a third approach, just because it's not simply a matter, Christianity is true, and say, Plato is true, and we integrate those two equally, or Christianity is true and Aristotle is true, and we integrate those equally. Or we say, no, Christianity is true, and that means all of the pagan Greeks are false, and we have to ban them totally. We start to see now a suggestion, yes, Christianity is true, but we're much more interested in humanism and a humanistic understanding and a naturalistic focus on the world, and that's taking us in a different direction.
- 3.77** Here's a sketch from one of Leonardo's books, also very important. Here, again, we have a human figure. It's not a monk, it's not a nun. We're interested in the human body and the proportions of the human body. It's interesting that he's drawn squares and circles. We're learning all of the mathematics and mathematizing the world, and we're studying human physiognomy and human anatomy and all of its proportions.

We're interested in the human from a purely naturalistic perspective, and this is informing my approach to art.

- 3.78** Quotation from same generation, a young man who died relatively young, about this point. This is now then to say, yes, we are interested in Christianity. We are interested in the truth of religion, but we're also interested in all the other religions. We're also interested in all the Greeks, all of the Romans. Anything we can get our hands on, we want to learn it. And that takes us, again, in a different direction because it's not a matter of Christianity and the pagans being equally integrated.
- 3.79** Once you say there are 20 important philosophies, we should study them all and mix and match from the best of them, well, maybe one-twentieth of the resulting mix is going to be Christian. We're off in another direction. Now, I want to save some time for us to plunge into a text from this period. Mirandola died young. He wrote some interesting stuff, but he didn't develop into a mature intellectual and develop his full potential, unfortunately. But we do have one manuscript I want to spend a little bit of time on.
- 3.80** But what this means is that education, by the time we now get into the Renaissance, the Middle Ages are over, the Renaissance, and on into early modernity, it does not look like it had looked for almost the millennium before. We start to see the development of universities. We start to see the Bible being translated into all of the vernacular languages. We start to have schools of natural science, naturalistic and humanistic art developing.
- 3.81** And the idea then is, if I am going to educate my children to lead the best possible kind of life, well, of course, it's probably going to include a significant amount of Christianity, but now there's a whole lot more. We have to have a much more expansive, ambitious program for education that's going to require a new mindset and some new institutions.
- 3.82** Just a history of education point here. It's common to say that out of this era came to be a formalized understanding of what basic education would involve, the trivium and the quadrivium, as they are so called. They're taken to be the seven classic liberal arts. If we now want to raise human beings who are going to be free, and it's interesting how that word "free" starts to pop up more and more as the Renaissance developed.
- 3.83** We're not so interested in developing obedient, disciplined children who will be dutiful subjects. We want free citizens, people who are liberated, in some sense. And education is going to be foundational to that, and the basic thing is people need to learn grammar. That's facility with their language, logic, and rhetoric. And then on the content side, the fundamentals are arithmetic, that's basic mathematics, geometry also, astronomy, and music.
- 3.84** We study those seven. Those are the foundation. Every child who's getting the best possible education is going to be exposed to those. But those really are seen as the

basics. Once those basics are underway, then there's a whole world for each individual child to explore, partly as a matter of that child's own interests and talents and so forth.

- 3.85** And that real education, once this basic quadrivium and trivium education has been instilled in the children, should be that you have created young people now who are able to take significant charge of their own education and pursue the whole rest of the world. The formulation here is to say, it's a real education. It's going to be a self-directed quest.
- 3.86** You're putting the individual human being back in the driver's seat, so to speak. And humans are interested in so many things, and there are so many potentials, that the world is open-ended in that sense. And this humanism, or education of "humitas" also becomes the Latin phrase, the educational model.
- 3.87** I want to spend a little bit of time with Alberti, who was a very colorful character, and wrote what came to be seen in the Renaissance as a manifesto of the new era. The title here is a Self-Portrait of a Universal Man. It was written over the course of the 1460s, so the Renaissance in Italy is well underway. Leonardo, just born. Michelangelo, not quite yet, but this is the world that they are being born into.
- 3.88** What's interesting, you take a self-portrait. I'm writing about me. That then is to say, I'm important. I matter. I matter enough that I'm going to write something about me, and I want you to read it. Now, the autobiography form is not widely practiced in classical history. There are some exceptions. Some argue that actually Augustine, whom we saw in the previous lecture, with his Confessions, initiated the autobiographical form, and it really was quite striking.
- 3.89** But there's a big difference here also that Augustine's Confessions is about how terrible and how awful he is, and how it took God's grace finally to correct him and set him on the true path to salvation. Whereas with this autobiography, what we have is Alberti, to put it very plainly, saying how wonderful he is, how accomplished he is, and how he wants to share his wonderfulness with everybody, with the expectation that people will say, "That's great. You've really done something with your life. I'm going to take some inspiration for that for my life and certainly model that for my children going off into the future."
- 3.90** And the second part of this, the universal man, instead of being a person who's focused just on one thing—you know one book, you know the word of God inside out, you don't know anything else—that is now passé. That if you're going to be a well-educated human being, you will know a lot of things about a lot of things. You will be accomplished in so many areas, and that proper education then for the youth will be a much more expansive universal project. We do have this phrase, "the Renaissance man," that has come down to us in English, and Alberti is a representative of that.
- 3.91** Let's plunge into the text, get a sense of his style, but also his themes. "In everything suitable to one born free and educated liberally," now pausing right there. You are born

free. Now, if you're an educated person, you will have read Plato, which means you will have read the Republic, which means you will be familiar with the allegory of the cave. And there the allegory says we are not born free. We are born in chains, in the bottom of a stinking cave full of illusions.

3.92 Exact opposite philosophical theme here. We are born free, not in chains, not enslaved. "And educated liberally." Education is not forced on us. I'm not dragged up the cage or cave to stand in the light. I'm not forced to sit in a seat and smacked with a rod on my fingers whenever I don't pay attention. I'm educated as a free person, in a liberal way, with lots of choices.

3.93 "He was so trained from boyhood that among the leading young men of his age, he was considered by no means the last." That's Alberti being about as modest as he possibly can be, because what he's trying to say is, "By the time I became a young man, I was something pretty special," even among a cohort of other young men who'd received good education and were pretty promising as young men. "And I was by no means the last," which means he was something else.

3.94 "For, assiduous in the science and skill," science is important, skills, physical skills, not just book-learning, being able to quote scripture, "of dealing with arms and horses and musical instruments, as well as the pursuit of letters and the fine arts." So physical training, as well as book-learning from letters and so forth. "He was devoted to the knowledge of the most strange and difficult things. He strove hard to attain a name in modeling and painting." So I learned how to do statuary. I learned how to do painting as well.

3.95 "His genius was so versatile that you might also judge, might almost judge all the fine arts to be his." So he was an ambitious young man. His parents had ambitions for him. We are going to expose young Leon to everything. He's going to have lessons in everything, and we're going to get him to a level of proficiency in all of these different areas. "Neither ease nor sloth held him back, nor was he ever seized by satiety in carrying out what was to be done."

3.96 Now, again, if we have read Plato, if we read much of formal schooling, we realize that kids, when they're supposed to be learning things, we have to drag them into the school room, tell them to sit, make them learn, punish them when they're not paying attention. Alberti's point is, this came naturally to him. He was full of energy. He wanted to learn. He had lots of energy. Nothing held him back from learning what he wanted to learn.

3.97 "For him, letters, in which he delighted so greatly," he loved reading, loved books, "sometimes seemed like flowering and richly fragrant buds." So now we're getting poetic. I have a book. I want to smell the aroma of this book. It's a delight to the senses, a delight to the eyes—perhaps the feeling some of us get when we go into a bookstore. Oh, so much wonderful potential just waiting for me here.

- 3.98** "So that hunger or sleep could scarcely distract him from his books." The exact opposite of parents and teachers telling kids, "Wake up, do your homework, do your studies." Here, he doesn't want to sleep. He is so enamored with learning; and then when letters began to be displeasing to him, he turned to music and painting and exercise. So I get tired of this, then I go and do something else. Tired of that, I go do something else. There's always more stuff, interesting stuff, fun stuff to learn.
- 3.99** "He played ball, hurled the javelin, ran, leaped, wrestled, and above all, delighted in the steep ascent of mountains," mountain climbing. "He applied himself to all these things for the sake of health, rather than sport or pleasure." Now, of course, these are sports. They're also pleasurable to do, but this is also serious business. It's not only that I have to look after my mind, look after my soul, I also have to look after the health of my body. This is a self-conscious value that I am committing myself to.
- 3.100** Self-directed education theme here. He learned music without teachers. You just give the kid a stringed instrument or some early forerunner of a piano, harpsichord or whatever. The kids start playing around with it. Maybe they hear some other more accomplished musicians doing it and they start to imitate, but then they try to learn to do that themselves. "And his compositions," not only learning how to play instruments, "his compositions were approved of by learned musicians. He sang throughout his whole life, in private or alone, but especially in the country with his brother or relatives. He delighted in the organ."
- 3.101** Getting more serious when he began to mature in years, he devoted himself for a while in entirety to letters, intellectual learning, the great books, and with some political ambition, canon and civil law. If you're going to go into politics, going to go into business, the law is good to learn. It's complicated, but it's going to take some dedicated study.
- 3.102** He mentions here that he started to have some health troubles. He also mentions one of the most difficult lessons he had to master in his young life, and that is dealing with other people, particularly people who are supposed to be your friends and relatives, but in fact are petty, backstabbers, and so forth.
- 3.103** "Since his relatives were neither kind nor humane to him in his illness, by way of consoling himself between his convalescence and cure, he wrote the play Philodoxeos, putting aside his legal studies, this when he was only 20 years old. As soon as his health permitted, he resumed his studies, intending to complete the law. But again, he was seized by a grave illness as he dragged his life along through labor and extreme poverty."
- 3.104** Finally, his doctors told him to back off, take it easy. "You're overextending yourself, Alberti." But there's a suggestion here that it's natural; like little kids, they're so enthusiastic about everything. You just want to learn everything. You want to learn to be able to do something. And if you are educated and raised properly, that will stay

with you throughout your life. Perhaps the error is going to be in the excess of that. Maybe at some point you will need a wise physician to say, "Dial it down a little bit, otherwise you're going to hurt yourself."

- 3.105** This is the exact opposite of, kids just want to sit in the cave and eat snacks and not learn to do anything and just become slugs, so to speak. "But in truth, because he could not live without letters, at the age of 24, he turned to physics and the mathematical arts. In the year before he was 30, moreover, he wrote in his native tongue for the benefit of his relatives, in order to be of use to those who are ignorant of Latin, some books in Tuscan."
- 3.106** An early sign of a new modern trend where educated people don't simply read Hebrew, Greek, and Latin; that we want education to be more expansive, have people more exposed to all of the things that learning has to offer. And that means, since they already know how to speak their local language, writing for people in their local language. So Alberti is a part of this trend.
- 3.107** This might sound like a lot of stuff that's very serious, but he also says he likes good jokes, he likes humor. So, "Besides these, he composed, also before he was 30, many dinner speeches, especially those facetious ones." Gives some examples here. And the idea here is that you go to dinner parties or whatever kind of party. Yes, you will want to talk about politics and religious and serious matters as well. But everybody likes a joke, everybody likes a funny story. Again, we are well-rounded or we're having a more comprehensive understanding of what a full life is about.
- 3.108** "He was always extremely circumspect in his habits," how he dresses, how he speaks, how frequently he bathes, and so on. "He wished by everything in his life, every gesture and every word, both to be seen and to seem worthy of the good will of good men." One of the interesting metaphors that starts to be developed here is living one's life as a poem. You start to see many of the poets who start to, rather than say, I live my life, and maybe I make a journal about what my life is all about, and maybe I will write an autobiography.
- 3.109** But if you write your life as a poem, then there's something more elevated about that, more artistic about that; but then in turn, to use that as a metaphor for your life. See your life not simply as prose, so to speak, but your life as a kind of poetry. You want to be poetry in action, to embody poetry as the best that the literary arts can offer, to do that in my life. I will become a poem myself.
- 3.110** And now the problem here, part of education is, of course, arts, music, letters, law, all of the physical activities as well. But there's also people, lots of people, great people. You can be friends with them, people who genuinely love you and so forth, decent citizens and so on. But one of the big lessons is, how do you deal with the jerks in life, and worse. "Although he was affable, gentle, and harmful to no one, nevertheless he

felt the animosity of many evil men and hidden enmities, both annoying and very burdensome."

- 3.111** Now, you're going to run into people who don't like you, who hate you. Some of them will be upfront about it. Some of them will be sneaky about it and willing to stab you in the back. How do you learn how to deal with this? Now, of course, part of it is strategies for dealing with them, what you say and so forth, but also part of it is attitudinal. What's your attitude to this sort of people?
- 3.112** "Even by these envious ones, he was received with honor face to face." So he knows and he learns this hard lesson; some people will be nice to you to your face, but really, they are envious and they're just waiting for a chance to stick the knife in. "But in truth, when he was absent, those who had pretended to love him most slandered him with every calumny, wherever the ears of the fickle and their like lay open, for they took it ill to be exceeded in ability and fame by him," they're envious, "far inferior to them in fortune, he had striven with such zeal and industry."
- 3.113** So what do you do in these situations? What's the lesson that we need to teach to children about this? This is one of the things that kids come home from school on a regular basis with. "I thought she liked me. And then I hear, this is what she was saying about me behind my back. I thought he was my friend. And then he backstabbed me and told these lies to the coach, and now I can't play the game on Saturday." Happens. What's the lesson?
- 3.114** Now, one thing is revenge. People are nasty, so you need to become nasty yourself in order to fight that battle in this world. Alberti is saying no, not that. "He bore injuries of this kind from his kinsmen with equanimity, more in silence than by indignantly resorting to vengeance or permitting the shame and ignominy of his relatives to be made public." Don't lower yourself to their level. Don't respond in kind. Take the high road, be the better person, forbear it. Yes, it hurts, but deal with it. Don't let them know. Don't do dirty laundry of the family in public. That's the right strategy.
- 3.115** Moving on. "When he had heard that a learned man of any kind had arrived, he would thus work his way into a position of familiarity with him, and thus from any source whatsoever he began to learn." So, I want to learn everything. And so part of my ambition, particularly as a young man, is anytime I hear in the area there's somebody who has a reputation for learning is coming through, I will find a way to meet that guy and see what I can learn from him. Self-directed education initiative, ambition for it.
- 3.116** Not only famous people doing fancy-schmancy, university degree-types of people and so forth, people who have anything to teach me. "From craftsmen, architects, shipbuilders, and even from cobblers," people who make shoes, "he sought information to see if by chance they preserved anything rare or unusual."
- 3.117** Self-education also in turn involves emotional education. "He was by nature prone to wrath and bitter in spirit." So he noticed in himself that he could get angry and start to

experience bitterness. So he introspects and realizes this about this. Realizes these are not good emotional states to encourage. They lead to dangerous routes. But he could repress his rising indignation immediately by taking thought.

- 3.118** I can assess. I'm getting angry here. How do I want to handle this anger? Should I act on it? Should I bottle it up? Should I save it? And I will make the best rational decision about what's going on emotionally and what my actions should be, and that is what I'm going to do. Another important part of education.
- 3.119** Life, physically, is hard. "He could endure pain and cold and heat," but these take some practice and training, to become physically tough. "When not yet 15, he received a serious wound in the foot, and the physician, according to his custom and skill, drew together the broken parts of the foot and sewed them through the skin with a needle. He scarcely uttered a sound of pain."
- 3.120** So, all right, this is going to hurt, but I'm not going to let the pain dominate. I'm trying to become a man. Man has to deal with a lot of pain, and I'm going to deal with this pain as a man would deal with it, a self-conscious part of the education. "His head was by nature unable to endure either cold or wind, but by persistence, he learned to bear them, gradually getting used to riding bare-headed in the summer, then in the winter, even in raging wind." So it gets too hot, it gets too cold, but I'm going to test myself, and get used to dealing with heat, getting used to dealing with cold as well.
- 3.121** And so the point of all of this, to draw it to a close, though, is to say, if you consider the range of things that Alberti says he set his mind to, and obviously his parents set before him, it is just about everything—all of the physical arts, all of the mental arts, all of the aesthetic arts, all of the scientific arts. The idea being that the world is open, the world is interesting, and it's interesting, and it's important that we be able to master as much of it as we possibly can, to become the best as individuals that we possibly can.
- 3.122** This ideal of the Renaissance, as expressed here by Alberti, becomes increasingly common. I want to end with this last line here, showing by example. The example is him. I can do this, but in one sense, I'm special, but in another sense, I'm not special, because we're all human beings, and basically anybody can do this. "Men can do anything with themselves if they will."
- 3.123** I wanted to ask, what do you think it is about human nature that causes us to feel obligated and responsible to force our perceived enlightenment on other people? And how can we prove that imposing that enlightenment contributes to the enhancement of that individual's life or society as a whole?
- 3.124** Yeah. So you're raising the question about forcing on. Is that the language that you use, forcing it on another person, the enlightenment?
- 3.125** Right, because of the ideas that we all think is correct, right?

- 3.126** Yeah. So I think there's an immediate fork in the road. Everybody is going to say, I have some idea about what the best possible life is for a human being. Partly, that's going to be general, because we're all human beings, so some of the same things are going to apply to all of us. At the same time, we're all going to make our own individual choices, so we're unique, so you want to leave some wiggle room.
- 3.127** I think your question then comes, once you've reached that point, you have some conception of what you think the best human life to strive for is, and you care about other people; suppose we add that. You want that for that person as well. How do I get them to do that? Here is where I think the fork in the road is going to. Do I think that what my conception is is something that fits with human nature, that it's natural to human beings? So I don't actually have to force it on them.
- 3.128** So, if we start with infancy, infants are going to naturally start to develop various things. They're going to start to naturally move their bodies and want to engage with various things and to stand up, to learn to walk and talk and so on. I will think all of that is great, but I don't have to force them to do that. Instead, my role will be to encourage them and provide the resources, so that they can naturally go in the direction that they go.
- 3.129** Now, probably also where the fourth part might come in is, you know that we all make mistakes intellectually in terms of our emotional reactions to things. We'll come to believe wrong things and try to do bad behaviors, maybe ones that involve hitting my sister or whatever, that I, as the parent, will be saying, my understanding of what it is to be the best human being involves not doing those things, but this kid is naturally doing those things, so I might have to intervene to stop. And that's going to be a kind of force. But it's a force where you are defending the other human being from the force of this person, the little kid who's misbehaving, so to speak.
- 3.130** But the default position there is to say that education and self-development is already a natural process, and my job is to assist it and provide resources for it. This is sometimes, this is not necessarily the best metaphor, called in education, the horticulture model. That if you want to raise a plant, you don't have to teach a sunflower how to become a sunflower. It's going to naturally develop its roots and sprout and do all of the things. And your job is to be a good gardener, to look after the soil and water, and protect it from various sorts of elements and so forth.
- 3.131** And so, force is not at all part of the educational process. If you think that metaphor is too limited, human beings are more complicated than plants, then sometimes people will use animal metaphors. Say you don't really need to teach a wolf cub how to become a wolf, or the amount of parenting that wolf parents have to do is pretty minimal. There's like, the wolf nature is going to assert itself, and their job is to provide and protect while the wolf naturally develops. So how far you want to take those metaphors is one, but that takes much of the force out of the equation.

- 3.132** The other fork in the road, though, I think where the force comes in, is to say, if you have a conception of what you think the best life for humans is, but you don't think it's natural to human beings, or you think that human nature is indifferent to it or in rebellion against it, then the only option you are going to have is to use force on that person.
- 3.133** So what you're going to then say is, nature is wrong, and you'll have to suppress that human nature, so that you can force the good thing upon the human being. Or if you think the capacity for good in most human beings is stunted, or it's just going to lie dormant unless it's acted upon by an external agent, and that's your job, then you have to use forceful methods to get that agent enacted, whatever that happens to be.
- 3.134** Then it might be the case to say, well, I don't think that reading comes naturally to human beings, but I think reading is good. Here's my kid, has no inclination to learn, so I'm going to force him to read. He wants to go play, but I say no. I use force to sit you down. He's not paying attention. I smack him. That activates his mind to pay attention for five minutes maybe. Then hopefully, he will get to the point, from this perspective, that maybe this reading thing is not so bad after all, and he will start to do it himself with less punishment.
- 3.135** But the justification, I think, boils down to, do you think the basic development is natural, in which case force is not necessary and not appropriate, or education is not natural, and therefore, force and lots of it is going to be necessary?
- 3.136** You mentioned the education is self-directed, and you also talked about the concept of being a free citizen. But earlier you said it requires censorship for child's education. Does that mean the parents should decide what kind of books should a child read? But doesn't that interfere with a self-directed education?
- 3.137** Yes, that's a very good question, and we've only glancingly touched upon that. So far, what we've had is some extremes, one extreme being represented by Plato and some prominent versions of the Christian tradition, where they will argue that much of literature, much of arts, is inappropriate for human beings in general, and certainly for children. So they are advocates of a strong censorship regime.
- 3.138** Now, if you use censorship narrowly in the sense of only governments or political institutions ban works, that's one thing. If you use censorship more broadly, just to mean you have some sort of authority figure who might be a teacher or a parent who curates the list and make some things accessible to kids and other things not, more people are more comfortable with that understanding of censorship, as an editor and curating.
- 3.139** Now, there the argument, and why people are more comfortable with it, just is to say that, while we want kids to be self-directed, kids, they don't know yet. If you just say, here's a kid, here's a library with 10,000 books in it, what are the chances the kids are going to get to the ones that you think are good immediately? The curated list will be

to say, well, here are the ones that are age development. Here are the ones that I think are more important. If you have decent parents or decent teachers, they will initially present a curated list.

- 3.140 But typically, the ones who are more in favor of the freedom side will also say that we're going to leave a lot of wiggle room in that list. If the kid has independently said, "I found this other book, can I read that, and it's not on your list?", that more likely your default position is going to be to say, "Yes. We'll try to make room for that in the curriculum."
- 3.141 If I say, I think you should be learning these seven subjects, say, and we're going to spend equal time on all of them, but the kid says, "I'm not very interested in this, but I'm very interested in this," then you would be open to the child re-weighting the curriculum, to say, "We're going to spend twice as much time on this as we are on this."
- 3.142 That's going to be self-directed, still within curated bounds. Then still within this tradition, the position is going to be, as children mature in their awareness of what the range of options are, and what they are interested in individually, there's going to be more and more of that self-directed and less curation of the list until, by the time they're adults, it's a completely open-ended thing.
- 3.143 So again, yes, tying into this question, the other side, though, is going to say, children don't know what's good for them, so we have to have a strictly curated list. In fact, we probably believe that children will naturally gravitate toward the books and interests that are bad for them. So we're going to have to be extra vigilant to monitor the children, to make sure that they're not going in that particular direction. And we're also going to be saying, as long as they are children, they really don't know what's best for them.
- 3.144 We are the experts, and so we're less open to the idea of any sort of wiggle room or customizing the curriculum. It's more likely to be a universal one-size-fits-all, according to the standards of the authority. Again, this is early stages. We're now just into the early modern world. But this issue of the degree of censorship and curation certainly becomes more and more pointed as we go along.

Lecture 4: Through the Scientific Revolution

At a glance — Education through the Scientific Revolution.

- 4.1 All right, let us start today in 1642, which was both a sad day for the world of science and philosophy, and also a great day for the world of science and philosophy. On that same year, rather, Galileo died, the great scientist of the early modern world, but also, importantly, a philosopher of science. In the same year, Isaac Newton was born, a great scientist, as we all know, but also philosophically attuned to the fundamentals of science and other important intellectual issues as well.
- 4.2 In the 1600s, it's clear that we are into a new world. We start to talk about the new world, not the new world merely geographically, but a new intellectual, cultural world. Everything is in change. In the old world, the old ways of doing things are increasingly challenged and being set aside institutionally and culturally. Now, we are familiar with this, now that we are well into the 21st century. But put yourself in the 1600s and think about what has happened just in the last century or so.
- 4.3 Often in intellectual history, and history more broadly, we will use the year 1500 as a broad demarcation between the modern world and the pre-modern world. Obviously, that's a little bit artificial, a little bit too precise, but think of the year 1500. It was only eight years earlier that Columbus crossed the ocean and back. Then suddenly the Europeans realize, and the Americans realize, and everybody realizes that the world is a lot bigger and a lot more diverse and a lot more complicated than anyone had realized before.
- 4.4 That means we're going to start thinking about ourselves as inhabitants of a much larger globe. We now use the phrase globalization, and it's starting to become a cliché, but we're actually now five centuries into modern understanding of globalization and all of the cultural impacts that have occurred.
- 4.5 Also, 1517, Martin Luther, with the great revolutions in religion. Obviously Western Christendom had been in turmoil off and on already for over a thousand years of its history. There had been a split between the Eastern Church and the Western Church, all sorts of sects within. But then this cataclysmic revolution that we call the Protestant Reformation and the reactions and counterreactions to all of that. And suddenly we are into doing religion in the modern world in a different way.
- 4.6 And it's a fait accompli that there are going to be multiple versions of Christianity in Europe and more broadly speaking, but there are also Jews and Muslims. And now suddenly we realize there are dozens and dozens of other significantly important religions around the world. So how do we think about religion in the modern world, differently from the way most people would have thought about religion, raised in mostly monocultures, everybody by and large believes about the same thing with respect to religion?

- 4.7 But now we are dealing with people all over the world and even in our own culture who are religiously diverse, and that requires a different way of thinking. We've already seen significantly the decline of feudalism, starting in the high Middle Ages, but this is accelerating as we get into the modern world. The idea that the vast majority of people are going to be serfs or peasants tied to the land, and that everybody is organized into rigid classes. If you were born into this class, you're going to stay in your class. You're going to do the work your dad did or your mom did, if you're born into the aristocracy and so forth.
- 4.8 All of those political classes are being broken down. All of the economic classes are being broken down. We're much more now comfortable with the idea that people will find their own way in the world, and that political power should not be concentrated only at the top, but more distributed and even some discussion, particularly now that we are learning and relearning the Greek and Roman models, about perhaps introducing more democratic, more republican ideas, into the modern world.
- 4.9 Then if we think of the industrial revolution, which is still going to occur, typically a century later, the great explosion with the invention of the steam engine, which was a big game changer, but already a century or two of an increased inventiveness, people working with their hands and making new machines. And it's striking that, if you look at agricultural machinery, which is, of course, extremely important, for thousands and thousands of years, agricultural technology looked pretty much the same.
- 4.10 But already by the time we get into the 1400s, 1500s, new kinds of agricultural machinery, more sophisticated agricultural machinery—and that's just agricultural machinery. Inventions of all sorts of clever things and innovation and people looking forward to new kinds of things that they can add to their household, add to their work lives, and so on, culminating a century later in an industrial revolution.
- 4.11 Then the birth of the modern sciences, the articulation of astronomy as a science. Then, of course, huge religious and philosophical battles over whether the sun is at the center or the Earth is at the center. So they're going to be important there. But anatomical studies starting to be done in the 1500s, to see the human body as interesting, as important, but also as a cause-and-effect system. If we can do it in anatomical studies, we can figure out why things go wrong. This is going to inform medical science.
- 4.12 Medical science is starting to transform itself significantly, with the expectation that we should be able to fix lots of problems that are internal to the body with this new anatomical science. Biology, chemistry, physics, the whole shebang of the sciences that we now recognize as modern sciences—the groundwork for those is being laid by the time we get to the 1500s and increasing sophistication.
- 4.13 So we're into a different world. We're doing religion differently. We're doing all of this science. People are much more free agents in their economic lives. Even the idea that maybe there's something wrong with this hierarchical understanding of the world, that

people are significantly born better than other people, that maybe more people should be born equal, and we shouldn't be looking down at serfs and peasants and keeping them in their place.

- 4.14 And even early voices starting to suggest with increasing frequency that there's something wrong with slavery. And this is a shocking thing because throughout all of human history, all over the world, the idea that slavery isn't perfectly natural, perhaps even divinely ordained, was for the first time being challenged increasingly. And that also is going to affect developments in the modern world.
- 4.15 It is striking to think, this is a shocking thing from our perspective, that if you go to most parts of the world, 500 years ago, somewhere between, depending on where you are in the world, but 70, 80, or even 90% of the population were slaves or something very close to slavery, serfdom, where you're tied to the land, owned by various sorts of people. That had been taken for granted all over the world for time immemorial. So, changes in moral thinking, changes in social thinking as well.
- 4.16 So the world is bigger, the world is more complicated. We're adding more responsibility to the individual to say, look, you're not just born into a class and then whom you can be friends with and whom you can marry and what your job is going to be, that's already been decided. You're just in a place in society, learn to fulfill your role in society. Rather, it's going to be up to you to make your decisions about who your friends are going to be, whom you are going to marry, what your career is going to be.
- 4.17 And to increasingly participate in the political process, to the extent that we think it's not just going to be the king and the aristocrats who make all of the decisions. As we start to reform the judicial system early in the modern world, we're going to ask people to be participating in juries and make important legal decisions and so forth. It then means we're going to need a different kind of education for young people to grow up in this new, more complicated, more responsible modern world, so that they are able to function in that world.
- 4.18 And what's that education going to look like? That's the question. I want to give a few quotations from major thinkers of the time. This is now a cross-European phenomenon. We find significantly major voices all over Europe starting to sound the same sort of themes about the proper kind of education. So this is Michel de Montaigne, a French thinker, 1575, from an essay, *On the Education of Children*.
- 4.19 If you think pre-modern education, whatever your stereotype of it is, it's going to be a certain strictness, a certain amount of willingness to use physical punishment and physical controls on kids, and the idea that there's a truth. The parents have it, the teachers have it, they lay it on the kids. The kids are meant to not question it, but accept it and absorb it into their own lives. So this is a shockingly radical thing for someone to be saying in the late 1500s.

- 4.20 "If the student embraces the opinions of Xenophon or Plato by his own reasoning, they will no longer be theirs but his." Now, think of that. Every student has the capacity for reasoning, and education should be about cultivating the student's capacity for reasoning. We're not talking about accepting on faith by whatever the authorities say, even if those authorities, the books you are reading, are Xenophon and Plato, great giants in the classics.
- 4.21 You don't read Plato and say, "Oh, Plato was a great genius philosopher. Who am I to question and judge Plato? If Plato said it, it must be true," or whoever is in your canon of acceptable authoritative books. We want the students to think through Plato or Xenophon or anything by the light of their own reasoning. It's not just that Plato said it, it's that Plato said it and Plato made these arguments. What do you think about those arguments? Do you agree with those arguments or not?
- 4.22 Of course, you know Plato is challenged by Aristotle and is challenged by the Stoics and Marcus Aurelius, so you're going to have to be willing to enter into different modes of thought and compare them with each other by the light of your own reason. The expectation here is that all human beings, all students, have this capacity. The capacity for reasoning is not limited to, say, a small elite number of people who should do all of the thinking and decision-making for the rest of society.
- 4.23 "Who follows another, follows nothing." Don't be a follower. You might end up going the same path as Plato or Aristotle or St. Augustine, but you are choosing to go down that path because you think it is the right path. You're traveling with Plato, say. You're not following in Plato's footsteps. "He finds nothing, and indeed is seeking nothing. We are not under a king. Each man should look after himself." Now we're getting political in the metaphor here.
- 4.24 On intellectual matters, as well as in political matters, you are not under a king. You are not a subject. The king says you do it. Plato says you believe it. You decide for yourself, you act for yourself, you follow your own path. You are a free agent, and a self-responsible agent. Strong claim. "Truth and reason are common to all men, and no more belong to the man who first uttered them than to him that repeated them after him." These are strong words, early modern world, and indicative of a new thrust and a new kind of education.
- 4.25 Francis Bacon, jumping across to England. Particularly those same themes about the importance of independence of thinking and the widespreadness of reason is really important to all. But a suggestion now, this is a very modern point, that the purpose of education is to make life better in this world. We pursue knowledge, scientific knowledge, economic knowledge, religious knowledge, because, however abstract and theoretical that knowledge gets, we want to apply that to our lives now, in the here and now, in this. We want to make the world a better place.

- 4.26 Now, that sounds like a cliché to us, five centuries later. But again, think of the dominant models of philosophy and therefore of education prior to the modern world. And two very common answers would be to say, well, why are we pursuing knowledge? Of course, knowledge is important, and we need to pursue truth and acquire truth and so on. But if you are a strongly religious person, where is this truth and knowledge that you are seeking? Well, it's not really of this world. This world is a shadowy, lower world, and it's not where truth and real knowledge lies.
- 4.27 Rather, truth and real knowledge are in another dimension. They are perhaps in the mind of God. And so, by seeking knowledge, what you're trying to do is get away from this world as much as possible, so you can go up and read and contemplate the mind of God. And then what are you supposed to do? Suppose you are successful at that. Well, that's the end of the story. You have achieved an intellectual communion with God. You know God, God knows you. That's it. You are in bliss, and that's the ultimate goal.
- 4.28 We're not interested, to use the Platonic metaphor necessarily, in reentering the cave, so to speak, or we're just trying to get as many people out of that Plato's cave as we possibly can. Or the idea might be, if you're not quite so religious, to say, unless we are seeking truth, we are seeking knowledge, and this is the phrase: We are seeking knowledge for its own sake. Knowledge is an end in itself.
- 4.29 We're not going to sully it or make it impure by asking, how can we make money off this? Or how can we use this to improve our bodies or to make gadgets and things that are going to make the world, the physical world, a better place? Knowledge is higher. It is pure. It is for the sake of contemplation. And early in the modern world, that is being challenged vigorously. Yes, of course, there might be higher knowledge, and that can be very important, but we are in the physical world. The physical world is important, and our bodies are important, and our success in this life as well.
- 4.30 So the point primarily of knowledge, however abstract and theoretical it is, is to see how we can use that knowledge to make our lives, the lives of people we care about, and the physical world, better. Knowledge is not an end in itself. It's a tool. It's something that makes us more powerful, like tools do. It augments our capacity to make our lives in the world a better place. And this is a very powerful statement being articulated by Francis Bacon, early 1600s in the modern world.
- 4.31 Now, all of this, of course, is very challenging. We have a new generation of articulate, brilliant thinkers who wanted to remake all the aspects of human life in the light of the modern world, challenging to long-established centuries and centuries and millennia and millennia ways of doing things. And probably the biggest one that has to be dealt with is this issue of religion in the modern world. What is the place of religion in the modern world?
- 4.32 If we wanted to say our bodies matter, well, we know that that challenge is a long-standing traditional way of looking at things. If we want to say that people should be

thinking for themselves and willing to challenge authorities, well, we know that that is challenging to some versions of religion that say, who are you to question? You should be faithful and obedient to what you are taught by higher authority and the various institutions that are built up around that.

- 4.33 Now we have this newcomer on the block, science, and not just prototypical naturalistic understandings. Robust self-conscious science is coming along and saying we have new tools, new methods, mathematical, experimental, and so forth, to figure out the way the world really works. And as we have, even in these early decades, have been doing this, we're finding that the answers that we are starting to come up with challenge many of the traditional answers that have come down to us through religion.
- 4.34 And what are we going to then do about the relationship between religion and science? Not merely as an intellectual matter—who has a truer understanding of the way reality really works, but also as an institutional and political battle? Because we know who owns or who has captured most of the important political and social institutions up of that time. Science is the new kid on the block.
- 4.35 Religions, the traditional ones, have been around forever. They've built up the institutions, they control them, and anything that they see as a threat, their track record shows, they will often react quite hostilely to. And we've already started to see significant amounts of that with respect to the early sciences.
- 4.36 So I'm going to take up this issue of Galileo Galilei, a few decades before he died, who is one of the great scientists of his time, the law of gravitation, all sorts of interesting experiments, things that we know have come down to us because of his discoveries. And one of the things that he had been interested in, since he was interested in just about everything, was, as a younger science, is the nature of the heavens, as we call it.
- 4.37 Now, notice when we call them the heavens, we're using a religious designation, generically. Everything up there is the heavens. So if we start to think more secularly, we might say the stars and the moon and the planets, and then perhaps God beyond that. But how do they actually work? And there's a traditional understanding of the way the universe goes, which is to say, the Earth is at the center of God's great creation.
- 4.38 It has to be at the center of God's great creation because God's greatest creation, the human being, has to be at the center of whatever God's ultimate plan for the universe is. God has an idea, has a plan, why he created the universe. He wants the universe to go through a certain, we might call it a drama of conflict and struggle, but it's going to reach an ultimate resolution. We are the main actors or the main agents for realizing God's plan for his creation. So, of course, we have to be at the center of everything, and then everything revolves around us, and then you get out to God himself.
- 4.39 When, a century earlier, Copernicus, Nicolaus Copernicus, had published the idea, he didn't have it published during his lifetime because he thought it was a little too

dangerous, but it was published, went to publication when he realized he was dying, and nobody could do anything to him for it, he published the idea that we should have a heliocentric model. That is to say, the sun is at the center, and see everything in orbit around the sun.

- 4.40 That idea had interesting traction in terms of pure physics and pure mathematical understandings of the world. It accounted for the data a whole lot better than the traditional Earth-centered or geocentric model of the universe. But it was shut down. And last time we mentioned that Giordano Bruno, an early enthusiast of the Copernican model, had been found guilty and executed for heresy. That wasn't the only heresy that he believed in, but this was dangerous stuff.
- 4.41 Galileo, when he was a younger scientist, wanting to make his career, not wanting to be burned at the stake and so forth, had decided, "OK, I'm just not going to pay attention to that set of issues for now. I am going to work on gravity and some other things," and so on. But nonetheless, he still had the astronomy bug. And in 1610, he's now older. He's seen as perhaps the leading scientist in, at least in Italy, but perhaps all of Europe. And if you're a leading scientist in Italy at this time, you're probably the leading scientist in the world at that point.
- 4.42 He renewed his attention, his interest, in astronomy. He'd heard about telescopes being made by lens grinding, some people in northern Europe. So he made his own telescope. It was, I think, a seven-power telescope. And he discovered something, you can see the fascination, that no one had seen before. He trained his telescope on Jupiter, and he realized and saw for the first time that Jupiter has moons. There are satellites that go around Jupiter.
- 4.43 You can see in his notebooks how excited he is at this brand new scientific discovery. He publishes this, but he immediately gets into hot water with the Vatican and the Vatican-friendly intellectuals all over Europe. The angry response almost immediately was, it is not possible that Jupiter has moons. Because we know, notice this strong word, we know that there are seven objects in the system. If you count them, there's Earth, moon, sun, Mercury, blah, blah, blah, you end up with seven.
- 4.44 If Jupiter has moons, that means there's more than seven. It's not possible that it could be more than seven. But notice the mode of argumentation that we are getting into here. We're seven. We say, why do we get so hung up on seven? The point is not that we are naturalistically, scientifically using our eyes, using mathematics and so forth to see and count what's actually out there in the world, and that's how we're going to learn how many things are.
- 4.45 Seven has significance. It has biblical significance. It has theological significance, and that's the most important kind of significance. How many days of creation were there? How many days in the week are there, correspondingly? How many deadly sins are there? And then my favorite one is, if you take God's greatest creation, which is the

human being, how many orifices are there in the human skull? There are seven openings. We have two eyes, two nostrils, two ears, and a mouth. Seven.

- 4.46 Seven is built into the order of creation in all of its fundamental and important features. We can know, a priori, to use a philosophical layer, ahead of any sort of reports from these pesky scientists that want to say there's more than seven. This is part of the nature of the argumentation that we have going on here. The new method that the scientists are discussing. Well, everybody has eyes and we should look at the natural world and we should go out and do math, do science, and figure out things that way.
- 4.47 That method of thinking, independently of the conclusions that they are reaching, is in conflict with the method of thinking, which is to say, we go by scripture, we go by authoritative interpretations of scripture, as it has come down to us in the tradition over the course of the centuries. So we have a great big collision.
- 4.48 Now, how are we going to overcome this conflict? And that is the problem that Galileo sets himself to, a few years later in 1615, when he publishes an open letter. He formally addresses it to the Grand Duchess Christina. But the expectation is multiple copies are going to be made and sent to everybody all over Europe. And this is Galileo saying, "Here is how, now in the modern world, we are going to think about religion, we are going to think about science, and how properly they should relate to each other."
- 4.49 So as we do, I want to go to the text for a little bit and Galileo's own words to see how he squares this circle. OK. So here we are. As a good logician, as a good reasoner, you recognize on controversial issues, you have your views, but also there are opposing views, and you present the opponent's views, and you present them fairly and squarely, and then you present your views, and you let the argument go back and forth. No cheap shots and so forth.
- 4.50 We start with, what is the argument for saying that the Earth must be at the center and it can't possibly be the the sun? "The reason produced for condemning the opinion that the Earth moves and the sun stands still is that in many places in the Bible one may read that the sun moves and the Earth stands still." Now, that's true, but don't take it from me. Go look at the Bible, and you'll find there are lots of places where that is stated explicitly or by close implication, that it must be that the Earth is stationary and the sun is going around the Earth. OK. But that's a premise, first premise.
- 4.51 "Since," another premise now, "the Bible cannot err," the Bible can't be wrong. But why not? Well, because it's the word of God, and God doesn't make mistakes, so second premise. "It follows as a necessary consequence that anyone takes an erroneous and heretical," so it's not just erroneous, it's not just a mistake. It's a religiously heretical mistake, which has sin connotations tacked onto it as well, not simply an intellectual matter, "an erroneous and heretical position who maintains that the sun is inherently motionless and the Earth movable."

- 4.52 OK, so that's the argument Galileo wants to take up. "With regard to this argument, I think in the first place, it is very pious to say and prudent to affirm that the Holy Bible can never speak untruth." So, we're not going to challenge the second premise. The Bible is correct. It never says anything wrongly. "Whenever its true meaning is understood." So, how do we know what the Bible's true meaning is? Here things get complicated.
- 4.53 "I believe no one will deny that it is very often abstruse, and may say things which are quite different from what its bare words signify." So what is he saying here? Sometimes what the Bible is saying is quite complicated, perhaps metaphorical, figurative, and so forth, but not to be taken strictly, literally. So we have to introduce a principle of interpretation here. Some things are taken literally, some things are taken figuratively. And how do we know in which case the Bible is speaking literally, in which case it's speaking figuratively?
- 4.54 But why do we need this distinction in the first place? Well, he says, because, "in expounding the Bible, if one were always to confine oneself to the unadorned grammatical meaning, one might fall into error. Not only contradictions, but even grave heresies and follies." What do we mean by this here? "Thus it would be necessary to assign to God feet, hands, and eyes, as well as corporeal and human affections, such as anger, repentance, hatred, and sometimes even the forgetting of things of the past."
- 4.55 So, examples. If we were to take the Bible literally, then we would read a phrase like this. "And God's eyes flashed angrily upon them." People are doing some sinful things, right, and God's eyes flashed angrily. So if we take that literally, what are we saying? God has eyes. But what are eyes? Eyes are a physical organ. They're responsive to light and so forth. Does God have eyes? Does God have a body? No. So this must be a figurative way of expressing God's reaction to the sin that's going on. It's not literally saying God has eyes, that he's a physical being.
- 4.56 Or "God smited them, or smote them." A lot of smiting and smoting on occasion, right? With his fist. What are we saying, that God has an arm? What, is he pretty ripped, works out? No, he doesn't have a body at all. So when we talk about God smiting and smoting with his fist or his arm reaching out, right, that has to be a metaphor. And there are lots and lots of places in the Bible where, in order to make a point, the writers of the Bible are resorting to this figurative language and it's not to be taken literally. So that's this point here.
- 4.57 Now, why is the Bible doing this? Well, it's because, of course, the issues often are very complicated, and the Bible is written for the common people, the vast majority of common people, who are not particularly well educated, not very comfortable with dealing with abstractions and high metaphysical, theological understandings of God.
- 4.58 So we are writing important biblical points about the nature of God and God's relation to human beings, but using language that the vast majority of people can use, and they

speak in a much more physically oriented language. So, metaphor and more broadly figurative speech. "It's written for the capacities of the common people who are rude and unlearned." Not rude in the impolite sense, but just rude in their overall, basic regular manners.

- 4.59 So that means that, as a good Catholic, Galileo would say, "It's necessary that wise expositors should produce the true senses of such passages." So we read these literal descriptions of God's eyes and God's hands and so forth, but we have to have authorized people interpret what those actually mean. What are they meant literally, not just figuratively.
- 4.60 Now, after some discussion of lots and lots of examples of this, and hopefully this distinction is in place, Galileo wants to draw the first big conclusion. "Hence, I think I may reasonably conclude that whenever the Bible has occasion to speak of any physical conclusion," that is to say, anything about the physical world, about physics, especially those which are very abstruse and hard to understand, "the rule has been observed of avoiding confusion in the minds of the common people."
- 4.61 So, anytime we're talking about the physical world, but a theological message about the nature of the physical world or what's going on in the physical world, we're not necessarily to take that literally. Further, what is the Bible really about? Is the Bible meant to be a science lesson about physics and chemistry? No, the Bible is primarily about your soul and becoming reunited with God, the salvation of your soul. That's its real message, and everything then is to be interpreted in a way to make clearer to people what is involved for them reuniting with God. It's not meant to teach them about the way science, physics, chemistry, and so on go.
- 4.62 "Speaking casually of the Earth, of the water, the sun, everything," so it mentions all sorts of these things, but that's not what it's really about. "I think in discussions of physical problems, we ought to begin not from the authority of scriptural passages, but from sense experiences and necessary demonstrations, for the Holy Bible and the phenomenon of nature proceed alike from the divine word."
- 4.63 So what Galileo is now doing with this groundwork is proposing a division of the turf and a division of the labor. So yes, there is the higher spiritual world and there is this lower physical world. What is the Bible really interested in? It is interested in the higher spiritual world and how we should be relating to the higher spiritual world. The Bible is not really interested in the natural world and the way the natural world works. It's not that important. It's just a temporary way station for all of us until we ultimately go to meet our maker.
- 4.64 And so it's not trying to do science, it's trying to do religion, properly speaking. But that then means that, if we are actually interested in the way the physical world works, we should not be looking to the Bible as a science book. We want to understand the way the physical world works, we need to use methods that are appropriate to

understanding the physical world. And that's what science is. We look at the world, we organize it in conceptual structures, we do experiments, we do mathematics and so forth. That's the appropriate message.

4.65 And so what Galileo then says, this puts it here, is in effect, God wrote two books. "The Holy Bible and the phenomena of nature proceed alike from the divine word." So God engaged in two acts of creation. He created the natural world. He created the physical world and all of the laws that govern the natural physical world. And that, as an expression of God's creation, is worthy of study, worthy of knowledge. God wants us to come to know his creation.

4.66 But he also wrote another book, and that's the Bible. He wrote scripture. And he also wants us to know scripture, to know the Bible, and to be interpreting it properly. But of course, the people who are interpreting the Bible properly are the people who are educated and authorized to interpret the Bible, and a lot of times that's very complicated. So we need expertise in understanding that book of God.

4.67 But we also need a different kind of expertise for reading God's other book, people who are very good at doing science. Now, these are two books that God wrote. Each has its own expertise. But of course, since they both come from God, they should be compatible with each other. And if we are interpreting the Bible correctly and we are interpreting the book of nature correctly, ultimately those two interpretations should mix. They should meld. They should come to this center. Two ways of understanding God and God's creations.

4.68 And what this then means as an implication is, along the way, of course, we human beings are relatively ignorant, relatively knowledge. We are engaged in this process of discovery. We're going to make mistakes along the way. We're going to interpret the Bible and we're going to interpret it this way. We're going to be interpreting the book of nature and interpreting it this way. But we make mistakes.

4.69 And one sign of those mistakes is going to be, at some point we say, "Oh, look, our interpretation of the Bible here is in conflict with our interpretation of the book of nature." Science and religion are in contradiction with each other. What should we do in that situation? Realize that we've made a mistake. When we realize we've made a mistake, what we should do is, we should go back and reexamine our interpretations over here, and reexamine our interpretations over here, to find out where the mistake is. Correct the mistakes, and then realize we actually will have a harmony.

4.70 What we should not be doing is saying, "Oh, we've interpreted the book of scripture this way and reached this conclusion, and we've interpreted the book of nature and reached this conclusion. They are in contradiction with each other. We need to start torturing these people, until they give up on that bad interpretation, because this interpretation is always the right interpretation." That's not how rational, modern, and religious people should act.

- 4.71 If you are a truly religious person, you will be open to science, you will be open to religion and ongoing interpretation, even altering and adjusting the interpretations until the best scientific theories and the best religious theories are in harmony with each other. This is a very powerful, very important argument to make early in the modern world.
- 4.72 Now, why should we be doing this? We made a metaphysical point. God wrote scripture, God wrote the book where He is the creator. Ultimately, the two should fit with each other. But he also made us, and he gave us certain capacities, including cognitive capacities. He gave us the capacities for observing the world with our senses. He gave us our senses. He gave us a capacity for reasoning, for doing mathematics, for figuring out complicated scientific methods and so forth.
- 4.73 Why did he do that? The answer is going to be, well, God doesn't make anything without a purpose. If God gave you a mind, he wants you to use it. If he gave you senses that are capable of figuring out or observing all kinds of interesting things about the world, he wants you to use them. And so we should be thinking fundamentally of science, that is to say, a systematic use of our senses to observe God's natural creation, to understand it, to tease out all of its secrets through developing mathematics. We are being intensely God-revering and using our God-given capacities to do science to come up with the ultimate explanation of God creation.
- 4.74 What this then means also by implication is a very harsh criticism of much of traditional religion, which is to say, all of those religions that have said, "You should not think for yourself. Who are you to question the word of God and these wise institutionalized people who have interpreted scripture? Don't you question. Don't you think for yourself. Don't think that you can go off and discover new truths. You should have faith, you should obey, even if it doesn't make sense to you. It seems logically contradictory. Stop being a troublemaker and just believe."
- 4.75 That, from the Galilean religious perspective, is profoundly irreligious. That is profoundly heretical, if we were going to use the religious language, because you're asking people, human beings who have these magnificent God-given capacities, not to use their God-given capacities. And nothing can be more sacrilegious than that.
- 4.76 Now, we do know the church institutionalized at that time was not happy with this argument back and forth, and there was then, the year after a publication of this codex that we mentioned briefly the last time, that considered the Galilean argument faulty, why it's faulty in the arguments back and forth. We can, some other time, go into all of that. But nonetheless, the argument was rejected, and still, for some centuries to come, the debate between some understandings of religion and science continued to conflict.
- 4.77 But one of the things that was important about Galileo and this argument is that it was convincing to most people who were in the scientific community, most people who

were not necessarily scientists, but wanted to be open to more humanistic, naturalistic understandings of the world, and a significant number of people who thought of themselves as true believer, orthodox, religious people as well, that they were willing to reform inside the church.

- 4.78** And so, this argument largely carried the day, over the course of the 1600s on into the 1700s. And the idea then came to be, well, what we need instead of torture and burning people at the stake and so forth is a kind of intellectual tolerance. The biblical interpretation, we've been doing it for a long time, and we still have lots of different interpretations of the Bible. So, rather than burning people for believing the wrong thing, what we need to do is appeal to their reason and have more discussion, more argumentation, and we will get closer and closer to the truth with time.
- 4.79** The same thing on the science side. Scientists don't burn each other at the stake because they have internalized the idea that we are going to appeal to the evidence, we're going to appeal to the math, we're going to appeal to the logic. But it's a long, complicated process. And of course, tempers are going to get frayed, and we're not going to like each other variously, but we're going to tolerate. Anybody can make any kind of argument and counterargument, and over time we're going to make progress in the sciences.
- 4.80** And that argument, instantiated by the time we get into the early 1600s, started to reshape dramatically the modern world. So, space for science has been made, but also accommodation with religion. It's not a rejection of religion, even if it argues that religion is going to need some reforming. At least, institutional religion is going to need some reforming.
- 4.81** So no censorship. This comes to be the brand new doctrine. Certainly no censorship on religious grounds. There are some people who have the religious truth, a lock on the religious truth, that they also should have political power to ban all other religions, or what they think of as faulty interpretations of their religion. Generalizing from religion, the idea that maybe what we should do, since truth and reason are common to all human beings, is have a completely no-censorship regime.
- 4.82** Let everybody think about everything. Politics, religion, marriage issues, friendship issues, business issues. Let's have a completely intellectually free society. John Milton, the great English poet, author of *Paradise Lost*, here's a statement from his *Areopagitica*, then the 1600s, one of the very early ringing calls for an intellectually free regime. This is going to have huge implications for how we do education, a lot more educational freedom for young people to explore and make up their own minds along the way.
- 4.83** "Though all the winds of doctrine were let loose, so truth be in the field, we do injuriously, by licensing and prohibiting, to misdoubt her strength. Let her and falsehood grapple; whoever knew truth put to the worst in a free and open

encounter?" You have a true position, but maybe we don't yet know that it's a true position, but somebody is arguing for it. There's all these other false positions that are out there. We let them all fight it out, but they are fighting it out by means of appeals to evidence, reason, logic, math, and so forth.

- 4.84 Who's going to win? Well, the false positions don't have logic on their side. They don't have the facts on their side. And so eventually, they are going to lose that battle. Truth has all of the evidence and all the logic on its side. Eventually, it will prevail. Truth is not this frail thing that we need to protect against falsehoods. It is strong, it is robust, and with good education and proper mental training, anybody can see the evidence, hear the evidence, sometimes taste or smell the evidence, and learn the reasoning that will lead them to see the truth. So, truth.
- 4.85 "She needs no policies, nor stratagems, nor licensings to make her victorious. Those are the shifts and the defenses that error uses against her power." It's the people who believe false things and want you to believe false things; those are the ones who have to be sneaky and make stuff up and resort to all kinds of illogical, manipulative fallacious devices and so forth to try to fool you. Truth doesn't need any of that. But if we let truth prevail and we train people's reason well with good education, then we will learn all of those sneaky tactics that the false positions use, and people will not be susceptible to them.
- 4.86 Now, marching along in the next generation, I have John Locke, one of the great, great philosophers, one of the two or three greatest, most influential philosophers of the modern era, picking up and extending these themes and applying them explicitly to the kind of education necessary for young people in the modern world.
- 4.87 Now, we're going to look only at Locke's views on education here. We have another course, Modern Philosophy, in which we spend a significant amount of time looking at Locke's metaphysics and his epistemology, his fundamental philosophy, and why that is so important, and also in courses on political philosophy and overview of Locke's very important contributions to political philosophy, particularly in North America with the Declaration of Independence, which is a very John Lockean doctrine.
- 4.88 But this is John Locke as a modern philosopher, a modern thinker, now turning to issues of education. What does this mean in specifics? I want to spend a little bit of time on Locke biographically, though, just to give you a sense of the temper of his mind, and why he is a modern philosopher and a modern thinker par excellence.
- 4.89 Now, we know him mostly as a philosopher and political philosopher, but interestingly, Locke did not train to become a doctor. He trained to become a philosopher. He trained to become a medical doctor. When he went to college, if you want to get ahead in the world, there's a limited number of majors, very hard to get a job in philosophy now, but certainly even harder to get a job in philosophy then.

- 4.90 And so, he didn't want to be a theologian. He didn't want to be a lawyer. But medicine seemed like it was, with all of the new sciences, an open and very intellectually interesting field that would allow him to pursue his intellectual interests and make a mark on the world. But when Locke went off to university, at Oxford, he was not impressed by the prevailing medical theories that he was learning, most of which he thought were stuck in the past.
- 4.91 They were a mixture of alchemy and astrology and little bits of early modern science and all kinds of weird theorizing going on. Not very much experimentalism, which is certainly a new modern thing. There's a reverence for old books that somehow, if we want to figure out what the truth is and what the right cures are, we have to read books that were written by Hippocrates 2400 years ago or Galen, 1200 years ago, and just more or less mechanically follow the recipes that had been written down in those books.
- 4.92 And Locke was, even as a young man, not at all impressed by that. Now, my favorite example of this, though, is the issue of anemia. And this would be the theory that young medical doctors or prospective medical doctors would have been taught, Locke among them. He mentions this is an example of an approach to medicine that he thought it was inappropriate. Now, I'm going to talk through this, but the issue is anemia.
- 4.93 So anemia is a condition in which people don't have very much energy. They feel listless, they feel weak. They might try to talk themselves into saying, "No, I really should get some stuff done, but I just don't have the energy. I can't get into it," and so on. Why are people anemic? What causes anemia? This will be a wonderful thing if we can figure out how to fix the problem of anemia. Let's process it. This is the theory that Locke would have learned.
- 4.94 Anemia is probably a problem of weak blood. We, at this point, have done a certain number of anatomical studies. After William Harvey, we now know that there's a whole circulatory system of the blood. Maybe this generalized lassitude all over the body is because the blood is weak, everywhere in the body. We've also discovered that if you give people who have this weak blood, or anemia, salts of iron, iron in a salted form, that they get better. It doesn't cure their condition, but it improves their condition.
- 4.95 We've got some empirical medicine and a hypothesis. Iron salts seem to work for anemia, and anemia, rather, is probably a matter of weak blood. But why? Why would iron salts fix people's blood and make their blood stronger and better? Now we engage in a chain of reasoning. We say, well, what do we know about iron salts? Well, they come from iron. And iron, compared to other metals, is quite strong. It's stronger than tin, it's stronger than copper, stronger than gold. It's a tough one. So it's a strong metal.
- 4.96 And that's why we use iron in making weapons. We don't go with copper swords. We make iron swords because they're tough. So we use iron a lot in war, and that makes

sense. We also know that Mars is the god of war. OK. So now we're mixing in a little bit of ancient Greek and Roman religion. Ares, renamed Mars, as the god of war, because the gods are probably involved to some extent. It can't be a purely naturalistic or physical explanation. So Mars is the god of war.

4.97 But also, we know from doing our astronomy, that Mars is the red planet. You look at Mars through one of those newfangled telescopes, and it's got this distinctively reddish hue. That can't be a coincidence. We also know now that blood is red. So that's why iron salts cure anemia. It is the strength of the iron, communicated religiously and astrologically through Mars, and the coloring is the vehicle for communicating that strength to the anemic person, thereby solving their problem.

4.98 OK, how impressed are we, with this cutting-edge medical explanation, medical theorizing? It's a whole mishmash of empirical stuff with Greek and Roman gods and astrology and so forth. So Locke says, no, this is not it. We need to reform the metaphysics and epistemology that underlies even early modern cutting-edge medical research.

4.99 Now, another indication of Locke's cast of mind, I want to suggest, is that he was experimental. It wasn't simply a theoretical matter. One of the things that enabled Locke to make his fortune as a philosopher was a medical procedure that he engaged in as still a relatively young doctor. This is the Earl of Shaftesbury, a bigwig, literally a big wig, but also very wealthy, part of the British nobility, politically connected, politically active and so forth.

4.100 But he had terrible ulcers, just debilitating stomach ulcers, very painful. None of the doctors could fix his ulcers. They tried all sorts of things. They read every single old book about ulcers possible. None of those recipes worked. But they had heard good things about this young John Locke fellow, and so they decided to give him a chance. They brought him in. He did a consultation.

4.101 Now, as I talk through this, just focus on how Locke's mind is working, how he is thinking about this problem. Ulcers in the stomach, the stomach is a biological organ. We know that the ulcers are caused by a lot of acidic stuff that happens in the ulcers because sometimes, when he vomits, it's extraordinarily acidic stuff that comes out. So, we have too much acidity in the stomach at the time. It's the first step.

4.102 Now, we can give him a purgative to get it out through his bowels, or we can make him vomit a lot, but all of that acid is really nasty for the throat and it's really nasty for the bowels, so we don't want to do too much of that. How can we get this acid out from inside the stomach? Locke said, you know what I'm going to do is, I'm going to go to a metalsmith, and I'm going to ask the metalsmith to make a very small spigot, the kind of spigot that you would use at the end of your garden hose, but much, much smaller.

4.103 So it has a little tube with a shut-off ballcock that you can turn open or closed. I'm going to have it made out of gold, because we know that gold is relatively clean, doesn't

tarnish, or he's a rich guy, so he can afford the gold thing anyway. Little tiny spigot. We take Shaftesbury. Get some big guys. We're going to lay him down on a table after having gotten him extraordinarily drunk, to the point where he's going to pass out, so he's not going to give us any troubles.

- 4.104** Then this is what I am going to do. I'm going to drill a hole into his side, directly into his stomach. Take the spigot, put the one-half of the spigot inside his stomach, put some sealing goopy stuff around, leaving the other part outside, and cross our fingers, or whatever else it is that we do at this point in medical history. And Shaftesbury wakes up after, next day, rather hungover, but he recovers, and it works.
- 4.105** Anytime his ulcers are flaring up, he's just got terrible pain, he can go off to the side, even if it's at a dinner party, a little bowl, open up the spigot, drain out all of the excess acid into a bowl. Instant relief. OK, it works. But think about the cognitive set here. I don't know who you think is a rich and powerful person, and you're a young doctor. I'm going to get this guy drunk. I'm going to drill a hole in his stomach. That's right; that is audacity. But that's experimental. That's innovative thinking. This is very modern. And it worked.
- 4.106** So, Shaftesbury was so delighted with this that he said, "OK, Locke, you are now our family." He's got a big family. "You're our family physician. You can come and live on our estates. Do whatever you feel like, and just whenever there's a medical problem, come and solve our medical problem." And that let Locke turn to his true love, which, of course, is everybody's real true love, which is becoming a philosopher. So he could devote himself to writing philosophy.
- 4.107** Now, he did go into exile for a while. He and Shaftesbury were involved in some political conspiracies and on the wrong side of changing British political reform. So he had to spend some time in exile in Holland, which was a remarkably tolerant place at the time. All sorts of major intellectuals and people escaping from religious repression in various parts of the world. So he wrote most of his major works there.
- 4.108** 1688, there's a big change of government back in England. It's OK for Locke to come back, and he starts publishing his backlog of major works very quickly, because of the brilliance of his thinking. And it being a very modern way of thinking, he becomes a leading philosopher. And with that, he turns to education, recognizing that education needs some reforming in this new modern world.
- 4.109** How are we going to educate new, modern free thinkers, scientists, medical doctors, who are willing to challenge the old dogmas? And in this emerging parliamentary democracy with a somewhat republican division of powers, how are we going to educate the citizenry to be the kind of people who can be citizens in a democratic republic and manage their own religion and so forth?
- 4.110** I want to spend a little bit of time looking at some of John Locke's thoughts concerning education. Number one, the very first thing. "A sound mind in a sound body is a short

but full description of a happy state in this world." Now, notice, right off the bat, how modern that is. We go back to Plato, we go back to Augustine, we talk about the mind and the body, and we see those as different and opposed things.

- 4.111 Our goal is to get our soul out of our body, to acquire pure knowledge, to look forward to the afterlife, and to pay not very much attention to bodily things as well. And that's been extraordinarily influential throughout the medieval period and on into the early modern period. So Locke is saying, no, it's not that at all. The human being is an integrate. We are a mind and a body; we are together. And we want to have a sound mind and a sound body. And we want that mind to be in the body, integrated, connected with. We are educating the whole person.
- 4.112 Next. "Men's happiness, or misery, is most part of their own making." Another strong one. Rather than saying, "I am helpless, I am weak, I am a slave to my sinful nature. Only higher authorities or only God can save me. I can't do anything," Locke, as an early modern, we have agency, we have power over our own thoughts, our own character, our own actions. And so we have a significant measure of control over whether we become happy or not in this life.
- 4.113 Second section, talking about children in specific. "I imagine the minds of children as easily turned this way or that way, as water itself, as though this be the principal part, and our main care should be about the inside." So this is Locke's famous blank slate thesis. When we think of the minds of people, we are not born one way or the other, necessarily. We are born with a blank slate, a slate being like an early chalkboard upon which you can write, and you can write just about anything. But it doesn't come pre-printed with any knowledge or any ideas.
- 4.114 It also is the case in your moral character. It doesn't come pre-printed with a bias toward the evil, or even necessarily a bias toward the good. It's a blank slate, and so it's going to be dependent on the environment and your teaching, and then the decisions about how you are going to use your mind and your body. So, another early modern challenge to the doctrine; in this case, the target is the doctrine of original sin.
- 4.115 We are not born sinful. We're also not born automatically good either. We are born with a blank slate and we can make ourselves into certain kinds of people. It's not that our life is a matter of, we are sinful beings and we have to manage sin and try to stifle our sin. And that is mostly what character education should be about.
- 4.116 I was going to then start with the body. And it's interesting because he says, mostly when you look at infants and young children, of course they have a developing mind, but it's their physical development that is most arresting and most important at the very beginning. "I shall consider first," as he says here, "the health of the body." Why? Well, "how necessary health is to our business and happiness."
- 4.117 We're going to be going off and doing business, engaging in all of life's business. And in order to be happy, we have to have a healthy body. So a big part of education right off

the bat has to be about looking after your body. Parents looking after the body of their children, but also children learning how to look after their own bodies—dressing themselves, hygiene issues, nutrition issues, and so forth. Fundamental part of education.

- 4.118** I want to jump a little bit, when he starts to talk about kids a little bit older, when they get off on their own, age three, age four, and so on, and they start getting a little bit wild and crazy and hard to control. Their minds are developing, their bodies are full steam, and they're just all over the place. At this point, we start thinking seriously about discipline and more systematic approaches to education and trying to get kids to focus on the things that we, the parents and the teachers, want them to focus on. There is going to be some discipline that is necessary here, but the way Locke talks about discipline is very characteristically modern.
- 4.119** "I have spoken so much of carrying a strict hand over children that perhaps I shall be suspected of not considering enough what is due to their tender age and constitutions. But that opinion will vanish when you have heard me a little further. I am very apt to think that great severity of punishment does but very little good; nay, great harm in education. And I believe it will be found that, *ceteris paribus*," that is to say, holding all other conditions equal, "those children who have been the most chastised, seldom make the best men."
- 4.120** We are coming out of a long tradition of education that is very strict, very top-down rule-imposed, very quick with the punishment. The expectation, since you are dealing with little monsters, who want to do all sorts of sinful, bad things, that you need to be very strict, and lots and lots of punishing. And Locke is arguing the opposite. As parents and educators, we should go light on the punishment, because punishment makes whatever naughtiness worse, and makes bad men.
- 4.121** And he's making an empirical claim here. Look at people who have been raised in a punishment-light regime and a punishment-heavy regime. By the time they are adults, which ones make the better human beings? He thinks it's clear from the evidence that one's raised in a more benevolent attitude. Well, how are we going to discipline children? How are we going to teach them their manners and proper behavior and so forth? Again, Locke has a very modern answer to this.
- 4.122** "Manners, as they call it, about which children are so often perplexed," with all these rules about everything. Where do those come from? Kids don't know them, "have so many goodly exhortations made them, by their wise maids and governesses, I think, are rather to be learned by example than rules." Set a good example. You're the mother, you are the father, you are the teacher. Set a good example. Be good yourself, and kids will learn by imitation.
- 4.123** Of course, there might be a natural predisposition that little boys want to be like their dads and little girls want to be like their moms, so they're striving to be. So, if you set a

good example, that's going to be most of the problem. The argument here is partly that little kids are like little scientists. They're watching and listening to everything and processing everything. There's a lot more going on in there.

- 4.124 So, if you are acting a certain way, they will observe very carefully and try to do that themselves. If you speak a certain way, the same thing will happen there. But then, of course, at the same time, there's the hypocrisy point, and this might be why a lot of punishment doesn't work, if I am one of those people who has standards for behavior and I'm smacking my kids around every time they misbehave, but then I turn around and I do the same sort of thing. Maybe I swear a lot, but every time my kids swear, smack.
- 4.125 Well, the kids are not stupid. They're aware of the hypocrisy, and then that is going to undercut the teaching of the lesson. So if I set a good example, I practice what I preach, chances are very good that's going to happen there. "Children, if they are kept out of ill company," this goes back to the curation issue that we talked about here, just, if you have a crass, crude, alcoholic, groping uncle, don't let them come to family functions. Keep the bad person away. Make sure your kids are friends with good people. They'll turn out OK. Just go light on the company.
- 4.126 Now, he starts to turn to, interestingly, some curriculum issues, and he has a developmental notion of how the curriculum should go. I want to just give a couple of indications here, partly because the very first thing that Locke mentions as a formal part of the curriculum is a shocker. The first time I read this, it is dancing. As soon as possible, we should teach children dancing. We want children to develop their bodies, strength, endurance, gracefulness. We also want them to develop their minds, strength, courage, gracefulness, have good manners, and so forth. And Locke says there's something very interesting about dancing.
- 4.127 "Nothing appears to me to give children so much becoming confidence and behavior, and so to raise them to the conversation of those above their age, as dancing. I think they should be taught to dance as soon as they are capable of learning it; for, though this consists only in outward gracefulness of motion, yet I know not how it gives children manly thoughts and carriage, more than anything."
- 4.128 So as we then think about physical education, we want the kids to develop their bodies, and so we have them do sports and physical activities and so on. Maybe we have them take gymnastics and so on. But there's something different about dancing, which also has bodily control, as gymnastics do, but it adds this element of rhythm and incorporates it with music. When you start to teach children formal dancing steps, it's interesting. You're not learning how to use your body, but also you stand up a little straighter.
- 4.129 Just that act of standing up a little straighter and striking this pose, what does that do to my mind? If I'm slumping along, what state is my mind in? If I'm standing tall, what

state is my mind in? If I have a little son and a little daughter, and I also teach them dancing, they like it. It's fun, too. It's pleasurable. But also you teach the little boy how to approach the girl and say, "May I have this dance?" You're also teaching manners. She says, "Oh, I would love to."

- 4.130** She takes his hand, and they then are not only becoming more elevated as individuals, the whole social interaction between the two is also elevated. If we then consider how modern this is, because if you consider traditional pre-modern, what has the attitude been toward dancing? Absolutely not. Dance is physical. It's boys and girls displaying their bodies to each other, and we know what that leads to. We have to forbid it entirely. So for Locke to give dancing its pride of place is significantly modern.
- 4.131** "Never trouble yourself about those faults in them, which you know age will cure." Another point about discipline. Kids are going to be kids. It doesn't mean that you're dealing with an evil serial killer wannabe, just because various bad things are happening. So again, light on the punishment. "When he can talk, it is time he should begin to learn to read." And he comes right back to this issue of discipline and the first very strong modern statement of a more free approach to education, instead of a strict top-down.
- 4.132** "Give me leave to inculcate again what is very apt to be forgotten. Namely, great care is to be taken, that it never"—it, this is his education now—"be made as a business to him, nor he look on it as a task. We naturally, as I said, even from our cradles, love liberty, and have therefore an aversion to many things, for no other reason, but because they are enjoined us." So pause right there.
- 4.133** Part of human nature is to be born free and to love your freedom, to like doing things for yourself, by yourself, making your own decisions. Much of traditional philosophy and traditional education has said, that is the enemy. Kids want to do their own thing, and we need to smack that out of them and get them to conform to our understanding, to be obedient, to be disciplined. And Locke is saying that is exactly the wrong thing.
- 4.134** Liberty is the human natural birthright, and education should work with the need for and predisposition for liberty, so maximizing the amount of choice in education. And that is immediately followed up by another extraordinarily strong thing from Locke as an early modern. "I have always had a fancy," that is to say, a fantasy, a dream, a kind of idea, "that learning might be made a play and recreation to children; and that they might be brought to desire to be taught, if it were proposed to them as a thing of honor, credit, delight, and recreation."
- 4.135** Education should be fun. Children like fun. They like to play. And again, traditional enemy, education has said, that is the enemy. Education is not supposed to be fun. It's supposed to be hard work, painful, and we eliminate all of the fun from it. Locke is again saying, as a modern, that's exactly wrong. We should be saying, life is fun. Life is a delight. It should be, ultimately, recreation. This distinction between work and play as

oppositions between school and fun, as an opposition, that has to be exploded. Life should be about happiness, and it starts from the beginning and so on.

- 4.136** All right, we shouldn't have to whip children into reading. Reading should be fun. If children don't like reading, don't like actually learning lots and lots of things about themselves and the way the world works, something has gone wrong. It's probably our fault, as the educators.
- 4.137** Locke carries on in this vein, and what we start to find is education, revolution, starting to percolate in a much more modern direction, against many of the traditional notions of authoritarianism in education, obedience and discipline, and punishment being central hallmarks, there being a very strict education, that you don't have any choice about what you are to learn, into the exact opposite. With that, we'll pause.
- 4.138** Nowadays, we know that the childhood, the kid's born with the cell phone in hand. How do you recommend to help the teachers and the parents to guide them and to explore the nature, and don't feel like, why I don't have a phone and my friend has? Kind of thing. What do you recommend nowadays?
- 4.139** So you're putting your finger on two issues, and partly, I'm going to defer a fuller answer to this question until lecture eight, where we'll be in the 21st century, and this particular piece of technology. So part of the issue is, phones are wonderful, but they're also also very seductive. And as parents and educators, we want our children to be exposed to the whole world, but at the same time, we don't want the whole world just to be their cell phone.
- 4.140** So, how do we say, yes, the cell phone is wonderful for some things, but we also want you to make a significant amount of time in lots of other things. So that's one issue. And then the other issue is the social issue. So these kids have cell phones, these kids don't have cell phones, or this kid has a new cell phone, this kid has a three-year-old cell phone, which is just devastating, if you can imagine, the three-year-old. So that raises the issue, and we can generalize that.
- 4.141** If we're doing our education in a social context, there's lots of inequalities among the children and learning how to deal with the different kinds of inequalities. Some people have, I don't know, they can afford to get their hair done more regularly or nicer clothes or whatever. Both of those issues then do need to be dealt to start with. I would say, and I'll try to put this in a 1600s context. Let's set aside the cell phone issue, so we can see this as a broader issue as well.
- 4.142** On the first issue, it is going to be the case that you, as the parent or the teacher, have a conception of now, a well-rounded education for children, that you think, from your adult perspective, that functioning adults in the world are going to need to know, say, these eight subjects. That's an artificial number. But you're always going to have some children who are only interested in one, not the other seven, or they're interested in two, but not the other six.

- 4.143 How then do you manage the difference between their individualized interests from their childhood perspective and your broader range of interests from the adult perspective? I don't think that there's ever going to be a recipe answer to this. This is a point actually that John Locke, also in his thoughts concerning education, stresses, and he actually ends with this point here.
- 4.144 Here is to say, if our goal as educators is to find a kind of recipe and a one-size-fits-all recipe that we can just apply to all children equally, we're doomed to failure because the fact is that every kid, while they are all human beings, and we are quite right to say they need to have health education, they need to know some math, they should know some science, they should know some history, all of those generic things, all of that is valid—nonetheless, there is going to be significant variation.
- 4.145 So a big part of education then, he's going to say, is to get away from one-size-fits-all education. We have the education system as a template. We bring the children in, and we put them through a strainer. They have to fit into the mold of the system. As much as possible, we should be trying to go the other way and say, I've got this generalized template, but I customize it to the individual child.
- 4.146 Examples of this would be to say, suppose we consider nutrition. Now, we want all children to learn about nutrition, and so we know that they need some proteins, carbohydrates, fats, and so forth. All of that is generic to the human being. But nonetheless, we know that some of the children are different. Some of them are athletic and some of them are not. Some of them like broccoli and some of them don't like broccoli. There are all kinds of individuating factors that have to go along.
- 4.147 If you put yourself in the position of being, say, the mother of a large family and you're trying to cook for all of these kids, you have to strike a balance between my time is limited, our resources are limited. I want to make sure that all of the important nutritional food groups are covered. That's my job. Within that framework, as much as possible, accommodate the individual interests of the children. There are a variety of ways of doing it, but it's also never going to be a recipe. It's always going to require judgment call in particular cases, instead of a one-size-fits-all recipe.
- 4.148 Some of the suggestions that then people will start to say is to say, well, I don't know, this might be a little bit too easy, but you might say, if I've got seven kids and there are seven days of the week, one of the things I might do is, I as the mother, if I'm responsible, I'm going to set the basic menu for the week, but each day one of the kids gets to be in charge of customizing the menu on that particular day. So that brings in their individuality, but it also brings in their responsibility. Wait, I have to think about this. I make some decisions. I'm making decisions for me and for the whole family. So you're building in a moral lesson as well. So something like that would be the general approach.

- 4.149 I very much like the idea of Locke about play and dancing. I hadn't heard it a lot before. I think I have with Peterson Academy for the first time, out of my own volition, more learning as compared to before, more like being a little bit forced or in a certain way. My question is, what do you think about that? Do you agree with that? And if you think it could be applied somehow into an online university, maybe course-specific, or any ideas about that?
- 4.150 Yeah. So you're talking about my opinion in particular and current applications. I think Locke is hugely important, not only for historical reasons, but also for philosophical reasons. I wouldn't call myself a Lockean, but I would say that Locke is someone from whom I learned a great deal and I agree with. But also, I would say at the same time that Locke is thinking about an early modern, more scientific, empirical approach to education.
- 4.151 One of the things we do know is that science often involves saying there's early versions that need to be significantly altered or added on to. So I would say this is a very good early modern, rough draft of a healthy approach to education. And then for me personally, I've spent most of my career teaching in liberal arts universities, but liberal arts in that modern sense, where the idea is that you don't have to go to university. It should be a free choice. And we give you lots of decisions about what courses you are taking and what your majors are going to be.
- 4.152 Even within my own philosophy department where I hang out, maybe this is actually a Lockean point now that you put it, one of the things that we always have arguments back and forth among the professors is what the requirements should be and what the electives should be. I have always argued that nobody should be required to take philosophy courses. That's always put me in a very small minority in the departments, but in the universities, because usually the incentive in universities is, you want to make the courses in your department required, so that as many people as possible have to take the courses and therefore your salary is secure.
- 4.153 You can build your empire that way. But yes, this is the issue. I don't think in philosophy, particularly, but even more broadly—you take philosophy courses or whatever because you are interested in it. You want to learn that. If you're not interested in that at all, there's lots of other good courses for you to take. I think certainly higher education should be all of that. That then ties into partly your question here. We're scaling up to that more and more, starting from infancy. So even children like to make their own decisions, but will be bound within those decisions are made.
- 4.154 Now, even at the liberal arts university, we are making decisions about what the bounds are. So we might have 40 different majors that we think are legitimate, but we have lots of things that we said, no, we don't think those are legitimate academic disciplines, so we're not going to offer those at our institution. At the same time, we would say—this, again, would be a very Lockean point—that we are open to the idea that we're always learning new things, and maybe there are going to be new disciplines

that come along, and if they can prove their worth, then we add those to the curriculum.

4.155 Some things get a little bit outdated and old-fashioned. We drop those from the curriculum as well. Again, it's not a recipe, it's an open-ended thing. Now, how this would apply to online education in particular? Let me, again, since that's a very 21st-century question, although it has historical roots with the idea of self-directed education and emerging technologies, in one way, it's an older question. But let me save that for after lecture 8, because then we will be in this generation and ready to take up that set of questions.

Lecture 5: Romantic Educational Visions

At a glance — Romantic educational visions of the long 18th century.

- 5.1 All right, let's start in 1778. The 1700s, sometimes the long 18th century, typically labeled the Age of Enlightenment. And by the time we get to 1770s, John Locke had died early in the 1700s, but his philosophy had transformed much of the English-speaking world, certainly in America, a new country in America.
- 5.2 And his influence was strong on the European continent, particularly in France, where one of his great disciples, Voltaire, also known as a great philosopher of the Enlightenment, and things are churning in France. Isaac Newton had also died early in the 1700s, but his magnificent Principia Mathematica, laying the foundations for a physical, mathematical understanding of the world, that was revolutionizing all areas of science.
- 5.3 Also, by the time we get to the 1770s, the industrial revolution is now significantly underway. Several centuries of tinkering leading to greater amounts of innovation, to outright industrial transformation, particularly with James Watt's steam engine coming online. Now we have what seems like an unlimited source of power that, with mechanical devices, can be transformed to all sorts of machines and revolutionize manufacturing.
- 5.4 We are now well into the modern, enlightened understanding of religion, where we are rationally thinking about religion. We are taking self-ownership of our own religious lives, extending that to our economic lives. Adam Smith's On the Wealth of Nations, first treatise on free market economics and early capitalism, published in 1776. Revolutionary transformations in politics. The United States of America, 1776.
- 5.5 The world has been transformed not just as a set of ideas now, more enlightened ideas, but also institutionally, all of our major institutions in religion and politics and economy. We should also mention the great battle against women being second-class citizens, starting to be enjoined. We start to see the beginnings of early feminism, arguing that women should have equal liberties with men, and increasingly, movements starting to be formed against slavery.
- 5.6 Not just a few isolated voices saying there's something wrong with slavery, but critical masses of people in England and France and early America, saying these are not just ideals, liberty and equality, but all human beings have the capacity for reason and self-governance, including slaves. So we should get our act together and start to do something about it.
- 5.7 This is the age of Enlightenment, self-described age of Enlightenment. We have finally figured out, intellectually, what was wrong with many of the ideas that held human beings back in ignorance and poverty and superstition for so long. We can, with these

ideas, put them into practice. Knowledge really is power, make the world a better place. And we set about doing so.

- 5.8 I choose 1778 as the particular year because, in that year, this individual, Jean-Jacques Rousseau, died. Jean-Jacques Rousseau is known as the bad boy of 18th-century philosophy and as a leading counter-Enlightenment, anti-Enlightenment philosopher, also who's going to be extraordinarily influenced on the course of ideas after his death, but also on the course of education. And several of the modern approaches to education trace their roots to Rousseau's philosophy in particular.
- 5.9 But he was first a philosopher, and he made a reputation for himself by winning an essay contest on the issue of the Enlightenment. It was in the middle part of the 1700s, and a learned society had posed the question, "Has the Enlightenment in the arts and the sciences contributed to the progress of humankind?" So are we actually making the world a better place, as a result of these new Enlightenment ideas?
- 5.10 And Rousseau, partly through conviction, because he thought the Enlightenment was a terrible thing, and partly through calculation, figuring probably everybody else is going to argue, "Yeah, the Enlightenment is great," I'm going to be the outlier, and I will draw attention to my essay by arguing the opposite, that the Enlightenment has been terrible. He wrote this essay, that it came to be published shortly after, called the Discourse on the Sciences and the Arts in 1750.
- 5.11 While we are here, I will say, he goes on to write on political philosophy, on moral philosophy, other issues, and he has a special treatise on education that he publishes in the 1760s. We're going to lay the groundwork, though, for that education treatise. It's not really a treatise, but it's more an alternative literary style, mixed and match in there, but it amounts to a fairly systematic understanding of education.
- 5.12 What is wrong with the Enlightenment? Where has the Enlightenment gone wrong? We're not going to be just old-fashioned conservatists and say the modern world is a mistake, we have to go back to medievalism or whatever. Although he's going to argue that we can learn from some of the more primitive peoples and try to reincorporate and reform modern society in the light of their ancient human wisdom.
- 5.13 But the problem and the root problem with the Enlightenment is that it relies first and foremost on reason. That we think that we can observe the world, that we have this rational capacity, and that we should give pride of place to it. Of course, the Enlightenment philosophers are all saying, that's what enables us to take charge of our religious lives, our political lives, our economic lives, to do math and science and so forth. They all think that it's great and wonderful, but—and there are a lot of buts that are going to come out from the Rousseau perspective.
- 5.14 Technology, and this industrial revolution that we think is so great, is in fact weakening the human being. Instead of human beings, the way they used to have to be, being physically tough and hardy, we were farmers, we were hunters, instead now we can

move to the city and develop all of these machines and gadgets that are going to start doing things for us. What this then means is that we are going to start becoming soft. We are undermining our own physicality, which is important to having a robust humanity.

- 5.15 One of the technological achievements is the achievements in agriculture. We are starting to be rational about it, instead of just whatever happens to come across that we can hunt or gather, we'll do that. We are systematically controlling the animals and controlling the crops. And that, of course, we say, is wonderful because now we have more food. And then Rousseau points out, yes, but we also get fat.
- 5.16 And instead of being tough and lean, now with all of this abundance of food, human nature being what it is, we can't resist eating too much. Again, this is leading to an increase in physical debilitation. We start eating luxury foods that are not good for us. Instead of eating healthy foods, basic foods, just like our forebears, we are starting to eat all of this imported sugars and various other sorts of things that really are bad for our condition. There is a dark cloud to every supposedly bright cloud that the Enlightenment wants to talk about.
- 5.17 One of the things that starts to happen also is, as we get wealthier as a result of this agriculture, urbanization, all of the new sciences, all of the new technologies and so on, we start to have some people who were perhaps luckier. They got a better rational education, or they're born into the right families or whatever, become very, very wealthy, very, very quickly, and lots of people are being left behind.
- 5.18 And all the property is being gobbled up by the wealthier families. All of the industrial opportunities are being gobbled up by a small few number of families, and we have an exacerbation of the traditional problem, rich versus poor. But now we have fewer and fewer rich people, and more and more poor people, at a subsistence level, and this leads to social disasters.
- 5.19 Social disasters go both ways. The rich people start to think that they are special and they start to look down upon the poor people. The poor people get filled up with envy and they resent the fact that the rich people, from their perspective, somehow lucked into their riches, so they hate the rich, and we start to undermine society.
- 5.20 Also, reason involves self-reflection on all sorts of things that we used to do more naturally and instinctually, we might even say. For example, as we become richer, as we start to become more rational, we start to explore the arts, and we become more discerning about the arts. And we want to drink better wines, if we can think of the art of winemaking; and we want to have nice paintings in our homes, and we become more discerning about what's bad art, this copying, schlock, kitschy art, and what's really, really good art.
- 5.21 We start to want to go to the opera, instead of just going down to the village pub and getting drunk and telling dirty jokes to each other for the night. We start to have more

elevated aesthetic standards. But again, this leads to some people becoming much more snobbish and a lot of other people still being stuck at lower levels, and that exacerbates rather social tensions, even to the point where all of these increasing inequalities involve social frictions over things like friendship.

- 5.22** I'm not going to associate with certain people anymore. And marriages—I'm not going to allow my children to marry into certain families and so forth. And the idea that some people then, as our increasing aesthetic standards develop—there's better-tasting wine, there's better ways of dressing, there's better paintings, better statuary—we start to apply that to human beings, and we say there are some lucky few people who are the beautiful people with wonderful hair and straight teeth. Then there's the also-rans, and there are lots of people who are ugly as well.
- 5.23** That gets more exacerbated because we have now more explicit aesthetic standards that we are applying to ourselves and to other people. He harkens back to all of the social frictions that we have overapplied aesthetics. For example, if you think back in the day, when you think about even before humans, or we take the animal kingdom as a model, how they conduct their sex lives.
- 5.24** Basically, if you're a male of this species or a female of this species, if you are in the reproductive mood, so to speak, basically any male or female of the species will do for you. So you get together, you copulate, and then you just do whatever you do, and then the next generation is born. But how do we do it now in the modern world? It's the exact opposite of that. Even in earlier tribal times, young men and young women, they would get together pretty easily. Just about any girl in the tribe will do, or just about any guy in the tribe will do. They might have some standards and discrimination, but it's within bounds.
- 5.25** But in the modern world, we've got such elevated standards, and we're also so possessive now about our romantic partners as well that we're fighting all of these duels, we have all of this jealousy and heartbreak, and all of these things, the argument is, is much more complicated and much messier in the modern world, precisely because of the Enlightenment. It would be better for us, less social friction, if we took all of the heat out of this high romantic understanding and highly discriminating standards about love, sex, marriage, and so forth.
- 5.26** Also, in the modern world and the enlightened world, we talk about things like the American Revolution and people being free and you should be able to achieve happiness and do your own thing in your life. From Rousseau's perspective, that is morally dangerous because it is putting the individual first. It's all about me, my mind, my ideas, my values, my freedom, my goals in life. And that leads me from Rousseau's perspective, to be less caring about you and other people.
- 5.27** It's less about "we", I'm a part of the tribe, or I know my place in the class structure in society as an organic whole. Rather, it's, I want to do what's good for me, and you, I

don't really care very much about you. From Rousseau's perspective, the Enlightenment is a disaster. His question is, what we are going to do about the Enlightenment, now that it seems to be a fait accompli by the time we get into the 1700s.

- 5.28** He doesn't think we can just go back to old-fashioned times. So we need a new kind of modernism, but not an Enlightenment modernism. And we need some sort of intellectual and social revolution that's going to take modernity in a different direction. Now, one indication of Rousseau's tenor of mind is going to be on the issue of religion per se.
- 5.29** We've seen in the modern world a decline of religion, a decline of certain kinds of institutional religions, that traditionally had a monopoly on ideas and culture and so forth. So there's now a pluralities of religions out there. But we're also starting to see, as this new rational and scientific mindset takes hold, an increasing number of people who are not particularly religious at all.
- 5.30** Agnosticism starts to become an intellectually acceptable position. I don't know if there is a God or not, because I don't know if the proofs really prove it. It doesn't seem like it to me, and the counterproofs don't seem like it to me, so I'm going to adopt, say, an agnostic position, almost unthinkable four or five centuries before.
- 5.31** And an increasing number of people who are atheists, who will say, look, we look at the evidence. There's not really evidence of an existence of God. There is no proof. We've looked at the proofs for the existence of God. We don't think they work. And so, I think the rational thing to do is follow the evidence, follow the logic, and it's a hypothesis set to fail. And so we just have to get past that hypothesis and look at the world in more naturalistic and secular terms.
- 5.32** And this also is, from Rousseau's perspective, a problem. He's not an old-fashioned Catholic. He's not an old-fashioned Protestant, but he thinks that we are throwing babies out with the bathwater by getting rid of religion altogether in the modern world. So what we need to do is, not go back to the older forms of monopolistic institutionalized religion, but have a new kind of religion. But we have to take this question seriously.
- 5.33** Now, his book on education focuses on a young man named Emile, who goes off and has various adventures, and he meets all sorts of people. And one of the most appealing characters he meets on religious matters is usually interpreted as being closest to Rousseau's own thinking on religious matters, is a vicar from the province of Savoy; and in the course of his discussions with Emile and his role is to speak to a proper understanding and being with respect to religion.
- 5.34** It's not to say, institutionalized Catholicism as it is in France at the time, is the right way to go, or institutionalized Protestantism, as it is, say, in Geneva, which was Jean-Jacques

Rousseau's hometown, is the way to go. Instead, none of those traditional approaches are correct. Instead, we'll get the flavor by these remarks from the Savoyard vicar.

- 5.35 "I am resolved to accept as evident all knowledge to which in the sincerity of my heart I cannot refuse my consent." This is a cognitive point or an epistemological point. How do I decide what I'm really going to believe? We have the metaphor of the heart, put foursquare in front. I need to consult my heart, that is to say, to consult my feelings and feel what seems to be most compelling. That's the right way to go.
- 5.36 What we have is the Enlightenment philosophers say, we should go by evidence, we should go by logic, we should be mathematical. Everything has to fit into a test tube, and they leave feelings to the side. You don't believe whatever it is that you feel like. We have this dichotomy that has developed between reason and evidence, and feeling, passion, emotion—the heart versus the mind.
- 5.37 And Rousseau is arguing, if the trajectory of evidence, logic, mathematics, and science is to get rid of religion or marginalize it, that's the wrong road. I feel it in my heart, fundamentally, that that misses out on a vital aspect of what it is to be a human being. "I believe, therefore, that the world is governed by a powerful and wise will. I see it, or rather, I sense it." I see it literally, so that's why he immediately corrects it. But I have a sense. It's not an empirical five senses, but I have a sense, in that broader sense, that there must be a powerful will out there governing the world.
- 5.38 It's not going to be the orthodox Christian God or Muslim God or Jewish God or so forth. It's going to be much more generic kind of God. Now, I am an educated guy. I'm a philosophy guy. I've been arguing with all of these Enlightenment philosophers, and I've read everybody. I know all of the arguments back and forth. All of this reasoning gets very, very complicated. I don't know, ultimately, whether any of the arguments work or don't work.
- 5.39 But I want to say that that is not how religion actually does work. It's not the case that you start off in this neutral position and you just apply your brain to it, like a super scientist for a certain amount. At the end, God exists, pops out as the right conclusion. Oh, therefore, I believe in God. That's not proper religion. I don't feel like I am up to that. I'm not Isaac Newton. I'm not that mathematically, scientifically inclined. I don't think the vast majority of people are Isaac Newton, are destined to be Enlightenment philosophers.
- 5.40 The right kind of religion is going to be something much simpler, less rational, more feeling-oriented. "Suffused with the sense of my inadequacy, I shall never reason about the nature of God," you're not going to trap me and trick me into arguments about the existence of God, "without being forced by the sentiment of his relations with me." First, I might have a feeling, a sentiment about God, within that committed framework. Yes, I believe in God. I will reason about various aspects of it.

- 5.41 But reason comes after feeling, sensation, sentiment, passion, emotion first. We need to reverse the Enlightenment epistemology. Now, this then spawned a number of directions, Rousseauian philosophy. One of them did concern religion directly, to say, OK, fine. It seems like, as a result of the Enlightenment, religion is on the defensive. So if we are going to have religion, that means we need to go on the offensive against the Enlightenment and go against it fundamentally, with respect to its emphasis that everything ultimately should be rational and scientific.
- 5.42 One of the giant names in this newer, we're going to start calling this Romanticism, as a broad movement. I'll have some caveats about the word Romanticism a little bit later. But the idea of feeling and passion being put to the fore, instead of what they will see as cold, calculating reason to the fore. The same thing is going to then apply to religion and a new kind of religious theology.
- 5.43 We don't just go by institutional, some things that have been handed down by the tradition. We can't get there by reason, but there are certain core feelings that are powerful, and one should be able to go inward and realize something fundamentally true, and feel these things as deeply true about yourself. That is going to be the basis of your religious conviction.
- 5.44 So Schleiermacher's point is, this is one in particular, "The essence of religion is the feeling of absolute dependence." You ultimately get to the point where you've realized you cannot rely upon yourself. You are not an independent, self-autonomous agent the way the Enlightenment thinkers would like you. You are dependent. You are weaker, and that you need to accept this and not try to cover it up by all kinds of devices. "I repudiated rational thought in favor of a theology of feeling." So, belief in God, religious attitude, and whatever more formal theology that develops has to be based on feeling, not reason.
- 5.45 Next generation, Soren Kierkegaard, another giant in this tradition, who said, when he consults his feelings, he feels overwhelmed by the nature of the world. It's all bewildering, shocking, and frightening to him. "The whole of existence frightens me; from the smallest fly to the mystery of the incarnation, everything is unintelligible to me, most of all myself; the whole of existence is poisoned in myself, or my sight, particularly myself."
- 5.46 So this is not merely a feeling of dependence and weakness, that I am not an agent with some power and competence over my life, but cognitively, that it's just too overwhelming, scary, frightening. That's how I feel about the world. And this is completely alien from the way the Enlightenment thinkers feel about the world, because they feel that they can figure stuff out and they're in charge and they're competent. They're going to be captains of industry and so on. It's an entirely alien, feeling-based psychology.

- 5.47 As a result, I'm not even going to try to prove the existence of God, or entertain the idea that trying to prove and show evidence for the existence of God is the right way to go. Because if you go down that route, you're going to end up being an atheist. Reason leads to atheism, so we need an assertion of the exact opposite of that.
- 5.48 The incarnation that he mentioned—God, a purely immaterial being, became a material being. That makes no sense to me. That's a mystery. God is supposed to be spiritual, but he became a body. He's supposed to be eternal, but Jesus lived for about 33 years. That's temporal; that's not eternal. There was a virgin birth. That's unintelligible to me. There's also the Holy Ghost floating around there somewhere. I don't understand how that works. It's all completely a mystery to me.
- 5.49 But I believe it and I feel it in my bones, that this is true, significant, important, and the Enlightenment is hostile to that. So I need to be hostile to the Enlightenment. And he has these rhetorically wonderful formulations. We need to take reason and "crucify" it, not exalt it. Crucify reason, your own reason, and then your feeling-based epistemology will develop.
- 5.50 Christianity is absurd. You need to "take a leap into the absurdity." The Enlightenment philosophers will say, no, you take the evidence, you take a logical step, another logical step, you work your way. You don't take any leaps into whatever. And if something seems absurd, you don't believe that. You don't believe in absurdities. Contradiction is the big no-no for rational philosophy. Here, the absurd is the only thing that strikes the deepest chords in you, that seems to give you authenticity and meaning, so you need to take a leap into it. Bypass your reason, if you haven't already crucified it.
- 5.51 Now, out of this Rousseauian Romanticism, there was this development in religion and theology that came to be prominent, late 1700s and 1800s. What it's obviously going to mean is, if we are educators, and we are raising our kids and we want them to learn about the most important things, well, science education, technology education, this idea of self-confidence and agency and so forth, all of those immediately become suspect in our curriculum and in our mindset approach to education. We're needing to recast education in a different direction.
- 5.52 Now, another important offshoot of Rousseauian thinking was among those who were convinced that the Enlightenment had gone wrong in some way, but they could not get themselves to adopt religion as the right road to go down. And so, we see a movement of those who say that, instead of God or religion being salvation, where I'm going to find my highest passions and my highest feelings and something that I can make a wholehearted, authentic commitment to, it will be art that is my vehicle of authenticity and creation.
- 5.53 The arts is the place where it's not cold, calculating, rational science and everything is a laboratory test tube and so forth, but rather, I express myself. I try to create an alternative reality that is wonderful in beauty and filled with wonderfulness in various

respects. And so Romanticism, as that label is used in our history, stems from this era and often harkens back to Rousseau as one of its forerunners.

- 5.54 Now, a few examples here. This is Caspar David Friedrich, the Sea of Ice. One of the things that starts to happen in the modern world is, we explore all of the world, including the extreme parts of the world. We're partly interested in exploration for scientific reasons, sometimes for political reasons, sometimes for religious conversion reasons, but also sometimes we just want it to be an adventure.
- 5.55 How exciting and extreme and awesome nature is. I'm going to put myself out there on the edge, realizing that I might die as a result of this great adventure that I am undertaking. But nonetheless, I am living authentic life to its fullest. I'm not a lab rat, scientist guy, trying to be safe and understand the world in that world.
- 5.56 So this is The Sea of Ice, from some early Arctic explorations, somewhat fantastical exploration there. And it's all these jagged ice peaks, and you have the sense of the titanic majesty of nature necessary to force those ice flows against each other into this kind of position. But you might notice, if you look closely to the right side of the screen, there is a brownish thing, and that is a ship that has been trapped in the ice and is now bow pointed straight down. It's a dead ship now, and presumably everybody in that ship has died.
- 5.57 And isn't this exciting and dramatic and tragic? But this is life at its fullest, being willing to go out there and even risk your life in nature, the nature of the way it is. This also is life. This is a tiger attacking a horse, and the art is very dramatic and swirly, and you have a great sense of motion. But it is life at its most raw and elemental, the hunter and the hunted; and which one is going to survive and which one is going to die? You want not everything nice and neat, laboratory test tubes and categorical, logical tables; you want life to be passionate and even dangerous like this.
- 5.58 Or this one here, of men fighting duels over points of honor and insult. So someone has won, but someone is dying. And again, it is a great tragedy. But then we add the element that this has happened at the theater, which is a place of drama and excitement and creativity and imagination and alternative lifestyles as well. So the victim is still in his dress from the night at the theater.
- 5.59 Or we might say, the Enlightenment world and the modern world, the industrial world is making everything processed and boring, and industrial regimentation. We want to go back in time, to these romantic old medieval castles, lords and ladies, and a certain understanding of religion. And so we will romanticize the past and try to bring that romantic understanding of the feudal past into the modern world and live in it as a fantasy world.
- 5.60 Now, more formally, what we have in this understanding of Romanticism is a philosophical and religious rejection of the Enlightenment, or a philosophical and artistic rejection of the Enlightenment. The idea is that, if we're going to be artists, for

example, the Romantics, sorry, the Enlightenment seems to want us to make everything mathematical. We want to have perspective, and everything is exactly in the place, and all the colors have to be exact.

5.61 It's like the best artist is going to be the artist who can fool you into thinking that you're looking at something like one of those new photographs that are just coming into existence in the early 1800s. You're looking very carefully and critically, but there's no imagination, there's no passion, there's no excitement in that. So what we need to do, if we're going to be genuine artists, is reject entirely that approach to doing art. It's not about formulas, it's not about copying, it's not about taking living things and making them seem dead.

5.62 We are mechanizing the world. The industrial revolution is a problem. In the scientific world, we try to understand living things, but what do we do? We take them and we kill them. Then we take them into the lab and we dissect them. You take all of the life out of things, under the pretense that that's going help you understand how living things go.

5.63 And built into Romanticism of this flavor, typically, is a lot of dualisms, that you have to be either a spiritual person or a material person, and they're plumping for the spiritual. You're either going to be passionate or you're going to be a person of reason, but it's cold reason, and you want to be a person of passion. You can be spontaneous or you can be a person of routine. You want to be really free versus there are now all of these scientific laws and rules that we have to follow and so forth. If it comes down to a dualistic choice on all of those themes, the Romantics choose one side, seeing the Enlightenment as the other.

5.64 Now, some of the Romantic poets, as they are described, and I have had some examples from painting as well, and some sculptors. But here, the poets often articulate the philosophical themes more explicitly, that we should be poets and have a poetic sensibility, and that is put against this modern philosophical Enlightenment approach. Let's read John Keats, one of the early Romantic poets, who also had the Romantic good fortune from literary history of dying young, which adds an additional cachet.

5.65 Oh, he was such a great poet, and he died young. It's such a tragedy. Isn't that terrible? But really, isn't that wonderful, to die romantically? You know what we're talking about; we get that feeling. This is from a poem called Lamia. "Do not all charms fly at the mere touch of cold philosophy?" There you are. Philosophy is cold. As soon as philosophy touches anything, the charmingness of it goes. You get all analytic and you just—

5.66 "There was an awful rainbow once in heaven." It was positive. When we looked up in heaven, we saw this rainbow and we were filled with certain feelings of, wow, off. "We know her woof, her texture; she is given in the dull catalogue of common things." Now,

we're very impressed with Newton and new optics, and you take the prism and you can take white light and break it into the prism, and you can do the mathematical formulas of it. It's now a picture in your science book. That's what the rainbow is now reduced to, an illustration in a science book. All of the magic has gone out of it.

- 5.67** "Philosophy will clip an angel's wings." No more flying for you, angels. We'll bring you down to Earth. "Conquer all mysteries by rule and line. Empty the haunted air and gnomed mine. Unweave a rainbow, as it erstwhile made the tender-personed Lamia melt into a shade." It's a lot of elevated language, but philosophy is doing a lot of really bad stuff. We're getting that sense. It's taking all of the fun out of life, basically.
- 5.68** Even more pointedly, "The bald-headed philosopher had fixed his eye, without a twinkle or stir, full on the alarmed beauty of the bride, brow-beating her fair form and troubling her sweet pride." What should life be like? Well, it should be like a lovely young woman, full of life, full of energy, full of vitality. But now we imagine the Enlightenment philosopher, the way he looks at her. He's old and desiccated, and he doesn't really see her in the fullness of what she really is. She's just now an object of study, and that is terrible, what the Enlightenment is doing.
- 5.69** This is an international movement. There's Goethe, the great German poet, author of Faust. But one of the lines, this is not quite what Goethe believes, but one of his strong characters articulates it, and Goethe is personally in this area here. "Gray, dear friend, is all theory, but green is life's golden tree." Theory, gray, dead. The tree of life and life itself is green, so we have that opposition. "Feeling is all. Name is only sound and smoke."
- 5.70** Name is what we do when we put things in categories, we conceptualize, we slap a symbolic label on it, and we offer a definition of it. That's just smoke and sound. Feeling. How does the thing feel? How does the world feel? Not, can I put labels on everything and have nice structures of definitions?
- 5.71** The industrial revolution itself, celebrated by the Enlightenment philosophers, but look at what it's doing to the landscape—cutting down all of the trees, we're taking iron, taking the dead trees, making the runners on which the iron is going to go. And then we have this great billowing thing going along, scaring all of the animals. And we call it progress.
- 5.72** Famous line from the American Romantic Thoreau and his Walden, "We do not ride on the railroad. It rides upon us." The Enlightenment philosopher is saying, isn't it great that we invented these railroad things and we can travel everywhere and haul all of this freight and make life better? We are selling our souls. We're selling our living identity as part of nature, by letting the industrial revolution take everything over.
- 5.73** It seems like there's no more magic in the world. We don't believe in fairies and gnomes and so forth and all of those spiritual types of beings, even the dark ones, the demons and so forth, and then the good ones. All of that is, well, maybe we believe,

but even as moderns, if we're believers, it's kind of a half-hearted believer. We're much more rational and scientific, and the world has been dematerialized.

5.74 William Wordsworth, some more poetry to give the feel of this. "The world is too much with us." This is from early 1800s, and let's just plunge into the feel of the poem, as well as the words. "The world is too much with us, late and soon. Getting and spending, we lay waste our powers." Now, as moderns, we all want to get rich, we want to get stuff, buy stuff. The industrial revolution is empowering all of this, but we're laying waste to our powers. That is to say, we're taking our vital human energy and putting it into all of that and not into things that, really, we should be using all of our human energies for.

5.75 "Little we see in nature that is ours. We have given our hearts away, a sordid boon! This sea that bears her bosom to the moon, the winds that will be howling at all hours and are up-gathered now like sleeping flowers; for this, for everything, we are out of tune. It moves us not." If we pause there for a moment, you imagine you're on the coast of England, you're up on a cliff somewhere, and it's a semi-dark, partly moonlit night. And you are 500 years ago or 1000 years ago. And you go and you look out at the sky, you look out at the sea.

5.76 What is that experience like? It's sublime. It's awe-inspiring. It's a little bit overwhelming. It's a little bit mysterious. What's really out there? What's in the sea? Am I going to live or die? You're literally on the edge of the cliff, but you feel that you're at the edge of life and profound mysteries. But now we are modern. We've got all this science and math and logic and industrial revolution.

5.77 We go, we make a trip to the seaside, and we stand on that very same cliff; and we look out and we say, "Wow, the ocean is really big." But that's about it. It's impressively big. The sky is really big. But it is a shadow of the experience that people had 1000 years ago. That's Wordsworth's point. We can't experience it, and it's our loss.

5.78 "Great God! I'd rather be a pagan, suckled in a creed outworn; so might I, standing on this pleasant lea, have glimpses that would make me less forlorn, have sight of Proteus rising from the sea, or hear old Triton blow his wreathed horn." Instead of a modern, enlightened man, I would rather be a pagan with a creed outworn.

5.79 Now, a creed is a set of beliefs, even an untrue one, one that we know is not correct anymore. It's an old-fashioned way of looking. I'd rather be that guy than what I am now, so that when I stand on this edge of the cliff, I have this feeling that maybe this godlike Proteus creature is going to rise. That could happen. Wow. But I can't. I've been tamed, I've been domesticated, I've been artificialized by the modern world.

5.80 And another one, getting closer now to education in particular, Wordsworth still, the tables turned. "Up, up, my friend, and clear your looks. Why all this toil and trouble? Up, up, my friend, and quit your books, or surely you'll grow double." So this is our image of the scholar, a young scholar. At her desk, hunched over the book, that unnatural position, reading this dead collection of processed tree with abstract words

on it. He's supposed to memorize it. And he has to stay like that for a long time. Or the adult scholar doing the same sort of thing.

- 5.81** What's it doing to your body? What's it doing to the whole rest of you? Are you going to live your whole life in a library, et cetera? By contrast, "The sun above the mountain's head, a freshening luster mellow through all the long green fields has spread, his first sweet evening yellow." You get out of the school room, you get out of the library, you go out into nature, and it is magnificent.
- 5.82** "Books! 'Tis a dull and endless strife. Come, hear the woodland linnet, how sweet his music on life. There's more of wisdom in it." All of the wisdom supposedly in books, all of the wisdom out there in nature, waiting for you to see and experience—it's all out there in nature, not in books. "And hark! How blithe the throstle sings! And he no mean preacher. Come forth in to the light of things. Let nature be your teacher."
- 5.83** Not those guys in robes, standing at the front of the classroom. Not dead philosophers and others who've written books that are in dusty libraries. Not cold, sterile laboratories full of bottled chemicals and dissected animals. Real nature, that's where the action is. That's where reality really is. That's where full, authentic human experience can happen. And that's what education should be.
- 5.84** Now, if we were to extract this as a positive doctrine, obviously all of the anti-Enlightenment themes there, but there is a kernel of this that says we should think of nature as something wonderful, inviting, sublime, even dangerous, but embrace that danger. People in their lives should be full of passion. We should be open to the exotic. Life should be an adventure. We should be action beings, not scholarly, studious passive beings. And so this whole positive conception of the Romantics is enormously attractive to many people. The tricky part, of course, is that they can join that with an opposition to all of the Enlightenment themes, saying that it's an either-or choice.
- 5.85** Now, I mentioned earlier that there are some caveats about the use of this label Romantic. Some Romantics, particularly in the English schools and in the German versions of Romanticism, very much bought into the dualism. It's either the Enlightenment or it's the Romantic. There are other subschools of Romanticism that have not seen it as a dualistic opposition, but rather Romanticism as a kind of extension of the Enlightenment theme.
- 5.86** That finally, what we have, as a result of the Enlightenment, is some sense that we can control ourselves, control the world, not entirely, that we now have a certain amount of freedom that human beings did not have for much of the human history. We now have basic wealth and basic education that most human beings did not have.
- 5.87** So all of that foundation that was laid by the Enlightenment really has freed us and empowered us, so that we can go forth and sometimes do science, sometimes do art, sometimes perhaps do religion in a new modern way. But nonetheless, we can think of

our lives as a great Romantic adventure, working on the foundation that the Enlightenment has provided.

- 5.88** Now, that is a minority school within the broader Romantic framework, but there are many artistic expressions of it. For example, chances are good that we have all heard of Alexander Dumas. We know the Three Musketeers story. We know about the Count of Monte Cristo. These are ones that get made into movies, big-budget movies every five years or so. There's a conception here of, life has this great romantic adventure, but the idea is that we have to build that life of adventure on a rational understanding of the way the world works.
- 5.89** And so, passion and reason can work together. It's that we can be free and nonetheless integrate that with a more scientific, naturalistic understanding of the world. And what's important in most of these is that these are modern stories and that it's not about religion. The heroes are not religion. They're basically secular stories about human beings engaging in human value quests in a very romantic way. So the opposition from their perspective would be inappropriate, stultifying earlier philosophies.
- 5.90** We should mention here the other great French Romantic author Victor Hugo. Chances are good we've all heard of the Hunchback of Notre Dame, multiple comic book editions of this, multiple comic strip editions, some movie versions, and then Les Misérables. Both of them engaged in the issue of how, in the modern world, we can achieve great human passion without at the same time abandoning the idea that we also are rational and self-governing creatures and moral agents with our own destinies.
- 5.91** And then finally, I will mention Caspar David Friedrich, a German Romantic painter, and this one strikes me as emblematic of another kind of Romanticism. It's not the case that we are going to be, as moderns, cut off from reality. If you look closely at this guy, he's wearing very modern clothes of the times, the 18-teens. He is wearing clothes that are actually quite well-made. He's a man of some prosperity. He is, that is to say, a modern Enlightenment person, and he has the leisure time and a sense of adventure to go and stand at the top of the mountains and survey the vast sublimity of nature.
- 5.92** But you have the sense here that this is a human being, that we're raising human beings to the highest heights as well and integrating the best elements of the modern world, and it's not that dualistic opposition. Another artistic form is that human beings, as great artists, we can create our most perfect reality. We can take nature and even improve upon it in the light of the highest of human ideals, rather than expecting other agencies to be creating our lives and giving us what we are supposed to do.
- 5.93** So here we have a somewhat fanciful representation where the young man, who is the sculptor, he wants to make the best possible world. So he's making this statue of a beautiful woman. What the dream is, and perhaps he will be able to realize this dream, is he's so good at what he does. We make such progress that we can create this

idealized, perfect woman, and she will come to life and fall in love with me and everything will be glorious. We can create the perfect world, or at least a much better world, building on the tools that we have learned in the modern world.

- 5.94** But Romanticism, if we generalize it from it, does mark a direction for formal schooling and how, in less formal context, parents will think about their education. Certainly, there will be none of this old-fashioned strictness and discipline and one-size-fits-all curriculum that everyone is supposed to engage in.
- 5.95** The argument here is going to be that if we want to emphasize in our formal curriculum the things that are going to be the most authenticity, the most creative-enhancing in the human potential, and we want to emphasize more than strictly logical, rational, mathematical thinking, then what will come to the fore in our curriculum will be not the sciences, not business, money-making and so forth, not technology studies, but we will give pride of place to the arts, to the humanities, to literature, more specifically.
- 5.96** Those are going to be, at least in formal education, the best vehicles for training human beings to be the best that they can be. Also, it has implications for where or the place that education is going to take place. Instead of the idea of removing people from nature or removing people from real life and putting them in these artificial places that we call schools and separating them off, instead what we want to do is engage with reality, engage with nature, engage with real human social life.
- 5.97** So increasingly, in Romantic philosophy-inspired educational studies, you will see the beginnings of a de-schooling movement, or the idea that if we have schools at all, they will occupy a very small portion and perhaps only at higher levels for certain kinds of specialized studies. Instead, we will be embedded in the real world, we will be embedded in nature. As Wordsworth says, nature will be the best teacher, and it is all there. We all have the human faculties, particularly our passions, and perhaps supplemented by some reason, that is going to enable us to achieve the fullest. All right, we'll stop there.
- 5.98** I'm just wondering if the originator of the nature versus nurture was Locke because he was blank slate. No, you're not born with anything.
- 5.99** I'd say, in modern form, Locke is the one who is most associated with the nature versus nurture, but it does have longer philosophical antecedents. If you take, for example, the doctrine of original sin, that is a religious take on that same issue. It's saying that evil or our badness or the sinfulness that is human being is not a result of nurture, because it's with you from the beginning, so it has to be a part of your nature.
- 5.100** You find the same debates even among the Greeks, trying to puzzle out to what extent we are born a certain way, that would be our nature, and to what extent we are much more plastic, though they wouldn't use the word plastic, but much more malleable. Then you can basically take a human being and, as a result of external forces, shape it into whatever it is that you want.

- 5.101** So on that score, to go back to our first lectures from yesterday, Aristotle is pretty Lockean in the sense in that he was much more cognitively and morally on the not-nature side. The tricky thing is that nature versus nurture is, in my view, overly dualistic because there is a third option, which is to say that—because nurture typically means all of the environmental influences upon you, your physical environment, your social environment. Your nature is taken to be your biology and the expressed physiology and so forth as a result of that. But the third option is agency or some sort of volition or free will.
- 5.102** So, to what extent then am I, as I end up being, say, as an adult, a product of my biology? We have to say, well, partly it's that. Are you born a male and born with a certain skin color and probably some predispositions to this, that, or the other thing? That is part of my biological package. Some of it has to do with, I was born in Toronto, in Canada, and I went to these schools and lived in these neighborhoods, and so that was all influential.
- 5.103** But at the same time, you would say there are lots of people who, say in my family, who are basically the same biological package and we're all born in the same environment, went to the same school, same parents. But just take my brother and sister, for example, we're very different people in lots of respects. So you might say, well, it's also, you have your biological package, but you make choices about what you're going to do with it, or you have certain teachers, or your dad says something to you, or your mom sets this rule to you, and we react differently, depending on the choices that we made.
- 5.104** So, how much of it is environment or nurture, how much of it is nature or biology, how much of it is just free will and the choices that we make is going to be there. Now, on that score, Aristotle was much closer to a nurture/agency position. And Plato is quite explicit that he was much more on the nature side of things.
- 5.105** He has one of the myths that we did not talk about yesterday. He also calls it the myth of the metals, where he says, basically, you take human beings, especially if you focus on the psychological side of them, we have a rational element. We have an appetitive element, where we have needs for food and drink and sex and so forth, our appetites. But we also have passions and emotions, and some of us, we know as little kids, are more spirited; some kids are more laid back, and so on.
- 5.106** So he has this tripartite division of the human psyche into rational, appetitive, and spirited, as he calls it. But then he goes on quite forthrightly, and says every child who is born has a preset mix of each of those elements. So some kids are born with a really sharp rational capacity, and they're going to be the ones who really get math and logic and can think abstractly very easily. Other kids never going to get it. They have a rational capacity, but it's going to be much more limited. And so, no matter of educational training is going to change that.

- 5.107** And some kids are much more laid back, and you can give them books, and they will be happily reading their books and contemplating life's mysteries. Other kids, they can't sit still for a minute. They need to get out and do stuff and so forth. So they're much more spirited. But the mix that each of us has is individualized, but it's also set, and it's not a matter of training.
- 5.108** What that then means, when we come back to the education issues, is that for the Aristotelian approach, you're really emphasizing having kids raised in the right environment, the right parenting, right schooling, hiring the right kinds of teachers, and teaching children to use their agency well and to take responsibility for their agency, with the idea that you can take pretty much any kid and they can become proficient in just about anything.
- 5.109** Whereas for Plato, the idea that you are born with these preset mixes of all three things, the implication for education then really is an early form of tracking. Basically, some kids are born to be farmers, and they're born to be— we'll just use that as one example. If you're going to be a farmer, you don't have to be that smart to be a farmer back in the day, or you need some rationality, but also farming is boring, and you're doing the same thing over and over again. So kids who are not that smart and not that spirited, but nonetheless they are appetitive, they will be suited for farming and related fields.
- 5.110** Then you've got other people who are a little bit smarter, but they're not super smart, but they can't sit still for a minute. They need a lot of action. They're going to play a lot of sports. Those are the kinds of ones who are not going to be suited for the farming life, but they will be better, say, going into the military. Because in the military, you've got to be kind of smart, but not too smart, because you still have to follow orders by the people who are making the policies. But at the same time, in the military, you're going off and you're doing all sorts of things. You have to be willing to fight and so on.
- 5.111** Then the third group is, once you identify the young ones who are really quick, really smart, those are the ones you track into a different kind of education. They will be the philosophers, the politicians, the leaders of men-types of people. And hopefully, yes, the philosopher, the philosopher-kings. So that issue about nature versus nurture has huge educational implications right off the bat.
- 5.112** It's partly why, when you mentioned Locke, why he is so important in the modern world, because he's rejecting so much of the nature, obviously he thinks we have a body and we have a certain needs and we have mental furniture, but he does think that a huge amount of that is tabula rasa and agency.

Lecture 6: Strict German Reform

At a glance — Strict German reform after 1806.

- 6.1 1806, a hugely important year politically for the future of Europe and hugely important for education, its subsequent educational philosophy and practice. The first, a little bit more obvious. 1806 was the year Napoleon, in ascendancy, recently had crowned himself Emperor, conquering all of known Europe. North Africa, defeated the Prussians and the Saxons at the Battle of Jena.
- 6.2 It was a huge, huge battle. Almost a quarter of a million men on both sides involved in this altercation, and it ended up being a decisive battle for the French under Napoleon. Devastating for the Prussians, for the Saxons, and huge implications for all of the other countries as yet not conquered by Napoleon, that dwindling number of them.
- 6.3 Now, our story is going to focus more on the reaction of the Germans to this devastating defeat at the hands of Napoleon and, from their perspective, the damn French. How humiliating to lose to those guys. It's also significant, though, that this battle at Jena was remarkable because Jena was, at that time, one of the great centers of intellectual life in the world. When we do intellectual history, we're often struck by specific times and places when there seems to be an unusual constellation of genius happening.
- 6.4 So we go back to Athens and we say, what are the chances that Socrates and Plato and Aristotle would all be teacher-student to each other, of all the brilliant people at that point? And we go to Renaissance Florence and we say, Michelangelo is there, and Galileo is there, and Leonardo da Vinci is there, Machiavelli is there, and the Medici, and so forth. Wow, amazing. Jena, in the late 1700s, early 1800s, was one such place, a center of remarkable intellectual ferment and activity.
- 6.5 I have a list here of names of people, if you want to know who the who's-who of German intellectual life in the 1800s, these are some of the biggest of the big names. So Hegel spent time there. Fichte was there. Goethe was just up the road at Weimar, but his good friend Schiller was, and Fichte, and they spent a lot of time back and forth visiting each other. Schiller, the great playwright. The Humboldt brothers, one of the great reformers, bureaucratic in education, one of the great adventurers, his brother. The Schlegel brothers, Novalis Schelling, and so on.
- 6.6 All of them knowing each other and interacting to a significant extent in Jena at the time. And many of them were in Jena at the time the battle of Jena occurred. And one of them, for example, Hegel actually was brave enough to go out and see where the battle was to happen, and he was of two minds about this.
- 6.7 On the one hand, it's the French, and we think the Germans are pretty special. And so it's rather devastating to our national pride that Napoleon happens to be French and is

beating us. But at the same time, he can't help partly, this is driven by the nature of his own political philosophy, admiring the kind of man Napoleon is and recognizing him as a new world-stage historical figure on the order of Alexander the Great, Julius Caesar. And these are Hegel's words upon seeing Napoleon just outside, at the Battle of Jena.

- 6.8** "I saw the Emperor, this world-soul, riding out of the city on reconnaissance. It is indeed a wonderful sensation to see such an individual who, concentrated here at a single point, astride a horse, reaches out over the world and masters it. This extraordinary man, whom it is impossible not to admire." And so seeing Napoleon not only as an individual, not only as a representative of the French political military dominance at that time, but as a world historical figure, and seeing his significance in philosophical, world, historical frame.
- 6.9** His colleague, Fichte, Johann Fichte, he is the one we are going to focus substantially on in this, because aside from being a first-rank philosopher in the German pantheon of his era, he is the one, more than any other, who put his mark on German education, the philosophy of education, over the course of the next two centuries, such that now, when we talk about Prussia and the Prussian model of education, those of us who know education history and education, that stands for something and we know what it means. And Fichte, more than anyone else, is the one who made it happen.
- 6.10** He also was at Jena at the time Napoleon and the German troops went through, and his reaction was more remarkably negative. He recognized Napoleon and the French for what they were. And like Hegel, he recognized that this had been devastating for Germany. And it wasn't simply a military loss; this was a philosophical devastation that had occurred to the Germans, and to a large extent, it was the Germans' own fault.
- 6.11** They had gotten their philosophy wrong and they had gone down a wrong direction, and it was necessarily a philosophical reform that had to occur, along with it a moral reform, if Germany was once again to get its act together, get rid of the French, and find a way to find its own path forward. Now, that's big words in reaction to one specific spectacular event in Jena, but let's put some words to it.
- 6.12** Fichte is an important philosopher. I don't see him as a first-rate philosopher. He did not see himself as a first-rate philosopher, because what he wanted to argue was, there was a philosopher before him who had, despite centuries and centuries of philosophers trying to figure stuff out and argue and not seeming to get anywhere, there was a philosopher who finally had gotten it right. It was this philosopher whom Fichte saw as himself as a disciple of, who was going to take that high theory of that philosopher and apply it to German society and, specifically, German education.
- 6.13** One man distinguished himself as a pedagogical giant in his own right, due to the complexity and thoroughness of his plan. So who is this man who got philosophy right? It is Immanuel Kant. Now, Fichte, when he was younger and as yet unknown, at that point, Kant was recognized as the greatest philosopher in Germany, having published

Critique of Pure Reason in the early 1780s, Critique of Practical Reason, Critique of Judgment, the great three critiques that are the fundamental elements of Kantian philosophy. Kant had conquered the German philosophical world by the 1790s.

- 6.14** Fichte comes along as a younger philosopher and publishes a book called Attempt at a Critique of All Revelation. When it goes out, it doesn't have the author's name on it, but it is immediately snapped up, widely read, and basically everyone assumes that this is a book by Immanuel Kant, a fourth critique written by the great philosopher himself, Immanuel Kant. Kant himself then eventually said, "Wait, that's not my book." But he had to look at it and said, "This is really good."
- 6.15** Partly it was really good because the language was imitative of Kant's style of writing, and it took Kantian philosophy and was applying it to a new set of issues. So Kant gave it his stamp of approval or thumbs up, and immediately then Fichte became an element of the first rank of German intellectual life. And eventually, he was able to become a professor at Jena, which was a well-regarded position.
- 6.16** I want to say these two names here, Immanuel Kant and Georg Wilhelm Friedrich Hegel. Those are two philosophers of the first rank, and their influence has been enormous. In my judgment, the two most important philosophers of the modern world are John Locke and Immanuel Kant, and in many ways, they are opposite to each other. A little bit more about that to come.
- 6.17** But the point here is that Kant's philosophy is forthrightly being taken as a model for the cultural transformation, the educational transformation, that the Prussians, as the leading state within the broader German region, need to, if they are going to reform themselves in the light of this devastation of Napoleon.
- 6.18** I wanted to say, we're not going to say very much about Kant and Hegel, particularly right now, just because of their great importance. We have extensive coverage of both Kant and Hegel in the modern philosophy course. If you want the lectures on the basic Kantian philosophy, Hegelian philosophy, I would recommend those.
- 6.19** Political philosophy, we have a lecture on Hegelian political philosophy—absolutely important to nationalism and early, more authoritarian forms of politics coming out of Germany in our philosophy of politics series. And for moral philosophy, Kantian deontological philosophy or his approach to moral philosophy in the philosophy of ethics course. So I'm not going to presuppose that you know all of that stuff, but if you want to know the deeper roots to what Fichte is taking and applying to education, that is the place I would recommend that one go.
- 6.20** But a few words from Kant. Kant, he lectured on education explicitly for many years. Those are collected into his lectures on education. You can read those. But in his moral philosophy, and he sees much of education as applied character development and could apply moral philosophy, he has this to say about what it is to be a moral person.

- 6.21** "To behold virtue in her proper shape is nothing other than to show morality stripped of all admixture with the sensuous and with the spurious adornments of reward or self-love." We're raising kids to be good kids, to be virtuous kids. That's the first thing education is supposed to do, and that means one must teach children not to love themselves. Self-love is the big disaster here.
- 6.22** We do things not because we're going to get a reward or because I want to do them. We do them because they are the right thing and they are my duties to others, duties to society. They have nothing to do with the sensuous world, that is to say the material world. They are pure moral commandments. They don't necessarily come from a God. We might like to believe that they come from a God.
- 6.23** Kant can't quite believe in the traditional God, but he nonetheless believes we have to have faith in something higher and use our faith in something higher as a set of regulatory principles, including our moral regulative principles, and that is what we are doing in education. So one can think of it, this is a little bit too easy, as traditional strict religion without God, or God as a regulative hypothesis. That's what moral education is going to be about.
- 6.24** And the proper religious lesson we should require is that the most fundamental point of character is learning to obey, learning to be an obedient child. Just as religion in the Kantian tradition said it had gotten this correct, we have Adam and Eve, or we should say Eve and Adam, back in the garden of Eden. What was the first sin? The first sin was disobedience. And there's a reason why that is mentioned first, because the first then virtue is obedience. And that's the first lesson of the Bible. That's the first lesson of life. That has to be the first lesson of education.
- 6.25** Now, in his groundwork on the Metaphysics of Morals, that's the one I just quoted from, he has a footnote, Kant, to a Swiss philosopher of education, Sulzer, who in the generation before had been a giant in educational philosophy. And Kantian views on obedience, you can almost see them as taken from Sulzer, applied in this tradition, carries on. So it's important to read Sulzer in his own words, because if you are at all Lockean or even Rousseauian, you will find this quite shocking.
- 6.26** "Obedience," Sulzer writes, "is so important that all education is actually nothing other than learning how to obey. It is not very easy, however, to implant obedience in children. It will be quite natural for the child's soul to want to have a will of its own. And things that are not done correctly in the first two years will be difficult to rectify thereafter." So we got to get them in the first two years, because they want to do their own thing, and that has to stop.
- 6.27** "One of the advantages of these early years is that then force and compulsion can be used." Lots of force and compulsion on little kids to get them to obey. "Over the years, children forget everything that happened to them in early childhood." We'll pause right then. When are your earliest memories? We all play that game, right? And we go back

and maybe we can remember stuff from when we were four, three, but everything before then is gone. So Sulzer is saying, that's when we have to get the right habits of obedience into the kid.

6.28 "If their wills can be broken at this time, they will never remember afterwards that they had a will. And for this very reason, the severity that is required will not have any serious, rather, consequences." Now, that is strong. Kant picks that up, quotes Sulzer approvingly in his groundwork and in his works on education. And here's just a quick point from Kant's *On Education*.

6.29 "Above all things," and when a philosopher of Kant's stature says above all things, this is number one, "obedience is an essential feature in the character of a child, especially of a school boy or girl." Now, the contrast. I mentioned Locke versus Kant, both of them are giants, and both of them are taking philosophy, modern philosophy, in very different directions.

6.30 And the abstract philosophical themes, as they get increasingly applied all the way down, you see that explicitly, opposition on every major point in their philosophies of education and education practice. Let me just do a couple of juxtapositions of these two here. We saw earlier in discussing Locke and the development of modern liberal education, really emphasizing the liberal part of liberal education.

6.31 "Great care is to be taken that education be never made as a business to him, nor he look on it as a task. We naturally, even from our cradles, love liberty and have therefore an aversion to many things, for no other reason, but because they are enjoined us." Tell a little kid, do this, and it's part of human nature. And Locke says this is a good thing about human nature. We're naturally little rebels. We don't like to be told.

6.32 And education and good parenting will work with that. You won't crush that independent, want-to-do-their-own-thing spirit in kids. And then as a consequence of that, "I have often had a fancy that learning might be a play and recreation to children." So we naturally like to do our own thing. We like to play and explore and have adventures and so forth. Education should build on that.

6.33 By contrast, Kant argues the exact opposite. "One often hears it said that we should put everything before children in such a way that they shall do it from inclination." This is a slightly technical term of art. What Kant means by inclination is, you want to do it, or you are inclined by nature to want to do it. So, if you're putting things before inclination, we're saying, oh, here, you can play with this toy or you can play with that. You can read this book, you can read this book. Which one do you choose? It's a free choice for the child. What's your inclination?

6.34 And Kant is saying no. We hear that that's what we're supposed to do, and he knows that Locke is the foremost advocate of this person. In some cases, that's all very well. "But there is much besides which we must place before them as duty." Duty is more important than inclination. Duty, what you are supposed to do, not what you want to

do; what you are obligated to do, not what you are inclined to do. We have to be very clear that it's about obedience, obligation, supposed to, not any of those other vocabulary.

- 6.35** He's saying this is in part because that's the way life is. What kind of life are we praying for? Well, we have a lot of obligations when we're adults. "This will be of great use to them throughout their life, for in the paying of rates and taxes, in the work of the office, we must be led not by inclination, but by duty." You have to pay your taxes. You have to do what your boss tells you to do. Have to, have to, have to. Be a responsible grown-up adult. Do your duty, fulfill your obligations. That's life. Education is preparing for that kind of life.
- 6.36** The issue of punishment. I'll put another juxtaposition here. We saw in the lecture on liberal education and John Locke, light on punishment from that perspective. Kids are going to be kids, and set a good example. Kids will learn. They naturally will want to become good people if you show that good example. The important quotation again from Locke here, "I am very apt to think that great severity of punishment does but very little good; nay, great harm in education. I believe that those children who have been most chastised, seldom make the best men."
- 6.37** By contrast, when you read Sulzer, and then when you read Kant, it is the opposite. Punishment is necessary and frequent. It's partly that both have a somewhat modernized understanding of the doctrine of original sin. There's lots of badness in us that only punishment can deal with.
- 6.38** But Kant is quite forthright in saying, "Every transgression of a command in a child is a want of obedience." Every single one of them, and you can't let any of them slip. " And this brings punishment with it." And he goes into some detail about all the different kinds of transgressions and a recipe of different appropriate punishments that would fit those transgressions and so forth. You need to be up to speed and ready to do so.
- 6.39** Now, Fichte is a disciple of Kant, and he is much more politically animated than Kant ever was. Kant was mostly focused on fundamental issues, had some things to write about political philosophy, but less so. Kant has inspired Fichte philosophically, and what Fichte wants to do is say, what are the broader cultural implications, and particularly the educational implications, of Kantian philosophy?
- 6.40** In 1808, this is the year after Napoleon defeated the Prussians and the Saxons at Jena. Fichte is now animated. He has spent a year writing what will be a set of addresses to the German nation, as he writes them, and he delivers these as public addresses, and they take Germany by storm. Everybody realizes something has to do. We need to reform everything, but we need to have the right philosophical perspective. And Fichte is seen as the man to do so, because he is the foremost disciple of Germany's greatest philosopher, Immanuel Kant, and he is now a respected intellectual, respected professor, telling us what, philosophically, we need to do.

- 6.41 I want now to go into the text a little bit and read some of the addresses to the German nation to get the flavor of Fichte's ideas and their consequences for education. Now, Fichte starts with the immediate concern. This is being written in 1807, and every German is asking himself or herself, "Why did we lose the war against the French?" It wasn't just technological issues, it wasn't just strategy issues. There are character issues, religious issues, philosophical issues. The Germans have not prepared themselves philosophically for what they need to do.
- 6.42 The first reason is self-seeking, as he puts it. Much of the modern world we know has become enthused over these ideas of the rights of man and the rights of women—life, liberty, and the pursuit of happiness. My life matters. I'm not a subject or a serf. It's my life. That's a self-oriented thing. My liberty, my freedom. I'll do with my life what I want. Nobody is the boss of me. You can't push me around. My property, my happiness.
- 6.43 From Fichte's perspective, that is profoundly immoral. And all of the philosophical traditions that have elevated the self, elevated individualism, and talking about liberty for individuals is an absolutely corrupting moral and political phenomenon, and the Germans have absorbed a certain amount of this from further West.
- 6.44 "Self-seeking has destroyed itself by its own complete development." The more self-interested you are, the more selfish you become, the worse you become as a human being, the more society starts to disintegrate. So we have to nip that self-interest at the bud. Absolutely not, morally speaking. Now, more broadly, self-seeking is just part of a broader package that has developed. That is the Enlightenment. Again, the Enlightenment is the bad boy. "The Enlightenment of the understanding, with its purely material calculations, was the force which destroyed the connection established by religion between some future life and the present."
- 6.45 This is the second problem. Self-interests, the Enlightenment. What did the Enlightenment do? It undermined religion. It became a materialistic, physicalistic, we're only interested in science and what can be proved mathematically. Anything that doesn't fit that, we are going to set that aside. What that means is, religion becomes weakened, religion becomes set aside, and without religion, again, we become susceptible to corruption, from the Fichte's perspective.
- 6.46 Thirdly, a more narrowly political point, the weakness of government. What has been characteristic of the modern world has been limited government. We see in the American model, for example. Yes, we are going to grant government some power, but we are jealously guarding the liberties of individuals to do all sorts of other things, and the government is going to get some power, and only this amount of power, and we're going to explicitly tell the government in the Constitution and in the Bill of Rights that they can't do certain things. There are lots of things that they're not allowed to do.

- 6.47 And we take these various powers of government because we're suspicious, given the historical track records of governments using and abusing all of their powers, to take those powers of government and not let just the government have all of those powers. So we divide government into different branches. We say, you have these powers, and you can have these powers, and you can have these powers, but we have them in check against each other. So, we want governments to do a limited number of things, but really, we want to have a free society for individuals.
- 6.48 From Fichte, that modern political philosophy, coming primarily out of Britain, originally, has spread its way not only across the Atlantic, but who cares about the Americans? It spread its way across Europe, and it started to infect Germany as well. We do not believe in strong, powerful government as much as we should. And the reason for that is that, with people with too much freedom, they do bad stuff, they don't do their duty, and they don't get punished whenever they don't do their duties. Bad morality, bad Enlightenment philosophy, bad politics, those are the problems that Germany faces.
- 6.49 "It follows, then, that the means of salvation, which I promise," I being Fichte, "promise to indicate consists in the fashioning of an entirely new self." That's a very strong statement. That then is to say, it's not that we're going to be reforming and tinkering around the edges. We need to take the German self and refashion it from the ground up, an entirely new self. And where better to start than with infants?
- 6.50 Now, we've tried to do this before. Some philosophers and educators have indicated that they want to do this, but never successfully, never as "a universal and national self, and in the education of the nation." Now, that's a bit of, again, some technical terminology, a universal self. It's not an individual self. You are part of a more universal being, the German people, the German nation, and the German nation has its self, and you are part of that nation. And so we need to educate the whole of the nation.
- 6.51 And then, as he says here in a word, it is "a total change of the existing system of education that I propose," and that's the only way we are going to save ourselves. What we want to do, "by means of this new education, we want to mold the Germans into a corporate body, which shall be stimulated and animated in all its individuals by the same interest."
- 6.52 The problem with Western, from this perspective, ideas is we think of individuals and individuals having their own life, their own goals, their own dreams, their own agency. And we're just, with our political system, providing this free zone of space so people can go off and do whatever it is that they want. Hegel, sorry, Fichte, as philosopher, is saying that that individualist understanding of what it is to be a human being is wrong. You are part of a collective. You are part of a nation.
- 6.53 And first and foremost, you are not an individual, but you are rather part of a body, a corporation, that is decided by your nation. So we want to join all of the Germans into

one corporate body, one nation, and everybody will have the same interests, animated by the same interests. Not that you have your goals, you have your goals, your own dreams. We all have one set of values, one set of aspirations. We are all doing our part to realize those German aspirations.

- 6.54** Another problem has been that education has been piecemeal. Partly, this is the traditional way of doing things, not very much education for very many people. Now, we'd want to universalize and make sure that every German kid gets an education, gets the same education, so that they are on the same page.
- 6.55** "Nothing is left for us but to apply the new system to every German without exception, so that it is not the education of a single class, but the education of a nation. In this, that is to say, in the training of man to take real pleasure in what is right, all distinction of classes will be completely removed and vanish. In this way, there will grow up among us, not popular education, but real German national education."
- 6.56** Part of the past was, we had hierarchy, aristocrats, commoners, there's various classes and so forth. A lot of them didn't get much by way of education or they had very different kinds of education. We get rid of all classes. Everyone is equally German. All of the Germans are part of a corporate national body, to get molded with the same education, to pursue the same general value.
- 6.57** Now, Fichte is aware. Some people say this sounds rather harsh and strict, and what about the individuality of human beings? What about their own choices? And Fichte will have none of it. "I should reply," in response to those sorts of criticism, "that the very recognition of, and reliance upon, free will in the pupil is the first mistake of the old system." We do not rely on free will. In fact, we try to override free will, as Sulzer had argued previously.
- 6.58** You do things not because you want to or you will doing, or we're going to leave it to a matter of free choice and liberty, you do it because you have to, because you are supposed to. "On the other hand, the new education must consist essentially in this, that it completely destroys the freedom of will in the soil which it undertakes to cultivate, and produces on the contrary strict necessity in the decisions of the will, the opposite being impossible."
- 6.59** You want a kid to grow up where, when he is given a command, "You should do this; this is your duty," it's unthinkable for him even to imagine not doing what he is supposed to do. That will be a well-educated kid. "Therefore, in place of that love of self," including that self that likes to play, that likes to do its own thing, have its own idea, "in place of that love of self, with which nothing for our good can be connected any longer, we must set up and establish in the hearts of all those whom we wish to reckon among our nation that other love, which is concerned directly with the good, simply as such and for its own sake."

- 6.60 So any hint of the question, "What am I going to get out of this? Why is this good for me? How is this going to advance my goals or my dreams?" None of that. You do it because it's the right thing to do. You are motivated not at all by any considerations for their impact on you. You're supposed to do it. What about the parents in the home? Now, here we have a problem, and Fichte says part of the problem we have is, there are parents and there are lots of different ideas about what religion should be, what politics should be, what it is to be a good kid, etc. So parents are a source of diversity.
- 6.61 If we have kids coming into school and being taught one thing by their teachers, proper things, they're then going to be going home and unlearning them as a result of their parents' influence. How do we solve this problem? "It is essential that from the very beginning, the pupil should be continuously and completely under the influence of this education and should be separated altogether from the community and kept from all contact with it."
- 6.62 If we really are serious about raising children properly, we have to realize that children don't belong to themselves. They don't belong to their parents and their family. They belong to the nation, the German corporate body. As such, if we are serious about reforming education, then we have to remove all of the corrupting influences. As a practical matter then, boarding school for children, as soon as possible and as much as possible, until they are properly educated is an implication here.
- 6.63 What will it be like inside these schools? "It will be fitting, too, that the legislation in the schools should consequently maintain a high standard of severity," it will be very severe, "and should prohibit the doing of many things. Such prohibitions, which simply must exist and on which the existence of the community depends, are to be enforced in the case of necessity by fear of immediate punishment. And this penal law must be administered absolutely without indulgence or exception." Here are the rules. There are lots of them. All of them are to be followed exactly and immediately. And if there's any breaking of the rules, it will be punished immediately. No exceptions, no mitigating circumstances. That's necessary.
- 6.64 Now, another problem we have is that children vary as individuals. Some of them will take to the system of education more easily than others. Some will be more rebellious, some are quicker, some are better at some subjects than others. The point here is that, as part of a body, it's not just that your education is, I want to get as far in my education as I can. This is "we" education. I need to get my whole class to the right stage. We are the next generation of Germans. We together need to go on.
- 6.65 So that lesson has to be taught earlier. So that then is to say, if you are in a class and you've understood something, but some other kids in your class are struggling, you have an obligation not to go as far as you can, because maybe you're really good at math, but to use your time to get the rest of the kids up to your level, so that everybody is at the same level. You should be doing this as a matter of duty, as a matter of obligation.

- 6.66 "He," that is to say, the stronger students, "is to expect neither reward for it, for under this system of government, all are quite equal in regard to work and pleasure, nor even praise." You don't praise them if they do this. "For the attitude of mind prevailing in the community is that it is just everyone's duty to act thus. Under this system of government, therefore, the acquirement of greater skill, and the effort spent therein, will result only in fresh effort and work, and it will be the very pupil who is abler than the rest who must often watch while the others sleep, and reflect while the others play."
- 6.67 If we do the job well, we will then be graduating students who have mastered whatever material, but of course, they will be disciplined and obedient. Here's Fichte's description of the ideal graduate. "Its pupil goes forth at the proper time as a fixed and unchangeable machine produced by this art." Now, this is national education. It's collectivist education. It's very severe. It's separated from the rest of corrupting normal life. We hope to do this over the course of a generation, to reform German society in the next generation.
- 6.68 But we haven't said enough about religion yet. And Fichte is concerned to say that religion must absolutely have a place in good education. We also need to teach religion this way. "The pupil of this education is not merely a member of human society here on this Earth and for the short span of life which is permitted to him. He is also, and is undoubtedly acknowledged by education to be, a link in the eternal chain of spiritual life in a higher social order."
- 6.69 So what we need to understand is that when we are talking about life properly, it's not me as an individual, my body and my spirit. And it's not even all of us Germans, collectively organized into a nation, into a body, this generation; it is this organic community, and I am a cell in this larger German body. It also includes all of the Germans of the past and all of the Germans of the future. And they do not have material existence, but they do have spiritual existence.
- 6.70 And we are the inheritors of the German spiritual tradition, and we are the ones who will pass it on to the next generation of Germans. We are part of the entirety of Germany spread out over time. Now, why are we emphasizing the German thing so much? One of the things in early modernity, and certainly by the time of the Enlightenment, had been this idea of globalization. And globalization was partly an economic thing. We're trading with people all over the world.
- 6.71 But increasingly, a recognition that, in many cases, the things that we have divided ourselves up into—I'm a member of this class, and you're a member of that class, or this race or that race, or this ethnicity or that ethnicity—are somewhat artificial. Maybe what we should do is, as part of globalization, is not trade with each other. I make stuff or we make stuff, and we trade with people on the other side of the world who make some stuff better than we make stuff. And so we take their stuff and we trade for their

stuff. So we end up with a cosmopolitan culture, where everybody has the best stuff from all over the world.

6.72 But the same thing about ideas and culture, that we will realize some other cultures do some things better than we do, and we do some things culturally better than other cultures. But if we have this global cosmopolitan culture, we're all going to learn from each other, and we'll change our habits, and they will change their habits, and we'll all, of course, have some regional differences and individual differences. But essentially, we will become cosmopolitan, and national borders might not matter for very much. We'll all be, so to speak, citizens of the world, in some important sense.

6.73 And Fichte thinks that is another corruption, a specific corruption. There is no such thing as generic, abstract human nature. Instead, every culture has its own identity, every cultural group. And these are not superficial things, like how they dress and what they eat and this, that, and the other thing. These are fundamental self- and community-shaping things at the core of one's being.

6.74 So the Germans are unique. The French are different. The Poles are different. The Russians, the Turks, the Japanese, the Koreans, all of the Africans, and the Americans—you have to see them as perhaps very abstractly, generically human, but they are all quite different subspecies. And you can't take, say, an olive tree and an apple tree and expect to treat them the same. They have different roots, different fruits, different destinies; and so we shouldn't be trying to mix the two from each other.

6.75 But the Germans, one of the things that is unique about them is that they are uniquely suited for the kind of education that Fichte is proposing, a severe duty- and obedience-based education. Maybe you'd have a hard time selling that to some of those other cultures around the world, but this is what is unique and proper for the Germans. And this explains why the Germans have not lived up to their potential, why they have declined, and the more immediate obvious example of their decline, the fact that they lost to Napoleon and they are now under the thumb of the French and not able to live as Germans.

6.76 How did they become so weak? They bought into foreign ideas, foreign culture. They let too much stuff that is not native to Germany into their culture. The way he puts it is here. "All the evils which have now brought us to ruin are of foreign origin." British Enlightenment, French Enlightenment, broader European Enlightenment ideas, all of them had their sources outside of Germany. All of them were attractive to some German intellectuals and imported, and that was an infection.

6.77 If we use the organic models, the infection model is the right thing. We are allowing the German body to be corrupted by infectious elements from foreign cultures. We are not able to handle it. So what we need to do is rediscover our Germanness, what is unique about the Germanness. And he gives a few examples. Obviously, Kant is the

culmination of German philosophy for Fichte. There's no better example of German religion than Martin Luther.

- 6.78** So we harken back to the example of German religion and why the Protestant Reformation started in Germany with a figure like Martin Luther, who is, as Fichte puts it, "the soul of a German man. Luther was possessed by an all-powerful impulse, the anxiety about eternal salvation. This became the life of his life, the strength with which Germans embraced Lutheranism." We see it became very powerful as a religious movement in Germany. This is the "proof of the German earnestness of soul." There's a distinctively German way of feeling, thinking, being, doing politics, doing religion, doing everything. That's what we need to capture.
- 6.79** Now, Kant, here we mention Kant as the one who figured out philosophically what we need to do. There's a famous line from Kant's Critique of Pure Reason, where at the height of the Enlightenment, Kant is arguing quite explicitly that there are lots of things and lots of important things that are closed off to reason. The Enlightenment philosophers think that, with reason and science and further investigation, we can figure out everything. Kant is saying no, reason is severely limited.
- 6.80** And he mounts some powerful arguments that have been enormously philosophically influenced, saying that reason must be kept in its place. And one of the great benefits from the perspective of Kantian philosophy is that that means religion will be safe from continued attacks by those who want to say, "Let's argue, let's talk about evidence for reason." Religion is outside the bounds of the zones that reason has competence to investigate. It is a zone of faith.
- 6.81** And so from Fichte's perspective, the great benefit is that Kant not only got basic philosophy right, not only got moral philosophy right, he saved religion from the Enlightenment, leaving it legitimately as a zone of faith. And so what we can now do with the German people is say, "Don't worry about arguments for and against the existence of God. Don't worry about evidence. This is your faith. This is the German faith. Believe it, and we can make it part of our education and use it to assist in our strict moral and political education."
- 6.82** It's also the Germans, he thinks, are special. While he does argue that each culture has its own genius, its own way of being, its own uniqueness, and each culture is going to go its own way, there is still nonetheless something special about Germany, that God has chosen some people to lead the rest of the world along the right path. This is a theme that's increasingly picked up by German intellectuals as the 1800s go on. There's something unique about the Germans and their way of doing religion and politics that will make them world leaders, and Germany ultimately should aspire to its first-rank place on the world stage.
- 6.83** "As Germany sinks, the rest of Europe is seen to sink with it." He does speak to the conservatives. He's very forward-looking, and he doesn't want us to say we should

simply go back to the Middle Ages, to the good old days. But he does say there was one time when the Germans were special. We look back and we think about the Germans, and that was in the Middle Ages. And so there is one thing that we can learn about this history incorporated into our present, in order to make Germany the nation of the future.

- 6.84** And what he mentions is that, during this period, this period of the Middle Ages, what made the Germans great was precisely their moral philosophy, and he describes it this way. "It was their spirit of piety, of honor, of modesty, and of the sense of community." There's no pride, no self-aggrandizement, no Alberti Renaissance man, "look how special I am, and I want to win all of the medals for arts and sports" and so forth.
- 6.85** "Seldom does the name of an individual stand out or distinguish itself, for they were all of like mind and alike in sacrifice for the common weal." That's what we need to rediscover, what was great from German past. That's our moral and cultural framework with the supporting religion. We need to expunge all of the foreign elements, particularly the modern Enlightenment elements coming from the French Enlightenment and Voltaire and the British Enlightenment and all of the Americanisms and so on, in order to keep Germany pure, on track to become the nation of the future.
- 6.86** Now, moral education is fundamental. This, I think, is the emblematic quotation from Fichte on this. This is where we will start with children. "There's only one virtue, to forget one's own person." Forget yourself. "And only one vice, to think of yourself." Don't think of yourself. Think of others. Think of the community. Think of the nation. You exist to serve the nation; you are a part of its whole.
- 6.87** Now, I want to close by putting some more dates to this. At the time Jena is conquered, Napoleon and the French, they take over. They continue, of course, to go further east. They have aspirations for Poland and then Russia, ultimately. And so they are under occupation. But at the heart of this occupation, the Germans have listened to Fichte and a few like-minded other intellectuals, and realized they really need to divert resources to radically reforming the entire education system in Germany.
- 6.88** Part of that is a matter of reforming all of the primary schools, all of the gymnasiums, and what we now call high schools, and also reforming German higher education. There was a huge amount of resources and energy devoted to dramatically restructuring the entire German education system and developing a government-run bureaucracy to support all of this, and to raise all of the taxes that are going to be necessary to do so.
- 6.89** One thing that's worth mentioning is that, while Napoleon then went further east and Germany was largely in the rearguard, the Germans were getting underway. One of the great signal events was in 1810, when the University of Berlin was founded as a brand new, modern university in the capital of then-Prussia, and it's going to become the flagship institution for the new Germany. This is where the new generation of German

intellectuals, the philosophers, and all of the others, including those who are going to train the next generation of teachers, are going to be educated.

- 6.90** In 1810, Fichte is appointed as the very first head of the philosophy faculty at the brand new University of Berlin. And in the next year, 1811, Fichte becomes the President of the entire brand new university with great control over who gets hired, a huge budget, and becomes the most important person for German education in the 19th century. Also, we should say, for those of you who have some knowledge of Hegel, a few years later, headhunting was going on, of course, and Hegel was brought to the University of Berlin.
- 6.91** One more quotation to end. This is the Prussian flag. This education, as we put it in practice, we can imagine what Prussian education will look like. It is now famous, after the 19th century, but this is "real German national education." And those are the concepts that are fundamentally important and philosophically charged in each case. Education in reality, our organic reality, collective identity as beings, part of a larger spiritual history. We are German, we are unique German. It's the nation that is first, and education should serve those purposes.
- 6.92** To make some connections here, we're obviously politicizing education in a certain direction. And this Prussian German model of education, enormously influential in Germany and further east, as it happened over the course of the 19th century, when, say, Russians wanted their children to get the best possible education, they would send their kids to German universities, and those ideas would then come back to Russia. Much of the course then of Russian education can be seen as directly influenced by what was going on in Germany with Hegel, Fichte, Kant, and the others.
- 6.93** Earlier, we have seen Rousseau and his influence on Romanticism. This is a bit of an overstatement, but the character of French education, with his emphasis on feeling and fire and passion and much less structure inside the formal school system and spontaneity—again, an overstatement, but the influence of Rousseau on the course of French culture and French national education, and our then-corresponding stereotypical understanding of what French people are like, as we get into the 1900s and the 20th century, a lot of that has to do with Rousseau.
- 6.94** And the same thing with respect to the course of English and British and then North American, especially, education and the result of the influence of John Locke, the greatest of the British philosophers, with this emphasis on liberty, individuality, lots of free exploration, and the idea of an open liberal arts education.
- 6.95** So, while these things all start with philosophers doing very deep abstract stuff, John Locke, Jean-Jacques Rousseau, Kant, and Johann Gottlieb Fichte, and so on, the influence of these guys can be seen that, two centuries later, we do recognize the stereotypes of Romantic French education, Prussian German education, and free

American education. So philosophy really does have influence over the course of the centuries. All right, now we'll really stop there for today.

- 6.96** My job is an educator of a school district, and as I've been going through these six lectures with you, many of the parents reflect a lot of the philosophies. Some parents wanting very strict ways of educating. Some want Sally to have a well-rounded individual education. Some wanting us to focus on this area and that area, very much like the philosophers that we've talked about. How can we accommodate all of these parents in the way that they want public education in the future?
- 6.97** That's a good point. Partly, what I've been emphasizing has been education in a top-down sense, where you start with the philosophers and the big-picture strategic thinkers about education. The assumption there is that they influence then the smart, educated people. Sometimes they are influencing the politicians and the lawyers and the people who are going to become teachers, who are by and large educated. So you will see top-down instantiations of all of these. Well, you're right to raise the point that parents obviously have their own ideas. Some of them are well-educated, some are not particularly well-educated, but nonetheless, they come from cultural traditions and they have their own ideas.
- 6.98** What that then means is, particularly in a country like the United States of America, where we are emphasizing a lot of freedom, and we are a nation of immigrants, we have people coming from all over the world with all of those cultural traditions, that there's going to be sometimes harmony and sometimes conflict between the established models that are taught, say, in the formal schools of education, or the ones that are politically in favor at the time, and so are being pushed down from the top, versus what the parents themselves want to do.
- 6.99** So I would say, where we are going to go forward is more of the same, because what is happening in the schools of education has a life of its own to a large extent. A lot of tenured people with secure funding, and they have their own ideas, and they are going to be successful at training some generations of teachers in their own ideas.
- 6.100** We should expect that education will remain a political football in a Democratic-Republican society. Every time we change governments, every four or eight years, with a different educational philosophy as part of the mix, each new party will come in and do as best it can to change as much as it possibly can, again, from a top-down perspective. We will also, though, because we have a lot of freedom in our society, have lots of homeschooling network, lots of private schoolings. You're raising the issue of online education, which is the new kid on the block. All of these experiments are going to continue.
- 6.101** I think we will continue to be bringing in people from all over the world, with commitments to all sorts of ideas about what the best life is for their kids and how they want their kids to be schooled. So I don't think we can make any predictions except to

say that all of those cultural forces are strong, prominent, and will continue. We should expect them to just to—I don't know if I want to say, plunge into the fray, because it really is somewhat chaotic and overwhelming. But we should expect, as professional educators, that that's going to be the reality.

- 6.102** So part of it will be, this is going a little bit beyond your question now, then there's a lesson for us as individuals to say, well, partly what we need to do is work out our own thinking on education. We want to be people who are more informed and thoughtful and strategic about education. We've actually studied the history, we've studied the philosophy. We're making this our profession.
- 6.103** We want to be, in our own minds, very clear about what we think the best philosophy of education is, and then be people who are able to argue for it as good representatives of that tradition, but not to expect that we are going to be in a position to dictate. So part of what we then have to do is, realize all the stuff that we've just been saying for the last five minutes, that's the reality on the ground.
- 6.104** So part of my job is to do all of this in a very pluralistic, diverse, sometimes politicized, sometimes ideological, sometimes culture-bound traditions environment. So part of my job then is to say, how do I deal with all of these people? How do I deal with the bureaucrats who are just wanting to maintain their budget? How do I deal with the ideologists who want to impose their ideology? How do I deal with the parents who have their own ideas about education?
- 6.105** So that means I'm going to have to not only know in my own mind, but also understand these people and where they are coming from, so that I can have a good conversation with them. And I think that's going to be part of our professional responsibilities.

Lecture 7: Pragmatism and Progressivism

At a glance — Pragmatism and progressivism in the 20th century.

- 7.1 Let's move firmly into the 20th century now, and the year 1913. Important event politically is the election of Woodrow Wilson as President of the United States. That's a big deal and seen widely as a victory for an -ism, progressivism, both politically, intellectually, culturally, and the groundwork for this movement had been laid for a long time earlier in American cultural development.
- 7.2 Now, progressivism is a bigish tent, a hard word sometimes to define, but it does involve a number of individuals who are clearly in favor of progress. That's their self-descriptor. The kind of progress that they think is necessary for the United States, as a new and progressive nation, needs to radically change some of what are taken to be the traditional roots of American cultural politics, which emphasize individualism, liberty, limited government, all of which the progressive think are wrong.
- 7.3 Now, some of the constituent elements of this big tent that we call progressivism are going to be people who think we need a much bigger government, more government involvement. Some of them will go as far as saying that we are socialist, although socialist has a hard time gaining traction inside the American context. Sometimes they will argue that they are democratic socialists or social democrats.
- 7.4 Some of them will be more friendly to Marxism as a political philosophy, though back away from the idea that we need a violent revolution, as Marxism suggests, to take a culture to the next stage. Evolution, perhaps, in a progressive direction, towards something like a Marxist socialism, but maybe not. So the tent is large. There are a number of terms that get kicked around, but for labeling purposes, and partly for marketing purposes, you want to have a good-sounding label. And progressivism is the label that is most used for this approach to politics and cultural evolution.
- 7.5 Now, before he became President, Wilson was a professor at several universities, most notably as he ascended the ranks at Princeton University, where he was a professor and a widely published author of some important works. I want to give you just a flavor of his political philosophy, argued from his initial academic perch.
- 7.6 But in 1887, he wrote a piece called Socialism and Democracy, and he's arguing that, obviously, democracy has a good sound inside the American context. It's a democratic republic. But sometimes democracy is taken loosely as meaning, we're just going to leave political power in the hands of all individuals, who can think for themselves and make their own decisions about their lives. Just as a decision procedure for politics, we're going to give everybody one vote, and then we'll tally the votes, and whoever has the majority or sometimes the supermajority, that's what we will do for the next four years.

- 7.7** What Wilson and the other early progressives want to do is that we need to understand democracy in a more collective, that we collectively have the power. It's not power distributed to all of the individuals, but rather the power resides with the people conceived of more as a collective, not quite organic unity. And that then brings it much closer to socialism, which wants to prioritize the social over the individual. And again, socialism has a number of variations.
- 7.8** But what Wilson is arguing in this piece is that the proper kind of political philosophy, and the one that he is urging as an American, is socialism combined with democracy, maybe democratic socialism. Here's his statement here. Rather than the more limited, focused only on protecting people's liberty space, a conception of government that's been prevailing in mostly in American history, we need to go in a different direction.
- 7.9** "Government should lay aside all timid scruple and boldly make itself an agency for social reform." It's a job of government to improve society, not just manage the government to preserve people's liberty rights, as well as for political control. One indication of the first strategic movement that Wilson thought would be appropriate was to have the government take over charitable and philanthropic functions.
- 7.10** The traditional view in the American context had been, by and large, individuals can be self-responsible. You can run your own life. We have the expectation that, if people as individuals take charge of their lives, they have the liberty, this is the land of opportunity, people can make it. They don't need charity.
- 7.11** There's somewhat of a stigma attached to failing to the point where you need to take charity, maybe to get you back on your feet and so on. But that would be provided more by benevolent private associations, religious groups, extended friendship and family networks, that we can figure it out in a private, voluntary context, how to deal with the people who have bad luck or fall between the cracks in various ways.
- 7.12** Wilson wants to argue that that is an inappropriate model of society. It's too private, it's too individualistic. Part of a legitimate government function should be, say, its job is to look after everybody, including those who have fallen behind in various ways. And so, as he argues here, charity should not be private, but made an "imperative legal duty of the whole."
- 7.13** We're all part of the American republic. We belong to it. It belongs to us. The government is the agency of the whole, and its job is to look after everyone, not give them merely the liberty space to look after themselves. So we want to move government out of respecting, say, private charitable institutions and encouraging them to actively taking over charity, welfare functions, and so forth.
- 7.14** Now, he becomes the President of Princeton University by 1902, and a prestigious university; embarks on some reforms there, and then, as we know, a decade later becomes President of the United States. Now, progressivism is more focused on intellectual, social, and cultural activism, but it's often closely allied with pragmatism as

a philosophical movement. And pragmatism is usually seen as America's distinctively, or distinctively American contribution to the world of philosophy.

- 7.15** And most of the pragmatic philosophers will also be progressives in their social and political philosophy. And most of the progressives, while they were not necessarily philosophical or even thinking about these issues at that level of abstraction, would nonetheless draw upon the philosophical works of the pragmatists. I want to say a few words about pragmatism. The two most famous pragmatic philosophers in this tradition are William James, eventually to become professor of philosophy at Harvard, philosophy and psychology, and that's important, at Harvard University, and then John Dewey.
- 7.16** Now, what is this pragmatic thing? Again, it's a borrowing of a Greek word, "pragma", which means works, functionality, getting the job done. And a big part of it is this capturing of what we think of as the American spirit. That, yes, we look back to Europe and we draw a lot of our cultural institutions from Europe, and that's great. But the Europeans have a long history of being very speculative, and their philosophy is a little bit weird and head in the clouds. We Americans, we are down to earth. We are practical people. We are "get the job done" kinds of people.
- 7.17** We're not interested in speculations and arguing just for the sake of arguing and so forth. It doesn't really matter if we have tidy, perfectly worked-out theories. Sometimes the theories can be just pretty good, and the only real question is, does it get the job done? The functionality, the workability of a theory is the important thing. So we borrow this Greek word about pragma, deed, work, action, and we just elevate it to an -ism, pragmatism.
- 7.18** Our first question is going to be about any more abstract philosophical idea is not, is it true? Can we know it for certain? Are we absolutely sure that we are right? We don't care so much about that. What we're going to do is just take that theory, put it to the test, and see what the results are. If we like the results, then that theory is pretty good, and we will continue to use it, until it doesn't seem like it's working anymore. We're not too interested in the theoretical niceties.
- 7.19** So, a phrase to capture this is one that comes from William James, this phrase of cash value. So that's again, another nice Americanism. Because we like to make money, we like cash. Show me the money, put the cash in my hand. So stop thinking so much about abstract philosophizing in theory and speculation, and show what is going to be the practical result. Will this make money? Will it put cash in someone's hands? It's a little bit narrow, but will it get the job done? That's what we're into. What's the cash value of this theory?
- 7.20** William James was also interested in education matters. As a practical person, you want to say, well, my pragmatic philosophy, I want it to work. One of the ways we do work is by teaching the next generation the new ideas, the better ideas. He did give a series of

lectures that were later published, called just Talks to Teachers. Much of—they cover a wide variety of themes. But the important point here is that William James, in addition to being a philosopher, is a first-right psychologist of the late 1800s, early 1900s.

- 7.21** When you take courses in the history of psychology, almost always he will be cited as one of the founding fathers, so to speak, of this emerging discipline of psychology. It is at this period, late 1800s, going on into the 1900s, that psychology, for the first time in human history, is starting to put itself on a scientific footing, on a practical footing. We're studying the mind, we're studying the soul, but we don't use the word soul so much anymore. We're studying the psyche.
- 7.22** It's only, say, after Charles Darwin, who published Origin of Species in 1859, William James is a young man at this point, that we start to think of the human being as an animal, a very sophisticated animal, not as a being that is a special creation of God. We have a bottom-up, more material, physical evolution, including to a complicated brain, with all of the things that go on in our psyche. But we need to start thinking of those in naturalistic, physicalistic terms.
- 7.23** It's not until we start doing that, James and the early psychologists are arguing, that we're going to actually have psychology as a scientific discipline, or even as a serious intellectual discipline. If we think of the psyche as the soul or the spirit, and in some way, it's not a physical thing, it's not a spiritual thing, it's a spooky ghostlike thing that happens to come from some supernatural being by a process we don't understand; if we think that's what the soul is or that's what the psyche is, then we're not even going to try to understand it scientifically, because from the get-go, it seems like it's not something that we can understand.
- 7.24** It's going to be another kind of thing. And so psychology is not going to happen as a serious discipline. But James is arguing we have gotten to this point. We are making strong progress in understanding what the mind is, what its relationship to the brain is, how the human being as an entire organic unity works, and all of this then needs to reform or inform, and lead to reforms of how we educate young people.
- 7.25** We want to, sure, educate people's minds and their bodies, but now we're focusing on a scientific understanding of the mind, and that means we're going to be approaching things differently. So good teachers need to be good psychologists, is one of the big upshot points here, and that's got to be a part of one's ongoing training, keeping up with the psychological literature.
- 7.26** Dewey, as the John Dewey, the happy coincidence of being born in the exact same year that Charles Darwin published Origin of Species, so he is entirely, once that book and evolutionary theory and all of the debates over evolution versus creation are just on everybody's minds, and everybody is tremendously engaged in it, that is the intellectual climate that John Dewey grows up in.

- 7.27 And he wants to argue that, yes, we do need to rethink what we talk about, what kind of words we use, the language we use when we talk about the mind. Less talk about souls and spirits and talk more about brains and the psyche as a psychological phenomenon. But John Dewey also takes some broader lessons from evolutionary theory, because what evolution tells us is that the world moves and changes as a result of conflict.
- 7.28 It is males fighting other males, in order to have access to the females. It is all members of a species or most members of a given species, sometimes fighting amongst each other for access to the food. And as a result of changing environments, species change to adapt to those environments. As the individuals, members of the species fight amongst themselves for better access to the food or access to mates, they also, as a species, change and evolve over time. So change, conflict, evolution come to be the most important concepts that we need to think in terms of.
- 7.29 And this also requires a kind of intellectual revolution because forever almost, when we start to think big-picture about the way the world works, we have a different understanding. We start to think in terms of eternal principles, because God, who created the world, is a spiritual being and he is perfect. And whatever ideas he has and whatever principles he thinks the world should operate in terms of, including our moral principles, they are fixed, they are eternal, they are unchanging.
- 7.30 And so if we want to then understand our place in the world, according to that traditional view, what we are looking for are fixed and eternal universal principles that govern the universe's operation and provide the rules to us for how we are supposed to behave inside that universe. But if the world is not fixed, it is changing. If there is nothing that is eternal, then everything is, so to speak, up for grabs according to naturalistic forces. So we have to stop thinking in terms of eternal truths, absolute principles, and so on.
- 7.31 We have to start thinking in terms of change. What seems to be working in the here and now, with the expectation that that's not necessarily going to work in the next decade, the next generation, and so on. Everything is subject to change. If we think about even moral principles and political principles, we have the traditional view, well, God laid down these commands. They are absolute. They are binding to everybody across time. Same moral principles 3000 years ago, 2000 years and so on, on into the future indefinitely.
- 7.32 But if what is good is relative to the species, what enables them to survive and flourish, then that's a material understanding of what is good, not a spiritual one. But we also know that these species themselves are changing. And so what's good for the species is likely to change over the course of time. Those then become the operative words. We are changing beings. We're interested in flexibility. We're interested in adaptability. And that means those are the habits of mind and practice that we need to instill in young people to prepare them for a different kind of world.

- 7.33** Now, again, this concept of democracy that we saw Woodrow Wilson sounding. These two gentlemen are of the same generation. Dewey, in a philosophical sense, will also say democracy is not to be understood properly as an individualistic decentralization of power, and then we just tally up arithmetically people's votes to decide how we're going to act together in various contexts.
- 7.34** Instead, democracy should be seen not as a political device, but also as a moral principle. I am part of a people. I am a part of a collection of people, and my values are wrapped up in everybody else's values. And even my thinking and my beliefs are wrapped up in what other people think and believe, that democracy is a process not only of deciding political issues, shaping moral character, but shaping our very beliefs and our core identities.
- 7.35** How do we decide what's "true"? I'm going to put true a little bit in quotation marks here, because we're not talking about absolute truths or objective, factual truths that are out there independent of us. Instead, we think of truth as something that emerges out of a process, a social process, of discussion and debate and decision. That's the truth, for us, for now. Then we will continue the discussions and we might change our minds.
- 7.36** In the next generation, we have different truths, or truth has evolved into a different state because that is what society of that generation has decided. So it's a social process of constructing truth. Philosophically, this is very different from the idea of saying truth is something that we discover out there in reality and we absorb it into us. We are making the truth up as we go along. The proof that it's true is that it is working to achieve our goals in the here and now.
- 7.37** Now, Dewey went through a very famous and influential book on education in particular, *Democracy and Education*. I'm just going to give you a few quick quotations from this one, partly because, advertisement for other Peterson Academy courses, we have in the postmodern philosophy course, entire lectures on the fundamentals of pragmatic philosophy, and we take up Dewey and James there. So if you're looking for greater depth on the underpinnings of all of this, that is one place where one can go.
- 7.38** Also, we have a course in the philosophy of politics, where, less on the education issues, but other aspects of pragmatic and progressive politics, we explore those issues as well. But let's focus in on some key themes from this 1916 work. United States has not yet entered World War I. All of Europe is wrapped up in World War I. The United States won't enter until soon, but nonetheless, that war is also on everyone's mind.
- 7.39** But just to give you a flavor of Dewey's understanding of what human nature is, in this more collectivized, somewhat organic, democratic way, this is how we understand the individual. The fact is that individuals come into existence, they live for a while, and then they die. We start with that natural biological perspective. That's important, and it's sad for some of the individuals involved, but the more important thing is the group.

Dewey wants to argue that the proper understanding, historically, culturally, of what it is to be a human being is to be a part of a group, and it's the group that is emphasized.

- 7.40 "Each individual, each unit who is the carrier of the life experience of the group, in time passes away, yet the life of the group goes on." So when we look at individuals, what makes this philosophically pregnant is, we don't say, here is an individual with his or her own ideas and own goals and own life. You only have one chance at your life, and so pack as much as you can into it because, sadly, then you are going to die. We don't think of it that way.
- 7.41 The first phrase there, your purpose as an individual is to be "the carrier of the life experiences of the group." In me, as an individual, resides the experiences of the group, and I am carrying them along and passing them along to the next generation. Yes, the individual dies, but the important thing is then the latter part of that phrase, "the life of the group goes on," and that is our focus. Less individual, more group-focused.
- 7.42 This has direct implications for education. What are we trying to do in education? Well, we are trying to make sure that the group maintains its identity, maintains its existence over time. That means that we need to take the new individuals who are coming along and educate them, inculcate them, train them in the ways of the group, so that the group's ways of being carry on into the next generation.
- 7.43 The contrast, again, would be those more individualistic approaches to education. Here's an individual, a unique individual. I'm interested in helping this person realize their individuality in various ways. That's not what we are doing. "Education, in its broadest sense, is the means of this social continuity of life." Each individual, and we're calling them units more than individuals, and that's an interesting word choice, has to be initiated, as Dewey says, "into the interests, the purposes, information, skill, and practices of the mature members."
- 7.44 This is vitally important. "Otherwise, the group will cease its characteristic life." So from each society's perspective, the most important value is, this is how we do things. This makes us us. And the preservation of that into the next generation is our top value. Education serves that social purpose. A very strong understanding of what it is to be a member of a group, a member of a community. So just read this through. So.
- 7.45 "Individuals do not even compose a social group because they all work for a common end." So what does it mean to be a part of a group? So we might say, well, there's one goal, and we're all working to make that goal happen, so we're part of a community because we have this common goal. And Dewey says, OK, that's fine, because we know a lot of times people have their own goals and they're off doing their own thing, but you come together because you have a common goal. That's not enough.
- 7.46 "The parts of a machine work with a maximum of cooperativeness for a common result, but they do not form a community. If, however, they were all cognizant of the common end," let's just say we're all aware of this common end, "we're all interested in this

common end, and we regulate their specific activity in view of it, then they form a community." What we are going for is to take all of the individuals in our society and say there's one goal for this society, or one set of goals for society.

- 7.47** You all need to become aware of it. You all need to adopt it as your set of values, and then to contribute your efforts toward the realization of those values in community with everybody else doing the same thing. Then we have a community; and anything short of that is not yet a community. Now, that's a flavor of the philosophical background to pragmatism. We're going to start applying this to education.
- 7.48** I want now to turn to some other big-name intellectuals and activists inside the progressive tent, who are more or less pragmatic and explicitly so in their philosophical approaches. Most of them, I would say, it is fair to say that they are more pragmatist philosophically than anything, but they're not professional philosophers, so they might be a matter of picking some elements of it, but not necessarily buying into the whole package.
- 7.49** What they do have in common is a shared agreement, whatever the philosophical basis of it is, that there's a certain kind of social philosophy, political philosophy, and activism necessary, that we were all on that broad same page, to work toward.
- 7.50** Two names I first want to mention are Jane Addams, slightly younger than John Dewey by one year, but of the same generation. Slightly personal because she was a graduate of Rockford College, which is the institution which then became a university, Rockford University. I was a professor there for many years. And so Jane Addams is one of the big names at that particular institution. And then W. E. B. Du Bois, slightly younger, again, PhD from Harvard University, spent some time studying in Europe.
- 7.51** And their more particular focus is then to say, we are interested in, broadly, this pragmatic, progressive transformation, reformation of society, but we do have more particular interests. I want to use Jane Addams as a sub-focus on what do we think about Americans who are recently arrived at this country, immigrant Americans? We're a nation of immigrants. We've had certain education or immigrant philosophies, immigrant policies in place by the time we're getting into the 20th century.
- 7.52** We don't think those are working or appropriate. We need to use progressive ideas to transform how we deal with immigration. And then, using Du Bois as an example of philosophy and activism with respect to Black Americans, the argument here being, obviously, we have lots and lots of Black Americans. They are Americans. They're part of this country. But we don't think that America has done a good job with respect to this important element of its society. So we need, again, transformation or reformation of how we are dealing with Black Americans in a progressive context.
- 7.53** Jane Addams is most famous. She won a Nobel Prize in 1931, a little bit later when she was older, for her work at the Hull House in Chicago, an institution that she founded. The idea here was that, right now, we have all of these immigrants coming to America.

We are a nation of immigrants, and we take some measure of pride in taking in people from all over the world. But the prevailing philosophy has been that you come here and you're on your own.

- 7.54 We'll provide you the liberty space, we'll provide you the opportunity, or it's a land of opportunities. There will be opportunities, but it's going to be up to you to make your way in this new world because we believe in freedom and self-responsibility. We don't believe in treating people as charity cases and so forth. And what Addams wants to argue is that that experiment has failed. That has been ill-serving immigrants, and what we need is something much more robust. And so Hull House was set up as an assimilation institution.
- 7.55 So many immigrants arrive in the United States. They don't have any money. They don't know anybody. They don't speak the language or they don't speak the language very well. And so they're just thrust into it, and it's very difficult for them to be able to do so. So the idea of Hull House was, it will take in immigrants. And initially it was privately funded, but it will help them assimilate—formal language lessons, learning the cultural ways of America and so on.
- 7.56 So it was a kind of halfway house between your old country, whatever that was, and then by the time you spend a certain amount of time in the program, you can go out and have a much greater chance of succeeding in society. But the idea here is that that older laissez-faire, you're-on-your-own model is out. And what we need to do is be more paternalistic, more maternalistic, and help these people on their way to becoming Americans in the full sense.
- 7.57 As an example of the kind of person that Jane Addams has in mind, I wanted to use Sidney Hook, another one of the big-name philosophers in the 20th century, who was a pragmatist philosophically. In graduate school, he was a student of John Dewey, so he got his PhD under Dewey's work, and so he is a pragmatist philosophically. But like many or most of the pragmatists and certainly the progressives, he was very politically active and was part of the broader ambit of progressive politics, so to speak.
- 7.58 When he was younger, he was a communist, so it was something stronger down the socialist road than the progressives were. But after the terrible events that were happening in the Soviet Union, all the stuff that Lenin did, all the stuff that Stalin was doing, he broke from formal Marxist communism. He broke from the idea that, in any sense, the Soviet Union is a moral or progressive model, but nonetheless remained a man of the left, of the pretty far left, but started to think of himself as a socialist, but more of the democratic type of socialist. So think Woodrow Wilson, think of John Dewey's writings, and so on.
- 7.59 But the point and the connection with respect to Jane Addams is that he was a first-generation immigrant to this country. And you read his biography, he's a very colorful, interesting, he had quite a life, but he gives a description of what his early life was like

in New York City in the first part of the 20th century. And this is exactly the problem that Jane Addams wants to try to solve.

- 7.60 "The physical conditions under which we lived were quite primitive. On Locust Street, where we lived, the toilets were in the yard. In other tenements, they were shared with another family." Oh, that sounds like fun, doesn't it? "All were heated only by a coal stove and boiler in the kitchen. Gas provided illumination, and the most common means of suicide." High suicide rate among first-generation immigrants.
- 7.61 "We froze in the winter and fried in the summer. Vermin were almost always a problem. The women worked like pack horses. Their work was never done. My father left for work before we arose, and he returned from work when we were ready to go to bed. More than once, he would fall asleep at the dinner table with the soup spoon in his hand, poised in the air." Imagine how tired you have to be, right, to do that.
- 7.62 So this is life for young Sidney, first-generation immigrant in the land of opportunity, and it doesn't seem like a very pretty picture. How do we solve that problem? And Addams wants to argue that it is progressivism that is going to provide the right kinds of moral, social, and political principles. Democracy properly understood.
- 7.63 "As the acceptance of democracy brings a certain life-giving power, so it has its own sanctions and comforts. Perhaps the most obvious one is in the curious sense which comes to us from time to time, that we belong to the whole, that a certain basic well-being can never be taken away from us, whatever the turn of fortune."
- 7.64 We belong to the whole. We all have obligations to all others to look after each other. I'm to be looked after by everyone else. I have obligations to look after all of you. So, the fact that there are thousands and thousands of young Sidney Hooks running around in the streets on their own, by and large, that is unacceptable. It's immoral. We need social reform and we need political reform to solve that problem.
- 7.65 Du Bois, a highly educated young man, received his PhD from Harvard University, went on to do a significant amount of extraordinarily influential writing, very clear and articulate writer of prose, and animated by a certain amount of moral passion. *Souls of Black Folk*, published in 1903, this is now 40 years after Abraham Lincoln's emancipation proclamation. The war is over, we're a generation past it, but it is still obviously on everyone's mind. The legacy of slavery, and the fallout from the great battle over slavery, largely over slavery, is still with especially black people in the early part of the 1900s.
- 7.66 Has America, with its live and let live, liberal, do whatever you want, you're on your own, self-responsible philosophy, solved this particular problem? The argument here is that, no, it has not; that what has happened is that America has evolved, particularly for Black Americans, into a dualistic nation. We are said that we are Americans, but not really. And certainly we don't feel like we are Americans in the full sense.

- 7.67 "One ever feels his two-ness—an American, a Negro, two souls, two thoughts, two unreconciled strivings, two warring ideals in one dark body, whose strength alone keeps it from being torn asunder." Imagine being in that circumstance. You are born of slaves, or slaves in your family. And even if you weren't born of slaves or have slaves directly in your family, how you are treated by the vast majority of other Americans, there's a hypocrisy built into the current system, and that just tears people up inside.
- 7.68 It feels like you want to, and it's only the strength, the character strength of any given individual Black person, who stops that from tearing that individual. It's even more difficult for Black Americans to become Americans in the full sense under the existing system, particularly in the south, of course. Du Bois writes this of the south. "The south believed an educated Negro to be a dangerous Negro." That's an interesting connection. Knowledge is power, education is empowering, but education in the wrong hands, from the perspective of many people, that's dangerous.
- 7.69 The implication then is that we don't want these people to be very educated. There are great cultural and political pressures still in the 20th century to keep them in that less dangerous, uneducated state. And the south was not wholly wrong. As a general point about the human condition, "For education among all kinds of men always has had, and always will have, an element of danger and revolution, of dissatisfaction and discontent. Nevertheless, men strive to know."
- 7.70 So this is Du Bois, in elevated language, putting America on notice. We are going to get educated. We will educate ourselves. We will become more dangerous. This might even amount to revolution. The words are in the document explicitly. And how are we going to manage this? Making progress to the idea that all Americans should be equal and part of what it is to be America. We have this problem to solve, and we have not solved it at all well.
- 7.71 Du Bois, a little bit later in his autobiography, wrote this about a decisive time period in his life. "From my 30th year on, I have increasingly regarded the church as an institution which defended such evils as slavery, color caste, exploitation of labor and war." So, as a progressive, we have to look back at some of the old traditional institutions and their track record. Du Bois is not a religious man, certainly not in the old-fashioned sense, but he sees religion and most of the established institutionalized religions as obstacles to progress, particularly as obstacles to the progress of Black people.
- 7.72 They have been, many of them, again part of the historical record, perfectly fine with the institution of slavery. For centuries, they never said anything about slavery being particularly problematic. There are biblical passages that seem to sanction it. Racism, as we get further on into the modern world and people start to mix, the church has not been anywhere close to being on the front lines of saying racism is wrong. Maybe it's a little bit "Johnny come lately," getting involved in that. But the track record is terrible.

- 7.73** Religion is and has been a conservative, traditionalist, backward-looking institution, and progress means we need to shunt it to the side. On the positive side, Du Bois also strongly attracted to communism, to Marxism as an American. Then he says, one of the great things then about the Soviet Union and why I am a fan of what's going on in the Soviet Union, or was a fan at this period in my life of what's going on in the Soviet Union, was that it was based on a materialistic philosophy, and it was doing as much as it possibly could to suppress religion and shunt it off to the side, as an officially atheistic political system.
- 7.74** "I think the greatest gift of the Soviet Union to modern civilization was the dethronement of the clergy and the refusal to let religion be taught in the public schools." An important reform then for American education, by implication, will be, we have to get religion out of the schools entirely. Politically, we already have a formal separation of church and state—first amendment, freedom of religion, and so on. But nonetheless, religion still is implicitly, in many cases explicitly, part of the curriculum, and that is something that also then has to go.
- 7.75** I want to put this phrase "social justice" out. It's a phrase that has a longish history, but certainly by the time we get to the late 1800s, on into the 1900s, progressives are part of the movement that, systematically, in its language, in its rhetoric, stops talking about justice and start substituting the phrase social justice. And the challenge they have here is that justice is an old and venerable concept, but it is too individualistic.
- 7.76** Justice in one of its manifestations is, you do something bad, well, you pay the price. We don't pass the paying of the price onto other people in your community, in your family or whatever. You do something good, you get the rewards for that as an individual. By contrast, social justice is a more communal concept. We need to socialize the concept of justice.
- 7.77** So we might then say, with respect to charity—and we don't even want to use that word charity anymore because of the baggage that is attached to it—if we think of the standard act of charity, suppose I'm not doing very well and I come to you and I feel a little bit ashamed, right? I say, "Can you help me out? Can you give me a hundred bucks to get through the month?" And you might be a nice person and you've got a hundred bucks to spare. And so you'll say, "OK, yeah, sure. I will help you. I'll give you the \$100."
- 7.78** But there's a little bit of, you're up here and you are reaching down to help this other person down there. And so it's a little bit humiliating for me and it's a little bit elevating for you. Now, you're a nice person and you're not going to say, "Oh, sure, I'll give you \$100." You're not going to do anything like that. You will be nice about it. But nonetheless, built into that is this slightly humiliating, shameful thing. I should be doing better in my life as an individual.
- 7.79** So, you're not giving it to me as a matter of justice, because I haven't earned that \$100. If I had created some product and I was trading with you \$100, we would say that's a

fair trade, a kind of economic justice. And we can both shake hands, or we can both feel proud and good about the transaction. Nobody is humiliated. We're dealing with each other as equals. Charity conceived this way is a little bit unequal, and it seems to have that old-fashioned "some people are higher, some people are lower" element.

7.80 What we want to do is take that out of the equation altogether and say we need a huge amount of wealth transfer, just to focus on money issues, from people who are doing better to people who are doing less well for whatever reason. But we don't want to call it charity anymore because that is humiliating to this group over here. It puts these people up on a pedestal, and then somehow they're functioning like the aristocracy, handing out favors to the people who are lower on the social.

7.81 Everyone is equal. We all belong to each other. So when we are transferring wealth, it's not an act of charity, it's an act of justice. It's an act of social justice. It takes all of the stigmas out of it. So this is one aspect of this concept of social justice. The core of it, philosophically, is less individual, more social, as the adjective right in front announces it. And we need to recalibrate all of our moral concepts and many of our political concepts in line with social justice.

7.82 Jane Addams, again. "In this effort toward a higher morality in our social relations," it's a moral issue, not this old-fashioned freedom and individuality morality, a higher social morality, "we must demand that the individual shall be willing to lose the sense of personal achievement."

7.83 I don't make myself. I don't earn my own way in the world. Whatever I accomplish, I don't take pride in it, in myself. I need to lose that sense of personal achievement. We make each other. I was made by you. I am contributing to the making of you. "And shall be content to realize his activity only in connection with the activity of the many." That's democratic, that's social democratic, that is social justice.

7.84 Shifting to Henry Adams, another huge name for self-published, a book called *The Autobiography of Henry Adams*, another very good writer, interesting life. It became a bestseller after his commercial publication and won Pulitzer Prize. As someone who's also extraordinarily interested in the leading ideas of the time, pragmatic ideas, progressive ideas, and how those can and should transform education.

7.85 Now, he was a pragmatist, and you see lots of pragmatic elements in the book. Put this on your reading list. It's an important part of American history, and certainly for professional educators to pay attention to. What is philosophy all about? Well, this is the reputation that philosophy has after 2000 years of philosophy. "Consists of, chiefly of, suggesting unintelligible answers to insoluble problems." Now, that's amusing. OK. Hurts my feelings a little bit. But what's the point of philosophy, if, for two and a half millennia, nobody's got the answers? Everybody disagrees with each other.

7.86 Philosophy is, whatever it is that it's doing, it's not working. It's not achieving whatever goal it's set for itself, which was to yield truth and certain answers to various things.

Let's put philosophy on the back burner. Let's not think too much about philosophy. If you're going to be practical, don't spend all of your time with your head in the clouds. But also, don't worry yourself too much about inconsistencies and outright contradictions. Philosophers almost always like to say, no, we have to get the truth; we have to get things right.

- 7.87** A contradiction is always a sign that you've made a mistake in your thinking somewhere. That means you have to go back and do some rethinking, to figure out where this contradiction stems from, fix your thinking, right, and so on. As a pragmatist, there's a strong element of impatience here. Don't worry about it so much. As long as whatever theories or principles you're working with seem to be getting the job done, they are workable, go with it.
- 7.88** "In practice," this is the thing, in practice, not in theory, "such trifles as contradictions," that's a trifle, "in principle are easily set aside." Just ignore them. "The faculty of ignoring them makes the practical man." Spend too much time in your thinking, you are going to get ensnarled in all of this weird stuff, and you're probably going to contradict yourself. Then a year later, you're going to change your mind. It's a waste of time. Just get the job done.
- 7.89** As for education then, when we get down to practical reality, looking at the way education is done in America at this time, early part of the 20th century, "Nothing in education is so astonishing as the amount of ignorance it accumulates in the form of inert facts." Now, I've highlighted inert facts here. So the idea is somehow facts are stable, long-lasting, enduring. It's another little pragmatist statement. We should expect facts to be evolving and changing.
- 7.90** And any teacher who's basically just reading to you out of the same textbook that he or she was reading 40 years ago when he or she started his teaching career, and that's the same one that his teacher was reading to him, that's a sign of an education system gone wrong. Things are constantly changing. Things need to be constantly updated. And as for how we do it, "The chief wonder of education is that it does not ruin everybody concerned in it, teachers and taught."
- 7.91** Now, as illustration for this, I want to go again back to Sidney Hook, who, in his wonderful autobiography, talks about his schooling. Even though he went on to be one of the leading philosophers of the 20th century, the schooling just sounds brutal. Public schools. "The discipline was exacting. Our teachers were little more than martinets." So you just imagine a person walking directly with a stick ready to hit the bad kids with, in a semi-military uniform, if you can get away with it. We know that type.
- 7.92** "Our teachers were little more than martinets. We had to sit erect, with our hands clasped on the edge of the desks or folded behind our back, in absolute silence." Children should be seen and not heard; we know that saying. "Everything was done at command, according to a rigorous and rigid schedule."

- 7.93** Now, I want to pause at this point here. One of the interesting things about American education is that, as America started to rise in power and wanted to have a stronger sense of its own prestige, it realized that it did not have yet first-rate higher educational institutions. So as with most developing countries, the practice was, particularly among those who could afford it, to send their kids to the best places in the world to get higher education. And that almost always, in the American context, has meant sending your kids to Europe.
- 7.94** Now, early in American history, the place you would send your kids would be to Britain, for higher education. Maybe they could spend some time at Oxford or Cambridge or Edinburgh and so on. France had its moment, where it became important to send your kids to France, which was then seen in some ways, particularly culturally, as superior to the English.
- 7.95** But by the time we get to the middle part of the 1800s, the reforms that Fichte, Hegel, and the others had instituted in German education had a vast amount of resources that had been poured into higher education in Germany, meant that German universities were something to behold, and they were starting to be recognized as the most important universities in the world. So you started seeing a significant shift of parents sending their kids not to England, not to France, but rather to Germany, for finishing school, for higher education.
- 7.96** And then those bright young people would come back to the United States with German philosophical, German educational ideals. This happens to coincide with a big wave of immigration from Germany, supplanting much of the bigger earlier waves of immigration from Britain. Italy was also a rising immigration nation as well. Partly what we see here is the impact of the Prussian model of education on America, the other side of the Atlantic, with duty and discipline and silence, and sitting erect, and the guy is ready with his stick to punish you as soon as you say or do the wrong thing.
- 7.97** "The slightest infraction of proper conduct—a whisper, a paper dropped on the floor, a shove, a pinch, " this one sounds more fun, "or the dipping of a girl's braid in the open inkwell—evoked withering sarcasm from the teacher, denigrating scolding, and corporal punishment—whacks with the end of a twelve-inch ruler on open extended palms." All right, so that is the kind of education Sidney Hook experienced, speaking for, again, tens of thousands of people in the American context.
- 7.98** That is what Henry Adams is talking about. This just destroys everybody involved, teachers and students alike. That dramatically needs to be reformed. The kind of reform we need is progressive education informed by pragmatic philosophy, and I would summarize it in this kind of phrase. If you are a teacher or thinking about becoming a teacher, what is your job? What is your mission? You are a transformation agent. We are trying to transform society, transform the next generation, for social justice in a rapidly changing social and technological world.

- 7.99** No stability, eternal truths. It's not a matter of trying to reach into the individual, to help that individual develop their inner potential. It's more social. I am thinking in terms of broader society with a certain social/political set of outcomes that we are trying to maneuver this generation into. Now, that's the teacher mission.
- 7.100** If we can join that with the progressive political mission, as announced, say, by President Wilson, then we will have huge amounts of resources by a government level, poured into the education space, along with a retraining of teachers in light of their new progressive mission. Then we will have a two-headed strategy, so to speak, for realizing our goals.
- 7.101** I'll just end this section with this quotation again from Wilson's Socialism and Democracy. "Government should lay aside all timid scruple and boldly make itself an agency for social reform." That's the important part of this quote, adding on to the old-fashioned political control materials. Transition.
- 7.102** We're closing in on the end of our seventh session on philosophy of education. It has been a fast tour, going all the way back to Homer and the Greeks, skipping over the Romans into early Christianity, mature Christianity, the Renaissance, Reformation, early modernity, the Enlightenment, a Romantic reaction, Prussian reaction, now coming into the new world with progressive philosophy and education.
- 7.103** So I thought it would be a useful moment to do some comparisons, because that's a lot to absorb. And after we become familiar with the history, with some of the big names, with some of different models of education, one of the important ways human beings learn is by compare and contrast. We notice similarities, but we also notice differences. Those are instructive both ways in making up our own minds.
- 7.104** The idea in part has been taking inspiration from Mirandola. Now that we have, as he said, well, you really have to look at all of the important philosophies, and they're all important for a reason. The reason why we are still talking about them is, they've got something going for them. If there's something going for them, then we should know about that, even if we are eventually going to reject it. Supposing now we have some familiarity with all of them, now we want to start the sorting and evaluative process, and that's hard.
- 7.105** I wanted to spend a few minutes doing an exercise where we say, suppose we take each of the seven that we've looked at so far, seven, eight, nine, depending on how you carve things up, and tried to boil it down to essentials. Obviously, education is important. It's not just curriculum and methodology; it's schooling and assessment and outcomes and funding and so on. We just ask of each, what are the most important elements in the curriculum?
- 7.106** Then, to make the exercise harder, we said no more than three. What would the top three most important things to teach be, according to this philosophy of education? And since the what and the how go together, not only what would you teach, but how

would you teach it, or where would you teach it? Where would this have to occur? I asked two fundamental questions about each of the "isms" as an exercise.

- 7.107** Now, this is meant again as a rough draft. Feel welcome to argue with it. And of course, we would have arguments about this. And to go through this exercise on your own and see if what you end up with, results, compares with what I'm saying here. So let's go back to Aristotle, and we'll now call it classical education. What would we say is most important to teach? Well, I would say, first thing would be literature. Education starts with children. You start teaching children stories, the great stories. It would be the first formal part of their education.
- 7.108** All of the Homeric stories. The stories about Arachne and Athena. Hesiod, the extant playwrights, the ones that have written the great epic plays that we go to see at the festivals. The Greeks love their festivals. Lots of theater and sports and religion all intertwined. But know the stories, and those stories are richly philosophical. And so your children will absorb a lot of that.
- 7.109** Natural philosophy. Aristotle was a great philosopher, a great early scientist. The word science is a Latin derivation, and the Greeks did not yet have differentiated what we call philosophy, physics, psychology, biology, and so on. For them, it was philosophy of nature, including the philosophical respect from that. Taking all of that as a pack, look at nature from a philosophical, critical, observational, rational way of doing so.
- 7.110** And gymnastic, because we're not just interested in moral character, we're not just interested in the mind, we're interested in the whole body. The human being, from the Aristotelian perspective, is an integrate of soul and body. So equal attention to all of the physical activities, physical training, some of them more practical, some of them just sports, but all of that essential. So that would be my essentialization there.
- 7.111** And how do we do this? Well, reading. Perhaps a lot of this would be cheating already, listening, because a lot of this would be in conversation, but reading. You want your child to read, and now we are a literate culture, and it's important to read. Observing, going out and observing the world, and observing carefully, learning how to use your eyes, your ears, even smelling things, tasting things, become very good at observing the world. And then practice, practice, practice, a lot of the huge physical component, and then a lot of it is going to be discussion-based as well. All right, so how is that?
- 7.112** We jump to religious education, version one. In the Western tradition, under St. Augustine. The pagans we know are bad, sinful, wrong. Homer is out. Aristotle is out. Some redemption, possibly, for Plato. But there's really only one thing we need to know and learn. It's the most important thing, and that is scripture, the word of God. That's our curriculum.
- 7.113** We're not to challenge it, argue with it. We listen to our parents, to the priests, the people who know. We read. If we are among the literate in our society, read it, supposing we've learned Latin or Greek, and we memorize. Those are the fundamental

cognitive modes of learning. And what's not on the list is going to be a lot of discussion, questioning, argument, debate. Because that's a challenge to authoritative truth that has come down to you.

- 7.114** Religious education, part two. Aquinas, eight, nine centuries later, expanding the curriculum because while, yes, absolutely scripture, absolutely important, absolutely true, it also is the case that the Greeks and Romans did some quite impressive things, and we can learn supplemental things that can be integrated with scripture. So the curriculum would be dominated by Greek classics, particularly Aristotle, since Aquinas thought he was the best of them.
- 7.115** And you're going to do a lot of reading. That means you're going to need to learn the language, hopefully not only Latin, but also Greek. And a lot of thinking, because the big Aristotelian point is judgment, thinking for oneself, not simply blind obedience, blind faith, unthinking acceptance of doctrine. Read it through, figure it out, see for yourself that it's true.
- 7.116** Renaissance education, using Alberti as our highest representative here. We find that scripture tends to drop out or fall much further down the list. You are interested in religious scriptures, but you're also aware that there are lots of religious scriptures out there, because it's a big world. And so you might be interested in world religions, among them Christianity, Jewish traditions, and so on. But pride of place is given to the Greek and Roman authors, the humanists, the "humanitas."
- 7.117** Gymnastic, again, this theme, mind and body, and it's important, too. Still have, since we are just coming out of the Middle Ages, the martial virtues—know how to ride a horse, know how to use a sword, how to defend yourself when necessary. But then all of the sports, all of the physical exercises, swimming, running, jumping, leaping, and so on.
- 7.118** But then also the arts, because one of the things we know from the Greeks and from the emerging Renaissance is that human beings can make wonderful things— great new literature, great new sculptures, great new music. And not just as a spectator or auditor of these things, to be yourself, a kind of artist, to become proficient at some level at making music and painting and drawing.
- 7.119** Locke, liberal education. The word liberal has a long history, but self-consciously elevating it and putting it to the fore in a modern understanding of liberty. Yes, we are going to learn from the classics. There's a lot to learn there. But we might not even put the classics first. We might then say, we are moderns. We are interested in this world, and we are now aware of the new sciences and the oppressive achievements already of the new sciences. Certainly, the natural sciences are going to be a part of our curriculum.
- 7.120** We live in the natural world, learn about the natural world, first and foremost. Difficult in Locke, but I would say languages. One of the things that's very interesting about

Locke is, as a contemporary, he argues, of course, as you start from whatever national perspective you're starting from, for him as an Englishman, you learn to speak English; you learn to read English. He then goes on to argue that you should also learn French.

- 7.121** It was at that point, 1600s England, the most powerful nation in the world, some jockeying back and forth between France and Spain, but was arguably France. France was also England's biggest trading partner. And so it's a political giant, but we now want people to be participating in the political process because we're becoming more democratic, we're more commercial, we're trading with people. So it's the modern languages that are more important to learn than the ancient languages.
- 7.122** And one of the very striking things he says, "First English," and then, when we say, "What should the next language be?"—of course, there's going to be a next language—everybody expects you're going to say, "Well, it should be Latin or it should be Greek." And Locke says, "No, French." He's a modern. We'll get to Greek and Latin, maybe, but further down the list.
- 7.123** A lot of reading, but also learning how to be experimental. He trained as a medical doctor. He did that experimental procedure on Shaftesbury. He generalized that as the mindset and attitude and a lot of conversation. You're going into business, that's a lot of talking, a lot of deal-making. You're going into politics, you're going into law, that's a lot of talking, a lot of arguing, a lot of discussion that goes on in the modern world. We need to be good at that.
- 7.124** Rousseau, this education for natural feeling and some strains of Romanticism that it fed into. What are the things that bring out our feelings most strongly, make us feel creative and authentic and fully alive? It's not sitting in rows, memorizing mathematical tables. It is the arts, first and foremost, and the inspiration that art gives us; literature, all those wonderful stories of character and conflict, elevated themes. And those can be applied in the theater, bringing them to life and so on.
- 7.125** Then more broadly, the humanities. Anything about the human being and what it is to be a human being in this world, exploring the human from all aspects so that we can go out and live a fully human life. Then the more radical thing here is not learning so much from books, not learning so much from authoritative teachers, certainly not old dead people; who cares about them anymore? I'm being a little bit sarcastic, but making the point here, we're interested in the world as it is, living life, real life, full life, living things, not dead things.
- 7.126** Go out into nature and experience nature as much as possible. Or theater, many love the theater, partly because it involves acting, but it also involves big themes, it involves direction. It's intensely social. It brings what is just words on a page to life. It makes them a little bit more real. And the origins of the unschooling movement that is still with us institutionally, you can see in this movement here. School is just artificial. We're interested in natural, not the fake.

- 7.127** National education, or nationalist education, so-called Prussian model, Prussia being the most powerful of the German states at the time. I think first and foremost, national history. Know the history of your people. You are a part of a people. Its history has shaped and formed you; and your sacred spiritual obligation is to know it, to become a part of it, and pass it on to the next generation—why the Germans are what they are, and why they are especially what they are in the German case.
- 7.128** Moral literature. We are training people to be moral in this strong duty, obedience, rule-following sense. That needs to be drilled into naturally unruly, disobedient children, who just want to go off and do their own thing. I've got this phrase civic religion. So Fichte is very interested in religion. He thinks Lutheranism, that strong, robust version of Christianity, is most suited for the German character.
- 7.129** But the civic part is that we don't have separation of church and state. Well, separation of church and state became important in the English-speaking world. It was mostly unpopular and argued against by the major philosophers, Fichte, Hegel, and the others. We need to see not only the spiritual world as an object of reverence, but also the political world as an object of reverence. We should worship the state as much as we worship God, and those two are integrated in function. So, civic religion is my best attempt to capture that phrase.
- 7.130** And the important thing here is schedule, strict schedule is everything, so children learn schedules, timing, when to start, when to stop. It gets built into the fiber of their being; and authoritative relations. Teacher is teacher. Teacher is the authority. Teacher is the boss. Students are down here.
- 7.131** Then finally, we've looked at progressive education. We're going to boil it down to three—this is, again, difficult curricular elements, the social, social studies. This might then include some civics, some of your nation's history, although history is very interesting from the pragmatic perspective because, if everything is changing, changing over time, history starts to become less relevant. That's the way things were, but here we are now; we have changed too much for old historical lessons to be particularly relevant to the way we are now.
- 7.132** Perhaps some economics, social psychology, anything that would go under the social studies rubric. We're emphasizing the practical. We're not interested in theoretical math, theoretical physics, theoretical psychology, theoretical anything. Those sciences, of course, we have some theoretical people doing stuff, but what we are interested in is the practical results and getting those into the students as soon as we possibly can.
- 7.133** And then all of this with a social mission built into it, to say, you are a part of a community, you have obligations, service connections; you are part of the whole. So everything is infused with social justice themes and literature that supports that theme. In terms of methodology, the progressives are interested, since they are much less

individualistic, some of them are strongly anti-individualistic. Methodologically, they almost always emphasize group work.

- 7.134** Don't have children do anything by themselves. Always team projects, group projects. From the very beginning, you are a part of a we, part of a community. It's we who does this. It's we who gets all of the credit. It's we, if we mess up, we'll get all of the blame, but again, more collective. The classroom should not be authoritarian in any respect, teachers and the higher establishment, and that we're afraid of them or we're just passively doing whatever it is that they tell them.
- 7.135** From the beginning, children are part of the community, and they should start to have the sense of being a part of a democratic community that collectively makes the decisions. The teacher is also a part of that community as well. But nonetheless, she's not the queen. She's not the duchess. She's not the dictator in any sense. She's perhaps the first among equals, but you want to get that sense in the children that they are equals, that they are all participatory, that they all belong to each other.
- 7.136** Hi, Dr. Hicks. I'm curious, why does there seem to be a lot of overlap between the Marxists and the pragmatists, when it seems like Marxism is among the most idealistic and least practical philosophies if you're concerned with human life and flourishing?
- 7.137** We have a century or more of actual experience of Marxist or close to Marxist forms of politics in practice. We have a huge body of knowledge that was not aware to people forming their political views in the late 1800s, early 1900s. Although, even on that point, there still is an interesting issue that we still have a huge number of people who think some sort of Marxism still can be made to work.
- 7.138** As the question then becomes, as an educational question, partly an educational psychology question, how much empirical reality and data matters to people when they're making up their political, ideological minds? But yes, if we then set that aside, we go back to the time period, we put ourselves in 1905, 19-teens, early 1920s, and we say, well, it is interesting that so many of these people who are now young, relatively speaking, they are pragmatists. They are emphasizing workability and flexibility.
- 7.139** But Marxism is very inflexible. It talks about the iron laws of human development and so forth. So how do you square that circle, psychologically? I think that is going to be one of the reasons why the pragmatists will say, in a generic sense, and the progressives, we like the ideals that Marxism is striving for, but we don't agree with the underlying metaphysics and epistemology, and even the need for violent revolution to get there. So same generic ideals, but not the same philosophical underpinnings to it. So they would see them then as fellow travelers.
- 7.140** So, as we know, almost all of these big political movements are big tents, and they will have different factions. And if you say what completely goes into an overall political philosophy is, say, 20 elements, you'll just have overlaps where this guy believes all 20, but these guys believe 18, and they have two different ones. But these guys believe 16

and have four different ones. And these guys over here, they also agree with 18, but it's not the same 18 as these ones. So you're performing an act of abstraction, or in many cases, an act of averaging inside that big tent.

- 7.141 But the important things would be that the Marxists are materialist, and they believe that the world is what it is. So they are small objectivists in that sense, independently of our viewpoints of it, and reality is just marching along through social stages, depending on economic circumstances. So it's going from tribalism to feudalism to capitalism to the need for a revolution. And what we humans think and do really has nothing to do with the march of history, so to speak.
- 7.142 So one of the big differences is going to be among the pragmatists, who have much more of a social construction of reality notion. The idea is that what we humans think collectively has a lot of influence on making reality. So reality is more malleable. And if we believe certain things, and if enough of us believe certain things, we can change history. And so it's not going to be as deterministic as in Marxism.
- 7.143 I think another big one might be that the Marxists, for whatever reason, are always very quick to give up on any peaceful transformation from whatever society you're in to Marxism. They are ready to bomb, to destroy, to kill whoever they need to kill to get there. Now, you might say this is starting from a psychological point. You've just got angry young people who just want to blow things up.
- 7.144 Or it might be, at least according to official Marxism, that they believe that this whole idea of arguing and discussion that's built into democracy, that we can just convince people of the right ideas and get them to vote for Marxist socialism to come into existence, that that's unrealistic. I might tie better directly back into that first point about the determinism that seems to be built in—there is actually built into Marxism.
- 7.145 Because if you think, as Marxists do, that what you believe ideologically, your philosophy, your religion, is dictated by your economic place in society, what that means for them is that if you were born in, say, this economic class, all of your thinking is shaped by being a member of that economic class. You, say, think like a poor person, like a poor working person. If you're born into a completely different class, say the rich elite class, then all of your thinking and beliefs are shaped by your being a member of that class.
- 7.146 And so those two cognitively can't understand each other. They can't think on the same terms. They don't even define what's the best life, the good life, or even what's true the same way, because they've been conditioned by their different economic circumstances. And what that then means for a Marxist is that you cannot expect them to say, well, let's all sit down and talk over our differences and reach some sort of democratic—because that just seems completely impossible, according.
- 7.147 And so the other side of that, then to come back to the pragmatists/progressives, it will be people who don't buy into that deterministic cognition as the Marxists do, that they

start, many of them are American. So maybe it's just that they've been bred to think that democracy is a good thing. And so in their bones, they have this great faith that, through enough discussion and conversation, we can get past our differences and reach some majoritarian and perhaps even better conclusion.

7.148 So that then would be to say, the other big difference is that they would agree with the Marxists on the eventual outcome, but still think peaceful democratic methods will get us there, whereas the Marxists are going to say, democracy is dead, we need a revolution, and it has to be a violent one to get there. I think that's two or maybe three ways of trying to sort them on important issues.

Lecture 8: The Montessori Method

At a glance — The Montessori method, from its 1907 origins.

- 8.1** 1907, Maria Montessori opened her first Casa dei Bambini, so I wanted to draw our attention to Rome. It was a distinctive, new, entrepreneurial approach to education, and it's proved enormously successful over the 20th century. But at the beginning of the 20th century, it did not seem very likely.
- 8.2** Montessori herself is a remarkable woman, wanted to go to engineering college. Being a young woman in the late 1800s, some family resistance. Nonetheless, she did do some engineering training, then switched to medical school. Again, resistance, but became the first, or one of the very first, women to graduate with a medical degree in Italy.
- 8.3** Found it very difficult to find a hospital to take her on. Didn't have the funds necessary to open her own medical clinic, but she was then assigned a job at a hospital, working with some kids nobody else wanted to work with, who were basically warehoused, intellectually disabled children, thought nothing could be done with these kids. But here's young Maria Montessori. We'll just have her look after these kids.
- 8.4** She did start to work with these kids, kept in a basically warehouse room, low light, empty for the most part. She started to realize that they were starved, not necessarily starved for food or drink, but starved for interaction. She started to bring some simple toys for them to play with and work with. They came alive as a result of these manipulatives, as she called them, things that you can use with your hands.
- 8.5** Then partly because of her medical training, partly because of her engineering training, she started to break down the human system, physically, developmentally, psychologically, developmentally. This is the era of the birth of psychology as a science as well. Started to see what she could do with these intellectually disabled kids, who nonetheless had functioning eyes, functioning ears, functioning hands, and so forth. And so started to bring in and design toys that would focus on each particular sensory manipulative apparatus.
- 8.6** At a certain point, she realized that her kids were doing well. They were learning stuff. And there were some standardized tests that all school kids across Italy would take, age 5, age 6, age 7, various grade levels as well. She started having her kids take those tests. The surprising thing was that her kids were performing at age level with the kids who were not intellectually disabled. In some cases, her kids were performing above grade level. And so this brought her to attention of everyone else. And also it was a boost to her own burgeoning educational ideas.
- 8.7** As a result of that, she was able to get more funding. The idea being, well, whatever Montessori is doing with these disabled kids, if it's working for them and they're

making great progress, what if we tried it with not disabled kids, fully able kids? Maybe they should be able to do a whole lot more. The more formal system, somewhat bureaucratic, hard to break in, but Montessori then went private and opened her own school.

- 8.8** Now, here is an example of one of the Montessori manipulatives. In broad history of philosophy terms, I think the best label is to think that Maria Montessori is an empiricist in the John Locke tradition. How do we come to know—this is our fundamental question as philosophers—know "real"? The empiricist answer to this question is that we don't receive divine revelations. We don't just accept certain truths on faith. We don't just listen to authority figures. We don't have innate ideas or instincts that just automatically operationalize themselves.
- 8.9** Instead, we start with our senses. We have five senses. They become aware of different kinds of energy patterns and structures in reality. The first stage of learning is to learn how to use your eyes, learn how to use your ears, your mouth, taste, smell, and then the sense of touch and then the manipulatives.
- 8.10** We also know these work together with each other. I hear something and then I look at it; or I want to taste something, I use my hand, and my hand is working. Eye-hand coordination, ear-eye coordination, hand-mouth coordination, and all of these coordinations also then need to be built up from the individual sensations.
- 8.11** We start to then form abstractions. When children start to learn words, we start to put the words into sentences. We start to put the sentences into paragraphs. We put the paragraphs into theories and start to tell a whole story. There's a whole cognitive development that the empiricist tradition works out. The argument then is that, psychologically, this is the development we go along. Of course, there's a parallel physical development; that's another part of the story. But on this cognitive development side, educationally, we should start at the beginning and work our way up.
- 8.12** Now we are doing this psychology as a science, now applied to educational matters. This is an example of a toy. You see there's a little guy there, and he's not sitting at a desk. He's standing up, he's moving around. What has happened is that this stack of blocks was in a tray, a little wooden tray on a shelf off to the side with a whole bunch of other pre-arranged toys. And he chose to go over, said, "I'm interested in this." So he gets it, he brings it over to a vacant space on the floor, and he starts to play with it.
- 8.13** And he's playing, and that's the important word there. But what he is able to see with his eyes is that some of the blocks are bigger than other blocks. And he realizes, as kids like to do, you can start to build them into a tower. So he starts to build them into a tower. Now, this is the end result of the process. But along the way, no doubt, there's been a learning, trial-and-error, experimental process that he has engaged in. So all the blocks are just randomly in the block.

- 8.14** This one looks the biggest, so he puts it there, he grabs another one. Oh, that one is bigger. I have made a mistake. I need to look for that, or put this one on the bottom and put this one second. And it's trial and error. The other thing that's important here is that there's no teacher showing him how to do this. The assessment is built into it by the manipulatives themselves.
- 8.15** Another thing in later generations, we'll see towers of blocks like this. You can get them at most commercial stores, but they will be all different colors, and that's fine. But Montessori's point is essentialization and focus. At the beginning, you don't want to have too many things for the child to attend to. We'll have other manipulatives that might teach them about the color spectrum. Start with violet, blue, green, and so on; but right now, what are we interested in? We are interested in relative size, only relative size.
- 8.16** So we make all of the blocks the exact same color. That is to say, we are abstracting away from color differentiation and focusing just on size dimensions. Then we have a manipulative that the kids have a fun time playing; they do it on their own. The assessment is built into the exercise, and the teacher at most then will be assisting. So that's one example.
- 8.17** Another example, I don't have a picture of it, but would be to say, children, as a step of growing up, this is partly more physical, you want them to learn to dress themselves, and they want to learn how to dress themselves. You don't want to be dressed by your mom as you're growing up. "I'm a big girl, I'm a big boy, I can dress myself." But clothing is complicated because we have buttons like this. We have zippers. We have things that have to be tied. Sometimes there are latches. And so, all of these.
- 8.18** So what we can do is, Montessori decides, is I will get a board and on the board there will be fixed—here's two pieces of fabric with an open zipper, but the kid can take the two pieces of fabric together and learn how to do the zipper. Here are some laces that are untied. I can then play around with it and learn how to do the tie. This one has buttons, two pieces of fabric. I can learn how to do the buttons.
- 8.19** I might not be interested in that one. Actually, I did that one yesterday. I want to do something different, so I get a different tray, and I learn how all the fasteners on clothing work, on my own until I know how to do that, then I can start practicing it on myself.
- 8.20** Another example. We are now interested in colors, say. We have a pre-set, pre-built tray of blocks. In this case, we're not interested in size differentiation as we were with the blocks. All of these are the same size. We're holding size constant and only varying the colors. And so kids can take these color cubes out, or not cubes, color palette units. I'm not sure what the right word for that would be. And then start to arrange them in various ways.

- 8.21** And if my goal is to say, well, these ones are similar, they're more what we would call pinkish, these ones are more reddish, then I'll put the pink ones over here, the red ones over here, in an immediate sorting, but they're not all equally pink. I can array them in ascending pinkness, ascending redness, ascending violet, and so on. And again, this is something the child has chosen. In this case, you can see that she has a mat. And the way Montessori schools will typically do this is, children won't have desks. There are desks available if the child wants to work on a desk.
- 8.22** But at the earlier ages, when we start, the sensory training and eye-hand coordination starts earlier, a kid likes to play on the floor, but each kid will get a small roll-up mat, can just find a place on the floor, roll out her mat, get the toy or manipulative learning exercise she wants to play with, take it over to her mat. Every kid has their own mat. They learn pretty quickly. You don't step on other people's mats. That's their space, unless you're invited. So they start to learn some social respect and so forth.
- 8.23** More sophisticated examples. Here we are learning about shapes. We have a triangle with threads, and the manipulative invites you to take the thread and fasten it in a certain way. Then you can do squares, you can do rectangles, you can find ones that are closer to circles, hexagons, and so on. You learn also to start doing some counting. If you pay attention to that, you're starting to learn some basics of geometry.
- 8.24** But also, in this case, it's not the teacher just drawing things on the board. You are learning it yourself by interacting with the prepared materials and so forth. The idea here is a principle of philosophy. We learn empirically and we learn cognitively by doing. We integrate our sensory organs in a hierarchical structure to more and more abstract things. But at the same time, all of our senses need to be learned individually and integrated with each other in the overall structure.
- 8.25** We break this all down as a matter of philosophy and psychology, scientifically, and then design do-it-yourself exercises for the kids to work on in the classroom. So this is now more sophisticated working-out of her original insight, working with the intellectually disabled kids, because they need special extra help in learning how to use and coordinate various things, but it extends more broadly to all human beings.
- 8.26** There's a quotation from Montessori. It's worth working through. "It is true that the child develops in his environment through activity itself." There's a developmental concept that's important. Children develop, they go through a process. We need to understand that, and some things need to be learned first before they're ready for second things, third things, and so forth.
- 8.27** But it's also not a passive thing. We don't put kids in chairs and talk at them or talk to them. It is active. We are naturally mobile animals with a very powerful brain-mind combination, and the interaction of our minds with the environment as we are moving through it helps us learn much more quickly than passive sitting as well.

- 8.28** "But he needs material means, guidance, and an indispensable understanding. It is the adult who provides these necessities." So it takes a parent to prepare the environment, to design all of the manipulatives, to create all of the manipulatives, and then put the child in a structure in which he or she can explore all of them. And you know the kids like to play, so these things are fun to do. Eventually, they are going to work through all of them, all of the things that you know ahead of time they need to learn.
- 8.29** At the same time, they're making their own choices at any given moment about what things they want to work on, for how long they want to work on them, until they think they have achieved the mastery level. Or maybe they say, OK, I'm stuck on this one. I'm bored. I'm just going to put it away until tomorrow and go work on something else. So her phrase for this is, "Freedom in a structured environment." Freedom in a structured environment. Structured environment prepared by the parents, but you want to have the child have maximum freedom in that environment.
- 8.30** And we can elevate this to a philosophical statement, reality is structured. It has all kinds of cause-and-effect principles and laws that scientists are discovering. But nonetheless, we human beings are free agents, and we want to create human beings who can act as free agents in that causal structure of the world.
- 8.31** "One of the great mistakes of our day is to think of movement by itself, as something apart from the higher functions. Mental development must be connected with movement and be dependent on it." All right, so this is Montessori drawing on psychological literature. That has experimentally, she thinks, validated the idea that learning works better if it's conjoined with movement. We learn by doing. It's not that we're doing things mindlessly. It's also not the case that we are active mentally but not doing anything physically. It is mental activity, physical activity, integrated. That's when the best learning occurs.
- 8.32** Standard classroom. We're all familiar with this. What does school look like? Everybody says, yeah, that's right. But then just pause on the geography of this. It's seats with a desk, and they're all in rows. They're all equally spaced from each other, and they're all oriented to face toward the front, the one focus of attention. Everybody will be focused on the one show that is happening at the front of the room. That is almost universal, even in many alternative schools.
- 8.33** This is a Montessori classroom. There are some tables, there are some desks, there are some child-sized chairs, but everything is movable. The kids can move those things around, and they're welcome to move the things around. You can see each kid has his or her own mat. At the back of the room there, there are all of the trays with the different pre-designed toys. The kids come in, they get their mat, they decide what they wanted to work on, and then they go find a space wherever they want. Do you want to be in the sunlight or not in the sunlight? Yeah, et cetera. So, choice, mobility, flexibility, interactiveness.

- 8.34** A follow-up question is, well, where is the teacher? It seems like we have now decentered it. The teacher is not the universal object of attention anymore. The teacher is present, but she is off to the side. I'll save that for a moment, emphasizing here the free choice. "Children have free choice all day long," in her school. She's making a general philosophical point. What's life like? Life is based on choice.
- 8.35** By the time you become an adult, nobody's telling you what to do. You choose your friends, you choose what book to read, you choose what job you want to do, do you want to further your education, where you want to go. "Life is based on choice." We start right at the beginning, making choices. "They learn to make their own decisions. They must decide and choose for themselves all the time." That's emphasized. We decenter the teacher from the process. She's worked ahead of time, preparing the environment, preparing the manipulatives. But then once the process starts, it's a seriously individual, student-directed process.
- 8.36** So again, there's the teacher. The kids are all sitting with their hands, proper posture, everyone looking, listening to the teacher. She's the one who's talking. They might make notes or they might memorize, and then the expectation is they will return that knowledge in the form of tests at some point. The teacher then will assess how well they listened to what she had to say. Very different model.
- 8.37** Here's Montessori herself in one of her early schools in India. She started to become a worldwide success, and so we see the same thing—kids working on their own projects, on their mats in different locations, and she's off to the side. She's observing. Of course, sometimes the kids will ask for help. You'll come and you'll give some assistance. Sometimes kids are kids. There will be squabbles, and little fights will break out. So she has to become policeman and diplomat for a few moments, to get things back to where things should be.
- 8.38** Here's another example, a more modern example, of a Montessori room, same sort of structure. And there we do have a teacher. But also notice she's not standing while all of the kids are sitting or kneeling. She's down working with the child, on the project the child has chosen. And at this point, the child will have asked for some assistance. "I'm stuck on this. Can you give me some pointers or some advice?" And she won't do it for the kid, but she will give some pointers, that the kid will then say, "Oh, well, maybe I'll try this," and the kid will try it and then make some progress. A lot of emphasis on the individual, individual choice and so on.
- 8.39** What about the social part of that? Part of the social part is learning to respect each other's space, not being a space invader, so to speak. So you have your mat, this is your space. The kid has over here her mat, those mats over there. You might be interested in what that child or the kid is doing. That looks like a fun thing, but you don't just go and grab it and take it. So you learn to respect that they've made that choice. They are using it now. Just remember that when your turn will come.

- 8.40** Or maybe the other kid will see that you are interested and invite you to work with him or work with her on that. But it's an invitation, and the kid does not have to do so. Or sometimes the kid will be proud to do so. I know how to do this one very well. It's a lot of fun. Maybe I've done 30 times before, but I like to warm up each morning by doing this one.
- 8.41** And there's a kid who's two or three years younger. And he says, "Wow, that's impressive. How do you do that?" And then so the older kid will show. It's good for the one kid, good for the other kid as well. But again, it's a voluntary choice to interact with each other. So you're teaching them respect for their space, manners, politeness, asking, working on win-win relationships. And again, it's not authoritarian. The teacher is not the boss, et cetera.
- 8.42** "Children must decide," Montessori says, "and choose for themselves all the time. They cannot learn through obedience to the commands of another." Now, partly this will be a matter of obedience and command and authoritarianism going against Montessori's personal grain for social and/or political reasons. But she's also making a point about psychology. When do kids actually learn? We now have enough experience in model.
- 8.43** If we have the command-and-obey model in education, with a certain set of material, and we have a freedom-of-choice model in teaching that same material, the command-and-obey kids don't learn as much. Goes back to Locke's point. Then again, this is very Lockean. We tend to resist commands of other people. Somebody says, "Don't do that," part of you wants to say, "Well, actually, I want to do that now." That's a natural human thing that should be encouraged.
- 8.44** Or if I say, "You have to do this," you start immediately figuring out how you can get out of having to do that. Again, that's completely natural. I can, if I'm the dad, with my dad tone of voice, or I threaten you, get you to do things, and stuff can get done. So kids can learn through command structures and by obedience, but not as much, if they make their own choices, and it's operating on their own interests and letting them go at their own pace. Now, that's an empirical claim that would be needed to be put to the test, but Montessori is happy to make that sort of claim.
- 8.45** The teacher involved, preparing the environment, keeping up on psychology. There also, of course, is the physical development of children that's going on. A lot of the activities have to be designed for physical development. Here, if you think about gymnastics, the way we think about teaching kids gymnastics, again, is very scientific. We say, well, these are all of the ways the body can move. These are all the muscles of the body. These are all of the tendons and ligaments that need to be kept flexible. These are the lungs that have a certain amount of endurance.
- 8.46** We break the human system down into all of the subsystems, and we design exercises to work on each of those subsystems in a gymnastic program. The idea being then, we break them all down, you become good at each of them individually, then you start

putting them together in increasingly sophisticated fashion. The exact same thing happening on the cognitive, mental side of things.

- 8.47** The teacher does the preparation in the background, keeps up on the physiological literature, the psychological literature. But the actual learning, teacher takes a secondary role. I noticed the metaphors that she uses here. "The teacher's task is first to nourish and assist, to watch, encourage, guide, induce, rather than interfere, prescribe, or restrict." So the latter three, bad, the other ones correct. Some of them metaphorical. But it's a terrain. That's a mindset.
- 8.48** "Do not apply the rule of non-interference when the children are still in the prey of all their different naughtinesses." OK, again, another John Locke point. Kids are going to be kids. They're going to be naughty. They are going to screw up. They're going to be rebellious and throw temper tantrums as well. OK, that's going to happen. It doesn't mean that they are evil little monsters, that we need to have the full force of the law weighting down upon them.
- 8.49** Rather, interfere. If it's your kid causing an immediate danger to another kid, stop it. It might require a physical intervention. Safety of the other child is important. To that extent, you are a police officer in the classroom. There are going to be the normal collisions among kids, as there are among adults. "You must interfere at this stage. At this stage, the teacher must be a policeman. The policeman has to defend the honest citizens against the disturbers." So this is just adult life, moral principles scaled down to children.
- 8.50** "Our schools show that children of different ages help one another. The younger ones see what the older ones are doing and ask for explanations. These are readily given and the instruction is readily available." Almost always when we learn something, we have mastered it. If you're a little kid, you'll say, "Mommy, look what I can do. Daddy, watch me do this." We all are proud of ourselves, and we do want the social acclaim that comes with that as well.
- 8.51** That happens among peers and relative peers as well. If you're worried about kids not being social enough, well, yeah, we do like to show what we can do. Sometimes we even like to show off what we can do, and little kids will do that. There will be plenty of that and opportunities for a lot of that in the classroom as well. We can let it happen naturally and organically. "The older ones are happy to be able to teach what they know."
- 8.52** So here's a cluster of concepts that I think are the key concepts guiding Montessori classrooms, and I've put them in a more abstracted philosophical way. Naturalism, the natural world. We're interacting with the natural world. It's cause-and-effect in various ways. This thesis of empirical reason. We are rational beings, but the rationality builds up from the bottom, from the empirical data, the data of the senses, so to speak. We

learn by observing, comparing, contrasting, and abstracting, and it's an open-ended process.

- 8.53** We are both mind and body. It's not two different things that happen to be together, not two different things in conflict with each other, but rather a mind-body integrate. So it should be cognitive and physical happening at the same time, working together. Choice is a capacity that we have, one that we love, right from the get-go, making our own choices, making our own decisions. And we are individuals. And while we're all humans, we also have our unique interests, aptitudes, and so forth. All of these are the raw material that teachers and professional educators take, aware of.
- 8.54** Freedom, in this case, is the social correlate of choice. If you respect people's choices, then you are giving them freedom or allowing them freedom. So we don't want social interferences overriding people's choices. Respect for property rights, that is modeled with, this is my mat, you can't take my mat. If I have chosen to play with this toy this morning, it's now my property, so to speak, for the morning. There's lots of other things for you to play with, and it will become yours once I return it to the proper place as well.
- 8.55** The concept, I think, is important that Montessori often emphasizes is that we're trying to teach children an optimistic or benevolent understanding of the world that they live in. We know many models of education emphasize fear, that you should fear authorities, fear the teachers, fear your parents, fear the police. And so you do what you're supposed to do, based on that sense that there are big, powerful, scary forces out there that I must not get on the wrong side of.
- 8.56** Montessori's is the opposite. We're trying to train for positive, social win-win interactions with each other, that we can trust each other, that we can like each other, that we can work with each other. And that has to be modeled from the very beginning. This is romanticism, not entirely in the sense of romanticism as we use it, being irrational, passion-driven, and so forth, but a romantic sense of life as an adventure, and the child is starting off on this wonderful adventure, and our job is to guide him or her through that process.
- 8.57** Now, Montessori started developing her schools in the early 1900s. At the same time, progressivism was becoming to be an intellectual and cultural force. One of the very interesting things, particularly in the United States of America, was how hostile the progressives were toward Montessori, and that for good philosophical reasons.
- 8.58** Partly, it was that the progressives had made major inroads in the public school system or the increasingly government-run school system, and they did not want to give up a share of that. Montessori largely went and became a private grassroots phenomenon around the United States and around the world. But the root of the differences are indicated by quotations from two of the principal figures here. Here's Montessori.

- 8.59** "The educator must be as one inspired by a deep worship of life and must, through this reverence, respect, while he observed with human interest, the development of the child life. Now, child life is not an abstraction. It is the life of individual children. There exists only one real biological manifestation, the living individual; and toward single individuals, one by one observed, education must direct itself."
- 8.60** So that individualism, and the emphasis on the individual with his or her uniqueness, their own choices, sometimes working independently with each other, that is fundamental to Montessori. But then again, by contrast, when we read John Dewey, "Education, in its broadest sense, is the means of this social continuity of life. Each individual," and this is the status of the individual, "a unit who is the carrier of the life experience of the group, in time passes away, yet the life of the group goes on."
- 8.61** So you have a much more group, social focus, deemphasizing the individual in pragmatism and progressivism, whereas Montessori is very robustly individual, and lots of social things will happen, but they will emerge by consent to interaction by the individuals who are involved.
- 8.62** The most famous of the progressive educators was Kilpatrick, Professor Kilpatrick, and he was a follower of John Dewey's models of education, a disciple of them. He also described himself as a democratic socialist, as fully in the progressive camp. He was the professor who was most critical in academic publications of Montessori's early writings, and one of the reasons why, at least in the public/government sector of American education, Montessori was nowhere to be found for most of the 20th century. Instead, she developed a network of private institutions around the world.
- 8.63** Now, the progressive educators who were then dominating were not fans of Montessori, but it is very interesting how immediately people in the entrepreneurial space, in American culture and Canadian culture, and quickly in European culture, saw in Montessori the answer to their dreams. In many cases, these are young people who hated school, who dropped out of school as quick as they possibly could, who were the rebels, but nonetheless went on to do something pretty amazing.
- 8.64** So, one example of this is Thomas Edison, who, very early on, he's the very prolific inventor of, I don't know how many things he invented, thousands of things he invented. Many of them are still with us today in increasingly improved form. But the thing that he liked most about the Montessori system is this concept of play. Again, this Lockean point, where Locke, some centuries before, said, "I've had this fancy that education should be playful and recreation and never be made like work, business, and so forth."
- 8.65** Edison is picking up exactly on that. That's me. I like to play. That's all I'm doing in the laboratory all day, is playing. It's hard work, but it's play. "I like the Montessori method. It teaches through play. It makes learning a pleasure. It follows the natural instincts of the human being. The present system [early 20th century] casts the brain into a mold."

We have the mold; we're trying to fit the kid's brain into the pre-existing mold. "It does not encourage original thought or reasoning."

8.66 That's the current system. Montessori system, play, fun. Originality is going to happen as a result of that. Interestingly, also Alexander Graham Bell, usually credited as the inventor of the telephone, another inventive entrepreneurial guy, loved the Montessori system. He and his wife started an early Montessori school, one in Washington, DC, where he spent some time, and another one in Canada.

8.67 It's also very interesting, as we assess the empirical evidence, and of course, Montessori would be entirely in favor of let the track record speak for itself, how many people of our recent generation who were Montessori-educated? The two guys from Google, Larry Page, Sergey Brin; Jeff Bezos of Amazon; Jimmy Wales, in Wikipedia, all of them were Montessori kids. All of them consciously will credit their Montessori schooling with cultivating in them that learning as play, learning as fun, inventiveness, giving them the basic skill set to be able to go on and do.

8.68 Now, these are just some of the famous names from a very long list when one starts to look at. Yo-Yo Ma, Joshua Bell, both from the world of music; Helen Hunt and George Clooney, from the world of acting; and Stephen Curry, basketball player. But it is interesting how many people from what we think of as the more creative sectors of society, the people who are doing original sorts of things, are Montessori. So that's one set of data, just a list of names at this point.

8.69 There are now, around the world, approximately 22,000 Montessori schools in quite a few countries. The divide between Montessori being a purely private school thing, in poor countries and rich countries all over the world, and the public sector or government sector not having any Montessori, is starting to break down.

8.70 Some public school districts have started to have experimental modules that will be Montessori-inspired. Partly, they say, well, we've tried lots of other things. A lot of them don't seem to be working. Why not try Montessori added into the mix, since it's in the private sector? That might be a divide that's going away as we go on.

8.71 But in part, reflecting on this list of extraordinarily entrepreneurial people who have come out of Montessori schools over the course of the 20th and early 20th century, in a final unit, I want to pick up on this theme of entrepreneurship and suggest now that we are into the 21st century, that while we are in the modern world, an entrepreneurial culture, only 500 years or so in the long sweep of history of human beings being around for hundreds of thousands of years, the math is quite stunning.

8.72 But nonetheless, we now have a respect for innovators and entrepreneurs and rebels and people who are doing things differently, and are starting to pay attention not only to respecting people entrepreneurially in the business and economic culture, as the ones who come up with the new ideas for business and try them out. A lot of them fail, but some of them are spectacular, successful, and we benefit from all of that.

- 8.73** To thinking about the mindset that goes into being entrepreneurial, and whether, if we are in fast-changing, and we use all of these clichés all of the time, everything is transforming itself on a regular basis; if the mindset of education that was developed some thousands of years ago and may be culturally instituted or in formal education instituted, we're doing the same things over and over again because that's how we've always done them, whether that needs a rethinking, especially in our current time.
- 8.74** Now with that, by way of a preamble, let's have a look at some data to think about this point. In this course, as we've been thinking about education, we've talked about religion, politics, history, psychology. What else have we talked about? All sorts of things like literature and the arts, and all of these things that possibly could be in the curriculum, or front and center in the curriculum, but we've not yet explicitly focused on the concept of work or a concept of business.
- 8.75** I want, in this last unit, to pay special attention to that sector of human life because, as we know, a huge amount of one's life is taken up by one's work. Even if you win the lottery, many people who win the lottery feel dissatisfied and they still want to do something with work. Even people who've made a million dollars want to develop themselves through their work. Of course, they want to provide for their families and provide better and so forth.
- 8.76** For most people, your choice of your work that's going to occupy much of your adult life is a huge part of what your life is going to be about; and education should be, at least in significant part, preparation for the kind of work that kids are going to do as adults. What I want to suggest is that, while we have always been workers, the way we work and the way we think about work in the modern world is different. It's revolutionary. That's the point of this first piece of data here.
- 8.77** The horizontal axis is a timeline going back 2000 years. It goes back to year zero. That would be, in religious history, the lifetime of Jesus; in political history, Rome has just stopped being a republic, and it's now becoming imperial Rome with all of the Caesars and so on. So 2000 years. They're then divided in 500-year chunks, 500, 1000, 1500, and now this chart stopped in 2015.
- 8.78** The vertical axis is GDP. That's a measure of productivity. How much stuff do we make as human beings, and what's the value of that stuff? And so this is then normalized to US dollars in the year 2015. So the economic historians have been super busy, crunching all over the numbers, and out pops this graph. What this tells us then is, if we go to year zero, the entire world is making less than \$1 trillion-worth of stuff, and it's actually significantly less than \$1 trillion-worth of stuff.
- 8.79** If you had gone back another 2000 years, 4000 years, there would still be a flat line. 100,000 years, it's actually a little bit lower, but it doesn't show up on this scale. So we will start here, and then we go to the year 1000. Nothing has changed. Human beings are not more productive. You go to the year 1500. This is Columbus, Martin Luther,

Michelangelo, Leonardo, Andreas Vesalius. Things are starting to happen. That line actually has changed a little bit. It's gone up by about 20%.

- 8.80** Now, that looks like almost nothing on this graph, but a 20% raise is nice, and it tells us that something is happening. Then we jump another 250 years. Now we get to 1750, 1800. This is the age of the Enlightenment. This is the industrial revolution. This is American Revolution, French Revolution. And that line suddenly goes vertical. Never happened before in human history. So the point of all of this is that we are spectacularly rich as human beings now, and in a relatively short period of time.
- 8.81** None of us thinks of ourselves as particularly rich, probably, because we're measuring ourselves against Jeff Bezos and Elon Musk and so on. Those guys are superrich. But we, if we compare ourselves to other parts of the world, we say, OK, now I know how rich I am, because we are rich Americans or rich Canadians or whatever. But even those poorest of the poor people, wherever you want to go in the world, are rich, compared to most human beings through most of human history.
- 8.82** And that's spectacular. The implication here is about how we think about our work. Now, we want to work, we need to work, I got to pay the bills, etc. But that's not the same pressure that has been on human beings for most of human history. Right now, if my concern was only getting 2,000 calories a day and protection from the elements, which is what most people through most of history had to spend their waking hours trying to do, just to achieve that bare minimum, and often not doing a very good job at that, that's solved by us so easily.
- 8.83** None of us worries, really, about being protected from the elements. Our problem is, "Oh, man, the air conditioner is a little wonky today." That's our problem. Or, "I'm not sure I can pay my cell phone bill this month." That's a problem. But in human historical scale, that's not a problem. That's first-world, rich people problems. That's how we are and we think about our work. That means to say, we have higher standards and different kinds of standards about our work.
- 8.84** Another thing we're thinking about is, that line suddenly went vertical. Why did it go vertical? Why did it not happen 1000 years ago, 10,000 years ago? Why did that line start to go vertical first in Western Europe and not somewhere else in the world? Then not only Western Europe, northwest Europe, and then more specifically, the Netherlands, and even more specifically in England and Great Britain, because we can do the history here.
- 8.85** And then this does mean, when we're thinking about work, we're preparing children for their work, it's not groups of subsistence survival that we are preparing them for, something higher standards, but also they are going to be going to work in a culture that has become spectacularly rich like this. It'd be good for them to know why that happened, so that they can learn the right lessons about that and make that part of

their identity and then pass it on to their kids as well. That's a different kind of question.

- 8.86** Now, we might start entertaining all sorts of hypotheses about what goes into this, the industrial revolution, the Enlightenment, scientific revolution, all of those things are happening, 1600s, 1700s, just during and before the great takeoff. Those are some big things. And earlier, Protestant Reformation had occurred. There's lots of disasters that had also occurred.
- 8.87** But there's also a difference between cultures that have a disaster and it basically wipes them out. A culture that has a disaster, it doesn't quite wipe them out, but they never quite recover from it, and they just limp along for centuries after the disaster. And other cultures that have a disaster happens to them, and 10 years later, they're right back where they were, or 20 years later. They rebound, they know how to recover from disaster. That's another lesson that might be worth learning as well. What is the right lesson?
- 8.88** On this point about how spectacularly rich we are, this is a relatively short period of time. If you go back to the industrial revolution, the American Revolution area, late 1700s, by current standards, about 90% of the world would be poor, terribly poor, and 10%, you might say, are comfortable or rich. Now, these are the more recent numbers, what percentage of the population is poor by those standards?
- 8.89** When I was young, maybe 27% of the world was living in poverty. This is defined as, you're barely having enough calories to keep your body warm and protection from the elements. How much is that? About 27% of the people were right at that line, sometimes a little above, sometimes below it. Then over the course of the last 50 years, whatever it is that we're doing around the world, we've gotten that number down to below 5%.
- 8.90** Now, we're still working on it because we now think, and this might be part of the answer, we now think this is a problem that we can figure out the answer to and solve, because that's part of our mindset. But that's a mindset that gets taught to people, instead of a mindset that says, "Oh, well, the poor are always with us." Poverty is the natural lot of mankind, or perhaps God has chosen some people to be favored and other people not to be favored. Very different mindset, and that takes us again into philosophy territory.
- 8.91** Now we are rich. And we think people being rich, we want fewer people to be in poverty, and we want our kids to be rich and richer than we are. Figuring out what made it possible for us to be where we are has to be a part of our own education and then filtered into our curriculum in some sense, so kids know how to keep doing whatever good made that possible.
- 8.92** Another element, though, of this is a shift in the kinds of work that have become more commonplace in the modern world as we progress. Now, if we go back, again, 1000

years ago, say, the height of feudalism, you're a young person, what is your work going to be? It's already been decided for you. You're born into a certain caste, right? Your dad was a peasant. You're going to be a peasant. Your dad was the duke. You're the first-born son, you're going to be the next duke and so forth.

- 8.93** You're the girl, then your dad's an earl. Well, you're going to be married off to somebody else in the nobility and you know what your work is going to be—look after your husband and start breeding the heir. That's your job. It's all set. Tribal economies, boys learn to do man things, girls learn to do woman things. It's all preset.
- 8.94** There's not, "I need to find myself," as a teenager, and "I want to do something different." No, this is not even an issue, but that is a modern issue. Partly because we are rich, we say, "I want to do work not only that's going to keep body and soul together, but is interesting, exciting, passionate. It's going to make a difference in my life and in the world." That's a very modern mindset. Where does that come from and how do we teach it to other people?
- 8.95** Some data about this entrepreneurial world that we are entering into. That's to say, instead of, somebody else has decided what your job is going to be, and you fit into a pre-existing slot in the work world, the idea of, well, I'm a free agent. Not only can I choose from a lot of different jobs, but I could also make my own job if I want, have my own idea for a new business, and I'll start that business.
- 8.96** Even if I'm interested in getting a job, chances are good that I am going to be working in a company that is pretty close to its entrepreneurial roots, written by people who said, "I don't like any of the pre-existing jobs. I want to do my own thing." They're looking to hire people who think like they do, as entrepreneurial types of people.
- 8.97** One set of data here, from a business professor at Harvard who's tracking entrepreneurship, I'll just read the quotation. "In the 1960s, one out of every four persons in the United States worked for a Fortune 500 company." All right, so go back to the middle part of the 20th century. We are educators, we are parents, we're raising kids. We're thinking, OK, well, I want my kid to have a good education, have fun and so forth, but I also want the kid to get a job. It would be great if my kid could get a job at one of the really big, prestigious Fortune 500 companies, make good money, really stability, because it's likely to be around here for as well.
- 8.98** So interesting that, though, even by the middle part of the 20th century, one out of every four is working for the Fortune 500 companies. That's the biggest 500 companies in the country, the really huge giants. And that's impressive. But it also means that three out of four are not working in these giant companies. They're working in smaller companies. In many cases, they're working in family businesses, sole proprietorships, and so forth. So the idea of working for a big corporation with all of its bureaucracy and so forth, however stable and so forth, that's that data.

- 8.99** But if we look at today, and this was now from the early part of the 20th century, there'd been, just in that 40 years, a shift, from one in four to one in 14. "Today, only one out of every 14 people works for these companies." That's a dramatic shift. That then is to say, it's gone from 25% to maybe 6% in a generation. That trend does seem to be continuing as we go on.
- 8.100** That means the idea of going to work for a big corporation that might be hierarchical, it's got an HR manual that's this thick and everything is rule-bound and so on, and you're reporting to managers who are reporting to managers who are reporting to managers and so on, yes, that is a reality. That is a big reality, but it's not the whole reality. Chances are good that most kids are not going to be in those kinds of jobs.
- 8.101** Another example, even if we look at those biggest companies in the United States, and this is a striking thing, also some Fortune 500 data from the year 2000 on the left, these are the top 10, the biggest corporations in the United States. Since the United States is the biggest economy at that point, these are some of the biggest companies in the world.
- 8.102** Number one, General Motors, making cars. Walmart, number 2. Exxon, that's oil and gas products, number 3. Ford, more cars. General Electric, appliances. IBM, computers, business machines. Citigroup, banking and finance. AT&T, telecom. Philip Morris, tobacco and related products. And then the Boeing Company, number 10, airplanes.
- 8.103** Now, what's interesting about those companies is, these are the big 10 in the year 2000. If you go down 20 years earlier to 1980, how many of those companies were in the top 10? Not many of them, but you can look at the data. If you go back to 1960, a lot of them were young companies and not in the top 10. 1940, 1920, and so on.
- 8.104** Then more strikingly, if we then say that's the year 2000, we jump 20 more years, we consult the Fortune 500 list, and here are the top 10. Walmart, Amazon, Apple, CVS Health, United Health Group, Berkshire Hathaway, McKesson, AmerisourceBergen, Alphabet, ExxonMobil. When we pause and reflect on that list, what's interesting is, eight of those 10 were not on the list in 2000. Eight of the 10 didn't exist, or they existed as young startups with a couple of guys in a garage, hoping to make it big. Then suddenly they've had this entrepreneurial idea, good idea. It exploded.
- 8.105** Then what are the two that are still on the list? Well, Walmart went from number 2 to number 1. Good job, Walmart, for being very entrepreneurial and adaptive and innovative continuously, not resting on its laurels. Then the other one is ExxonMobil, which is, on the 2020 list, number 10. It was number 3, so it's in relative decline. Now, here's the question, in 2040, which is now only for us, 15 years away, will Exxon still be on the list? Well, we don't know. How many of the other companies will still be on the list? We don't know.
- 8.106** What in part that means is that the companies that are currently the big companies in the world are very close to their entrepreneurial roots. They're not thinking in terms of,

we want to hire people who are safe and dependable and know how to follow orders. We're going to be doing the same thing, following the recipe for the next 30 or 40 years until you retire and we'll give you a gold watch and then forget about you after six months. That's not what they are looking for.

- 8.107** They're looking for young people who think and act like they do, like entrepreneurs, because they are an entrepreneurial company and they want to stay on the top-10 list and more. But also, if history repeats itself in this churning, innovative, rapidly changing economic world that we have been creating in the modern world, the companies that are on the list in 2020, many of them weren't there in 1980, weren't there in 1960, weren't there in 1940, '20, and so on.
- 8.108** There's constant rising up and coming down, and it's the most successful and entrepreneurial and innovative and hard-charging of the companies that make it. Now, if that's the entrepreneurial reality, then what that means in part is that the kids who are graduating right now, if they're going to go to work in the big firms, or they want to go to work in the small firms that want to become big like that, what kind of education is going to enable them to get a job at that kind of firm and be successful in that kind of firm? That's what we need to be thinking about, if we are responsible professional educators.
- 8.109** The problem gets harder, I wanted to say, because if we do the question I floated a few minutes ago and ask, "What's the list going to look like in 2040?", the right answer to that is, "I have no idea." Nobody has, really, any idea what is going to be the top 10, and what kinds of things they're going to be making, and what kinds of things they are going to be doing.
- 8.110** But if we are taking children right now who are entering into the school system or whatever model of education going right now, and they're going to be with us for the next 10, 12, 15 years or so, and then they're going to be graduated, and we're going to give them the stamp of approval—here's your diploma, kid; you're off into the world—go find a job in what kind of company? Doing what kind of skills, with what kind of knowledge? So whatever we're going to do for the next 15 years gets you ready for that. And that's a very hard educational philosophy question.
- 8.111** Now, we know more specifically, we're in the midst, not only of an entrepreneurial revolution in the mindset, but technological. The industrial revolution has not stopped. You can see all sorts of artificial divisions. Well, first it was steam, and then it was electricity and electronics and so on. You can parse out that, along that. But it's been this continuous, new kinds of technology, new kinds of energy sources driving it and so on.
- 8.112** The robots that we are talking about now, the new—it just seems like it's at an inflection point of increasingly sophisticated robotics, but nonetheless, that's going to continue and perhaps even at an accelerated rate. How will this impact the work

environment and the children that we are preparing for work? Now, a century ago, we might say, oh, yeah, industrial revolution. We're all about the industrial revolution.

- 8.113** But I know I can get kids who are not necessarily college track, but they like to work with their hands, and I'm going to prepare them to go and work. They can always get a job at Ford, at General Motors, making cars or going, working in the big factories. That will be great for them. Well, this is a car manufacturing facility. This picture is now a couple of years old, but we know that just 10 years ago, there would have been 50 people working there, in this photographic frame. Now all of those people are gone, and the robots are assembling all of the cars.
- 8.114** We multiply that. That's not just the automotive industry. It's all of the other industries, and it's happening on a regular basis. Our question then is, as professional educators, all of the young people who like to work with their hands, who like to make stuff with their hands, who are doing what we sometimes call blue-collar work, what's that going to look like in 20 years? And how do I educate them so that they can, in this case, say, outcompete the robots? Whatever outcompeting the robots is going to mean.
- 8.115** And then the other big revolution that we are in the midst of—information processing. Now, again, we can say, this has been continuous. Gutenberg's press revolutionized information processing. We started hooking it up to steam machines and mass-producing all sorts of books and newspapers. Then we developed telephones and radio and television and then that internet thing and so on. All of those are information processing revolutions, and they've changed the economic landscape and the kind of work, creating all sorts of work by destroying certain old jobs. Now we are doing the same thing with artificial intelligence.
- 8.116** We know that that's just going to continue to be progressing. But then suppose we are thinking now about our children and our students, who want to have a career, want to have an interesting career. Maybe they're not so much, they don't want to work with their hands. They like thinking about stuff. They want to have what we call maybe a white-collar kind of job.
- 8.117** But in our recent generation, we had thousands and thousands of people employed as translators. Will we need translators now that we have Google Translate and the various other devices? All kinds of people who would be transcriptionists—every single court case that goes on, there is someone who is making a transcript of the record.
- 8.118** Every single surgery that has occurred, the surgeon is speaking into a recorder, everything that he or she is doing through the course of the surgery, and then that recording is transcribed for the hospital records. But then we know we have AI voice transcription, so we don't need the translators, we don't need the transcriptionists.
- 8.119** If I'm a lawyer, for example, and I got a new case, and I'm part of a big firm, and millions of dollars are at stake, and I say, let's plunge into research, and I've got 20 paralegals, and I send them into the library archives to find all precedent cases. These

are smart, young legal graduates, and they come back after a couple of weeks of work and present all the precedents that are relevant. Do I need 20 paralegals anymore, or can I have ChatGPT or Grok do a search of all of the legal archives and pull out the relevant? So what happens to those paralegal jobs?

- 8.120** And insurance adjusters, and you go through the list and list and so on. Again, millions of people in the blue-collar sector, millions of people in the so-called white-collar sector. Nonetheless, we know that the historical record is that pretty much every new technology, for every job it destroys—McKinsey did a study of this—it creates three and a half new jobs. That's why we have more and more people working, but they're working at things that we would not have imagined 30 years ago, 50 years ago, 80 years ago.
- 8.121** One of my favorite anecdotes at this point always is to talk about my mother, who, when she was a modern, progressive young woman going to high school, first person in her family ever to graduate from high school, she was taught, like all of the other girls of her age, you need to learn typing. Typing is the path to success. If you don't want to be a teacher or a nurse, learn how to type well and you will always have an office job. Well, we don't want to train typists anymore. And chances are very good that a lot of what we are doing right now is training people in the equivalent of typing in 20th century.
- 8.122** Computer programmers. Everybody, 20 years ago, if you had a computer programming job, will we need computer programmers in the next generation? We need to think through all of these things. Here's the challenge for philosophy of education and professional strategy about education, as we look into the future. Chances are very good that our kids are going to be working at jobs that don't exist yet.
- 8.123** Camera men, sound editors, video editors, we're surrounded with them here. 100 years ago, nobody knew what those jobs were. Airline pilots, et cetera. The list is long. We're constantly inventing new jobs as a result. We're preparing them for those jobs, but those jobs don't yet exist. So how do we do that? That's our challenge. I want to make some suggestions. This is only suggestions.
- 8.124** It's only a partial answer because this is a great, huge project that requires lots of smart people who are committed to education with their own areas of expertise to work it out in their areas; and then for all of them to start coordinating with other parents, teachers, professional educators, including those who are going to be education entrepreneurs and start the new schools that will have education for the future in it.
- 8.125** This is a, from an article, now 10, 12 years old, but the data are timeless, of senior human resource managers and senior executives who hire a lot of people. Typically, what they will do with these major companies is, every spring, they will go to all of the major universities and try to recruit students. Of course, students will go to the job fairs

and try to get jobs at these companies. The question was, to the people who are doing the hiring, what are you looking for? Initially an open-ended question.

- 8.126** So you just gather all of the answers. And of course, everybody's looking for everything great. They want the perfect employee who can do everything instantly and perfectly and so on—all of these traits that are wonderful. And on the list are things like, well, they need to be able to write, they need to know how to use new technologies, they need to be good at working with teams, they need to have some quantitative skills, they need to have media literacy.
- 8.127** Chances are good, we're going to ask them to solve some problems; we want some creative thinkers, of course, et cetera. We have this long, long list of things that we want people to be able to do. What's interesting is, you talk to thousands of people in HR, and the lists are very similar to each other, and that tells us one thing. Now, it's still an unstructured list.
- 8.128** Then the researchers ask some follow-up questions. All those things on your list, how would you rank-order them, from most essential to least essential? You got to have this one or we're not going to hire you. This one would be nice, but maybe not necessary. So rank-order your list, and that is the horizontal axis here. Things on the left, nice, moderately important, very important or essential.
- 8.129** The other follow-up question that was asked was, when you are interviewing hundreds or thousands of young people, employee candidates, how easy is it to find those traits in the college graduates coming out of usually good universities? And some things are, well, basically all of the kids, all the young graduates can do it. Some of them can do it. Very few of them can do it. So, how frequently you find those ones. Then you plot those results and you get this two-dimensional graph here; and the results are interesting.
- 8.130** If we take, for example, at the left end of the chart, media literacy, so what does that mean? You can read the newspaper, you can go on a browser and do the search engine and find your way around to major news of the world and report what's going on in the world. OK. Pretty easy to find. Most kids can do that. I'm calling them kids. They're college graduates or pretty darn close to it, but not very important.
- 8.131** What about proficiency with new technologies? Now, there is a difference in mindset between people who say, "New technology is scary. I resist having to learn these new things," versus "Oh, new tech, what can I do with this? I want to take this home and play with it on the weekend and come back on Monday. Do you know it can do this, that, and the other thing?" That's a range. Not very essential and moderate. Most kids are able to be comfortable with the new technologies.
- 8.132** Oral communication. Essential, not super essential, but essential. Can I talk coherently? Have a conversation, listen well, understand, make my point intelligibly, concisely, craft a clear email, run a meeting, etc.? Pretty important, but again, pretty easy to find also.

Teamwork, essential, and pretty easy to find. OK, so that's going along one vertical axis. Now we started going along the vertical or the horizontal. Now we go up the vertical.

- 8.133** Finding people with writing skills. Moderately important, but hard to find. Can you write an email? Can you write a paragraph? Can you write a 10-page report? Can you write a dissertation? Can you write the HR manual? Can you summarize these legal precedents in 100 words? And not all jobs require writing. It seems very difficult to find people who can do those things.
- 8.134** Creativity. Not quite as important, interestingly. A lot of the works, especially at the entry level, are not going to be particularly creative. But it's hard to find even college graduates who are able to think creatively. As I say, they might be able to solve the problems that their professors have solved for them, and solve problems that are just like that; but give them a new problem that they've not seen that kind of problem before, they're not sure what the answer is.
- 8.135** Do they know how to come up with two or three hypotheses, play around with them, mental models, assess for and against, and come up with a likely new solution? Very rare to find people who can do that. Then even more so, math abilities. This is a sobering one in one respect, but it seems like hardly any graduates have strong math skills. What this probably then means, even in this incredibly high-tech scientific world that we are living in, it seems like maybe 5% of us are doing all of the math for everybody else. If you find a math person, and that matters to your company, snap that one up. Rare, but again, not as essential.
- 8.136** Now, the really interesting part that might be a guide to answering our question, how do we prepare young people for jobs that don't exist yet, and we don't even know what those jobs are going to be, are the things in the upper right quadrant. Critical thinking and problem solving. Those are the only two that emerged as employers saying, these are absolutely essential for the kinds of people we are looking for, and they're hard to find.
- 8.137** What this then indicates is, if one can graduate with those two capacities, that's going to give you a competitive advantage against your peers, but it also is going to be your most likely chance of remaining competitive with the robots and with our artificial intelligence. Because problem solving and critical thinking are the cutting edge. Those are the most, perhaps distinctively human higher cognitive powers.
- 8.138** Now, what do we mean by problem solving? Part of it is tied into creativity, being able to solve new problems, ones that don't have the answers at the back of the textbook, or it's not in the boss' head. You just say, "Hey, what's the answer to this, Bob?" Or you can't go on your browser and just do a Google search or Bravo search or whatever and get the answer to it because it's out there. It's a new problem that needs to be solved.
- 8.139** We know, in an entrepreneurial economy, we're making new things. There's always new problems that need to be solved and people who need to know how to solve new

problems. That's what we want. Before AI can take the new solution to the problem and put it in its database for crunching and doing whatever it does, somebody first has to solve that problem.

- 8.140** Critical thinking. What do we mean by critical thinking? It means partly creativity, being able to come up with a new idea, but also being able to critique that idea. Sometimes people come up with lots of new ideas, but which ones are good ideas? It's easy to fall in love with the first idea that you come up with. I made this idea; it's wonderful. You want it to be true, but to be able to step back and say, well, maybe it's not very good. What are the strengths? What are the weaknesses? Do the weaknesses outweigh the strengths or the strengths outweigh the weaknesses?
- 8.141** To have that critical attitude with respect to my own ideas and be able to say, well, actually, that's not actually a very good idea. I have to reject that idea and then start over again and come up with another idea. But I'm committed to that process and I know how to do that; and to do that in a social context because, as we know, there's a huge amount of teamwork that we also are looking for. I can come up with an idea and here we're having a team meeting, we're trying to solve this new problem, and part then of critical thinking is going to be in a social context.
- 8.142** I say, well, what do you think about this idea? You say, well, nice idea, but I have this objection, and you articulate your objection, and I don't immediately hate you because you criticize my idea. Instead, we say, actually, that's a good criticism. Good point. I didn't think about that. That's part of critical thinking. Being able to have my feelings bruised because that was my pet idea. But nonetheless, I am critical with respect to it and I appreciate your contribution, if it, in fact, is a good criticism.
- 8.143** Then you, of course, being able to do the same thing, to be able to take the criticisms without getting defensive. Then to the extent that we are managers of the organization, setting up the social environment so people learn that it is OK to put your own weird ideas out there and also to welcome criticism and to feel comfortable with criticizing other people.
- 8.144** Now, that is a managerial skill, in a business-social context, but that chart tells us this is something that we teachers and parents need also to be figuring out how to do in our classrooms or around the dinner table, having the environment where people can put ideas out there and they can be discussed critically, but nonetheless, benevolently.
- 8.145** This was, I think, very nicely articulated by Eric Schmidt, when he was the CEO of Google. "You have to have a culture that is explicitly tolerant of the crazy idea, the criticism of it, and then its renewal." So this is a successful entrepreneurial culture. If that's indicative of the future of the work world, the companies that are going to be the most successful, how are we going to do that as parents, as teachers, and other educators in our classroom?

- 8.146** I'll end with a little piece of humor right on this point. I hope it's humor. Well, we say, particularly in modern times, we want our kids to grow up to be independent and creative and cooperative and have a sense of adventure about their lives, so I'm going to raise my kids that way. We're going to have classrooms that are that way and so on. Of course, we do know the challenge is, that is an invitation for us for some anarchy and some frustration. So, I do like this joke.
- 8.147** When the mother said, "When I said I wanted to raise my daughter to be independent-minded, strong-willed, and filled with her own opinions, I meant after she moved out of the house." There is an occupational hazard for us. We don't want to be the understandable hypocrite teacher, because if our thing is, no, we're going to have obedient children who follow the rules in house, that's not going to prepare them for the real world. The same thing in our classrooms.
- 8.148** We have to start at the home. We have to start in the classrooms if we're going to go there. If the message is, you are the entrepreneur of your work life, and we generalize that, you are the entrepreneur of your life, then our question as philosophers of education and practitioners of education is, how do we operationalize that, now that we are well into the 21st century and have a few centuries of entrepreneurship behind us?
- 8.149** Perhaps the biggest theme of this course, eight lectures on the philosophy of education, is how much education strategically is informed by philosophical decisions. What I have done is, I've selected, in the eight lectures, eight very different approaches to education, each of them having behind it very different understandings philosophically. We've now, for well over two and a half millennia, been arguing about everything in philosophy, and there are very articulate versions of all of those philosophies out there.
- 8.150** There are philosophical trends when one or the other goes into decline and others ascend, and much of tracking history, interestingly, can be seen as philosophies working themselves out, being successful or not, and then being replaced by other philosophies with historical impact. But if we think of education for life, and what we want is to live a meaningful life, to lead the best life that we possibly can, those are all questions that we have to ask for ourselves, and they very quickly become philosophical questions.
- 8.151** But if, as educators, we want to be teaching other people, then we have to be extra clear in our thinking. It has to be not only this is what I believe and I operationalize it in my life, I have to know how to make it real and compelling for other people who are my students in various ways. What we've seen is a full range of philosophical questions, from metaphysical questions about reality, to epistemological or cognitive questions about knowledge, questions about human nature, our identity as physical and psychological beings, questions about morality, questions about social philosophy, and questions about politics.

- 8.152** Do we think the world is a natural world, or also there's a supernatural dimension beyond it? That takes us in very different educational directions. Cognitively, do we want to train people to be rational, to be emotional, to have faith, to follow authority, informing their core beliefs? Again, major junctions in the road in terms of our educational strategy.
- 8.153** Do we think that children are born neutral, morally, neither good nor bad, but with capacities that can be developed? Or do we think that people are naturally tending to the good or naturally tending toward the bad? What's our raw material in education, so to speak? We have to understand what we take our raw material to be, so we know what we are doing with different forks in the road.
- 8.154** Do we think human beings are primarily individual beings or social beings? There's, of course, a continuity here. We can say people are primarily individuals, who then voluntarily form various sorts of social networks. We are both. We're not just lone wolves. Or do we think we are social beings? Perhaps our individuality is an illusion or secondary, but really, we are part of a larger organic whole. Again, if I think I'm teaching a social being primarily or an individual being a fork in the road for education strategy.
- 8.155** Then all of those then, jointly, will help us answer the big questions. If I want to say, time is limited, resources are limited in my education space, what's the core? What's the most important knowledge or the most important skills or the most important character traits? Then I've got my mission and my formula and why I believe those things, and then I can hopefully get on with the fun and challenging product of education.
- 8.156** Dr. Hicks, we started out in lecture one, where you were asking us what the ideal schooling would look like. I could tell you personally, I went from my schooling and improved on it because I have frozen in time my experience of education. But then we looked at all the philosophies through the different lectures, and I am, as an educator, as a governor of a school district, seeing all of those parent groups follow those philosophies.
- 8.157** Then we end it with the idea that our kids probably need critical thinking and problem solving. As educators, how do we implement that in an area of dealing with different parent groups that fit in each of these philosophies that are pretty much frozen in time of what they think education should be like?
- 8.158** So chances are good, if you're the governor of a pre-existing school system, then you are beholden to certain structures that make decisions. You have to report to them, get their approval. You also are in a situation where there's a pre-existing institution that has its own inertia, going in a certain direction. As you say, since you have constituents who are representative of all of the different philosophical approaches, that's the reality on the ground.

- 8.159** Beholden to power, how much realistically can we change, given the inertia, and then all of the diplomatic skills that are going to be necessary to deal with all of the different constituencies? So I think that's very hard, and I wish you well at doing so. But I think probably the best that you can do in that circumstance is, be yourself a decent, thoughtful, diplomatic person who has thought through all of these issues.
- 8.160** To the extent that you do that, you have confidence in your own mind. Then that means you will present, when you're dealing with all of these other constituencies, an articulate voice for what you think the right set of answers are. But also, you will develop a reputation among all of those constituencies for being that kind of guy. You will be more than other people, say, who are not so diplomatic or who are more confused in their thinking, or are just jerks in various ways. They will be less successful at educating or instituting the kinds of reforms that you want.
- 8.161** Now, that's speaking only to that one particular context. If the context is, the ideal context, suppose you had an unlimited budget and you're starting your own school, well, at that point, you've got the power, you've got the resources, so you don't need to worry about a lot of those things anymore. What you then need to know is, how confident am I that I've thought through all of these issues, that I'm in a position to start from scratch a brand new school? That would be the other end of the spectrum. I think short answer would be, how to answer that question would become more specific and tailored to the specific context that an education reformer finds himself or herself in.
- 8.162** If the skills summed up that would be most important for the future are critical thinking and problem solving, would it not be correct to say that going towards the Montessori-type style would be probably a better approach and more an individualistic-type approach? Because I see those two traits are very much so more to the Lockean than progressive.
- 8.163** I think that's a good implication. If we then say that we take that data seriously and we say, so the top educational values are going to be problem solving and critical thinking, then we can take the eight or so different philosophies of education that we looked at and evaluate them in terms of how well we think they will operationalize and succeed at doing those.
- 8.164** Then you, I think quite rightly, would say, clearly Montessori has a leg up on that because, right from the beginning, it's cultivating the independence and creativity and the basic skills that lead to that sort of stuff. They're doing problem solving right from the beginning. Yes. Then we might then also say, well, maybe the Prussian model would be at the opposite end of that, because that's all duty and structure and no deviation, and the answers are already set and so forth. Then array the others along that spectrum.

- 8.165** Now, one caveat, though, is that that data was, in one sense, a subset of the broader question I think we education strategists need to think about. That is, surveying people who are hiring for work in existing organizations. That's what they are looking for. We might say that's a lot of students in our cohort, but a lot of our students are not ever going to be, we might say, looking for jobs. Maybe they're going to become entrepreneurs, or they're going to maybe go in some other career path. There would still be an open question how much problem solving and critical thinking are needed for those other students as well.
- 8.166** It's also a somewhat nested claim in that it's talking about people's work life. We want to say, well, work life is a big part of life, but it's not the whole of life. There's also friendship and romance and people's leisure time and what they're going to do when they are not working. And part of our education strategy has to be focused on developing for that. And then the open question would be, how much do we think problem solving is important for that, or creative thinking.
- 8.167** You say, well, maybe I don't plan to go to work for someone else. Maybe I want to have a big family. I want to have eight kids. And so that's a perfectly legitimate career path for someone to follow. Now, we might then say, well, actually, that's going to involve a lot of problem solving. So yes, problem solving still fits into the mix. It might involve a lot of critical thinking and creative thinking, because we know kids ask all kinds of oddball questions and so on. So it might be that those two traits survive multiple domains and so on.
- 8.168** But I think, yes, at least in terms of this lecture, and I think I would make the claim that probably it's fair to say that Montessori, of all of the ones we looked at, is the one that is most along the path of maximizing things in that upper right quadrant as we speak of it. I just add one more point, it might just be that Montessori put forth, as a scientific approach to pedagogy, such as we're learning human biology, we're learning human psychology, that it will continue to evolve, but that it has absorbed, say, the best parts of the Lockean model.
- 8.169** Some of the Lockean model has already absorbed parts of the Aristotelian model as well. It might be that this is a continuously updated, more basic viewpoint as time goes on. All right, well, let me thank you for your attention. It was a lot of fun for me, and I hope so for you, too. All right?